Prediction Risk After Observation Compression

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Observation sketching compresses n observations into m linear combinations while retaining all d features, thereby reducing the number of equations used for fitting. We study how to estimate the prediction risk of the resulting predictor, defined as its mean squared error on a new observation while treating the fitted predictor as fixed. We propose two estimators based on residual information from the original data. The first uses residual directions not captured by a data-independent sketch. Under Gaussian data, it is conditionally unbiased and yields confidence intervals for positive risk. The second applies an observable residual correction, whose accuracy is characterized using Gaussian–Stein identities and trace calibration. For Gaussian data and scaled orthogonal sketches, we derive a finite-sample comparison of the estimators' mean squared errors and identify a sketch-size threshold at which their relative accuracy reverses. Neither method requires refitting the model. We evaluate the estimators using row-sampling, weighted-Haar, and dense-Gaussian sketches under Gaussian and uniform errors. With Gaussian errors and $m < n$, both estimators follow the mean true risk more closely than generalized cross-validation. For weighted-Haar and dense-Gaussian sketches, they are also closer to the mean true risk than Bellec's formula at larger sample sizes.

1. Introduction

Observation compression reduces the number of equations used to fit a predictor. In linear regression, an observation sketch replaces n observations with m linear combinations while retaining all d features. When $m < n$, the resulting regression problem has fewer rows but the same feature coordinates. A sketch chosen independently of the data determines which combinations of the original observations enter this smaller fitting problem. While compression can make fitting more parsimonious, it can also change the prediction risk of the fitted model. This raises a natural question: after fitting a predictor from compressed data, how can we use the original observations to estimate its prediction risk on a new observation?

Clarification (symbol m):

m is the deterministic number of rows of the full-row-rank sketch $\boldsymbol{S} \in \mathbb{R}^{m\times n}$, with $1 \le m \le n$; it may vary with n in asymptotic sequences. In Theorem 3.1: $m < n$, so the complementary subspace has positive dimension $n - m$; no restriction $d < m$ is imposed by the complementary-law construction. In Theorem 4.2: The noninterpolating proportional regime has $d < m \le n$ and $(d/n, m/n) \to (\gamma, \rho)$ with $0 < \gamma < \rho \le 1$. In Theorem 4.3, Theorem 4.4: $d + 3 < m < n$. In Theorem 4.4: $(d/n, m/n) \to (\gamma, \rho)$ with $0 < \gamma < \rho < 1$; the local crossover definition treats $\rho_n = m/n$ as continuous, while the actual sketch size remains integer. In Theorem 4.1, Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10, Lemma B.11, Lemma B.12, Lemma B.14, Lemma B.15, Proposition B.18, Proposition 40, Lemma B.20, Lemma B.21, Lemma B.22, Proposition B.23: The standing noninterpolating range is $1 \le d < m \le n$, and for all sufficiently large n the balanced-sketch dimension gap satisfies $m - d \ge \kappa n$ for a fixed $\kappa > 0$.

Clarification (symbol d):

d is the deterministic positive integer feature dimension at each sample size, and may vary with n. In Theorem 4.2: The dimension sequence satisfies $(d/n, m/n) \to (\gamma, \rho)$ with $0 < \gamma < \rho \le 1$. In Theorem 4.4: The dimension sequence satisfies $d + 3 < m < n$ and $(d/n, m/n) \to (\gamma, \rho)$ with $0 < \gamma < \rho < 1$.

A difficulty is that the training residuals reuse the responses that determined the fitted coefficients, so they cannot generally be treated as independent assessment errors. Row sampling provides a

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simple example of how to avoid this dependence: observations not selected for fitting can be used to evaluate the fitted predictor. Dense sketches are different because they can mix all n observations, leaving no individual observations unused. Nevertheless, when $m < n$, the sketch leaves directions in the original observation space that are orthogonal to its rows. These directions do not enter the compressed fit, and residuals along them can therefore provide information about prediction risk. This observation reveals a tradeoff controlled by the sketch size: increasing m provides more directions for fitting but leaves fewer directions for assessment.

Existing work on sketching has largely studied what is gained or lost in the fitting step. In numerical linear algebra, random projections and row sampling can approximate the full-data regression objective or coefficient vector by solving a smaller regression problem [1]. The surveys of Mahoney [2] and Woodruff [3] describe the computational savings and matrix operations underlying these methods, while Pilanci and Wainwright [4] studied iterative sketches for obtaining accurate solutions to constrained least-squares problems. Statistical work has gone further by asking how compression affects estimation and prediction. Ma et al. [5] analyzed the bias and variance of regression estimators based on statistical leverage-score sampling. Raskutti and Mahoney [6] distinguished residual efficiency from fixed-design prediction efficiency, showing that a good numerical approximation need not imply good statistical prediction. Dobriban and Liu [7] derived limiting efficiencies for parameter estimation and prediction under several sketch families when the sample size and feature dimension grow together, and Chen et al. [8] studied prediction risk and sketch-size selection for unpenalized regression, including regimes in which downsampling improves generalization. Together, these results describe how the size and structure of a sketch affect the predictor obtained from compressed data. Here, we focus on how to estimate the risk of the particular predictor obtained from an observed dataset and sketch.

This question is closely related to the broader problem of estimating prediction error from the data used for fitting. Generalized cross-validation (GCV) adjusts training residual error for model complexity [9]. More recently, Bellec [10] developed out-of-sample error estimates for regression with convex penalties under Gaussian features, and Patil et al. [11] estimated the conditional prediction-error distribution for high-dimensional ridge regression. Results specific to subsampling and sketching have also appeared. For subsampled estimators, Bellec et al. [12] developed corrected GCV and risk-estimation bounds; their full-data criterion for a single row-subsample fit that exactly matches the sampled responses agrees with the estimate based on unused observations. Patil and LeJeune [13] showed that ordinary GCV can be inconsistent for finite observation-sketch ensembles, including a single sketch, and described a correction using both sketched and unsketched information. These developments motivate a more direct question for observation sketching: what residual information remains after compression, and how accurately can it estimate the risk of the compressed-data predictor?

We answer this question in two ways. The first method uses residual directions orthogonal to the sketch rows, extending the idea of evaluating a row-sampled fit on unused observations to general observation sketches. The second uses the full-data residual but corrects for its dependence on the fitted coefficients. Both methods evaluate the same predictor fitted from the compressed data and require no refitting. They use the original observations, or quantities sufficient to compute the relevant residuals. The two approaches also use the available residual information differently, which allows us to study how their relative accuracy depends on the amount of compression.

Our main results characterize these two approaches. Under Gaussian features and errors, we first derive an exact conditional residual law for any linear predictor constructed only from the compressed data, provided that the independent sketch has linearly independent rows and $m < n$. This result gives a conditionally unbiased estimator of prediction risk and a confidence interval for positive risk. For least squares with $d < m$, we then derive an observable correction to the full-data residual, together with a reference based on the population feature covariance. Under the stated assumptions, their expected absolute errors are bounded by a constant times $n^{-1/2}$. For sketches with independent Gaussian entries, we further establish consistency under Gaussian features and errors and quantify the discrepancy between ordinary GCV and prediction risk when the dimen-

sions grow proportionally. Finally, for scaled orthogonal sketches under Gaussian data, we obtain a finite-sample comparison of the mean squared errors of the two risk estimators and derive a limiting sketch-size threshold at which their relative accuracy changes. These results show that compression determines not only the predictor being fitted but also the residual information available for assessing it. In particular, they clarify when that remaining information is sufficient for accurate risk estimation and how the sketch size affects the accuracy of different assessment methods.

2. Compressed regression and prediction risk

For $i = 1, \dots, n$, let $y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta}^* + \varepsilon_i$, with a fixed coefficient vector $\boldsymbol{\beta}^* \in \mathbb{R}^d$. The feature vectors are independent and identically distributed, with mean zero and covariance $\boldsymbol{\Sigma}$, whose eigenvalues lie between fixed positive constants. The errors are independent and identically distributed, independent of the features, with mean zero and variance σ^2. Write $\boldsymbol{X} = (\boldsymbol{x}_1, \dots, \boldsymbol{x}_n)^\top \in \mathbb{R}^{n \times d}$, $\boldsymbol{y} = (y_1, \dots, y_n)^\top$, and $\boldsymbol{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_n)^\top$. A sketch $\boldsymbol{S} \in \mathbb{R}^{m \times n}$, $1 \le m \le n$, is independent of the data and has full row rank almost surely. The compressed data $(\boldsymbol{SX}, \boldsymbol{Sy}, \boldsymbol{S})$ are then used to produce the fitted coefficients $\hat{\boldsymbol{\beta}}$. The corresponding fitted predictor maps a feature vector $\boldsymbol{x}$ to $\boldsymbol{x}^\top \hat{\boldsymbol{\beta}}$.

Clarification (symbol β^*):

$\boldsymbol{\beta}^* \in \mathbb{R}^d$ is the fixed coefficient parameter in the response model, with its value specified under the response law being used. In Lemma B.21, Proposition B.23: Under $\mathbb{E}_{\text{aux}}$, $\boldsymbol{\beta}^* = \mathbf{0}$; the original response law retains its original coefficient parameter. In Lemma B.15: The additional fitted-coefficient moment bounds assume $\|\boldsymbol{\Sigma}^{1/2}\boldsymbol{\beta}^*\| \le B$; this restriction is not imposed for the coefficient-error and residual moment bounds.

Clarification (symbol Σ):

$\boldsymbol{\Sigma}$ is the population feature covariance at each sample size, with $c_{\boldsymbol{\Sigma}} \boldsymbol{I}_d \preceq \boldsymbol{\Sigma} \preceq C_{\boldsymbol{\Sigma}} \boldsymbol{I}_d$ for fixed constants $0 < c_{\boldsymbol{\Sigma}} \le C_{\boldsymbol{\Sigma}} < \infty$ independent of n.

Clarification (symbol σ^2):

$\sigma^2 = \mathrm{Var}(\varepsilon_i)$ is the population noise variance in the original observation model. In Theorem 4.2, Theorem 4.3, Theorem 4.4: $\sigma^2 > 0$.

Clarification (symbol $\boldsymbol{x}_i$):

$\boldsymbol{x}_i \in \mathbb{R}^d$, for $i \in \{1, \dots, n\}$, is the feature vector of the ith original observation and the transpose of the ith row of $\boldsymbol{X}$.

Clarification (symbol ε_i):

For $i \in \{1, \dots, n\}$, $\varepsilon_i = y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta}^*$ is the structural observation error and the ith coordinate of $\boldsymbol{\varepsilon}$.

Clarification (symbol $\boldsymbol{y}$):

$\boldsymbol{y} = (y_1, \dots, y_n)^\top = \boldsymbol{X}\boldsymbol{\beta}^* + \boldsymbol{\varepsilon}$ is the response vector for the response law under consideration. In Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10: Under design differentiation, $\boldsymbol{\varepsilon} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}^*$ is held fixed, so $\boldsymbol{y} = \boldsymbol{Z}\boldsymbol{\Sigma}^{1/2}\boldsymbol{\beta}^* + \boldsymbol{\varepsilon}$ varies with $\boldsymbol{Z}$. In Lemma B.21, Proposition B.23: Under the auxiliary response law, use the auxiliary response $\boldsymbol{y} = \boldsymbol{\varepsilon}$ when recomputing response-dependent quantities.

Clarification (symbol $\boldsymbol{\varepsilon}$):

$\boldsymbol{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_n)^\top = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}^*$ denotes structural noise rather than fitted residuals. In Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10, Lemma B.14, Lemma B.15, Proposition B.18, Proposition 40, Lemma B.20, Lemma B.21, Lemma B.22, Proposition B.23: Design derivatives are pathwise with $\boldsymbol{\varepsilon}$ held fixed, together with $\boldsymbol{S}$, $\boldsymbol{\Sigma}$, and $\boldsymbol{\beta}^*$; consequently $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta}^* + \boldsymbol{\varepsilon}$ varies with the design. In Lemma B.16: The field is differentiated with respect to $\boldsymbol{Z}$ conditional on $(\boldsymbol{S}, \boldsymbol{\varepsilon})$, so the structural-noise coordinates are held fixed.

Clarification (symbol $\boldsymbol{S}$):

$\boldsymbol{S} \in \mathbb{R}^{m \times n}$ is the observation sketch acting on the left of $(\boldsymbol{X}, \boldsymbol{y})$, with $1 \le m \le n$. In Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10, Lemma B.11, Lemma B.12, Lemma B.14,

Lemma B.15, Proposition B.18, Proposition 40, Lemma B.20, Lemma B.21, Lemma B.22, Proposition B.23: $\boldsymbol{S}$ denotes the normalized representative $\overline{\boldsymbol{S}} := \{n/\operatorname{Tr}(\boldsymbol{A})\}^{1/2}\boldsymbol{S}$ constructed from the original sketch, so the reused symbols satisfy $\operatorname{Tr}(\boldsymbol{S}^\top\boldsymbol{S}) = n$. In Theorem 4.1: The statement permits the original sketch scale; its balanced-sketch proof replaces the sketch by the trace-normalized representative, using invariance of the fit and corrected risk estimates. In Theorem 4.2, Theorem 4.3, Theorem 4.4: $\boldsymbol{S}$ retains its natural sketch scale; a scaled orthogonal sketch satisfies $\boldsymbol{S}\boldsymbol{S}^\top = c_n\boldsymbol{I}_m$ with $c_n > 0$.

Clarification (symbol $\hat{\beta}$):

$\hat{\boldsymbol{\beta}} \in \mathbb{R}^d$ denotes the fitted coefficient vector produced using only $(\boldsymbol{S}\boldsymbol{X}, \boldsymbol{S}\boldsymbol{y}, \boldsymbol{S})$, hence is $\mathcal{F}_{\boldsymbol{S}}$-measurable. In Theorem 4.1, Theorem 4.2, Theorem 4.3, Theorem 4.4, Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10, Lemma B.11, Lemma B.12, Lemma B.14, Lemma B.15, Proposition B.18, Proposition 40, Lemma B.20, Lemma B.21, Lemma B.22, Proposition B.23: $\hat{\boldsymbol{\beta}}$ denotes $\hat{\boldsymbol{\beta}}_{\boldsymbol{S}} = \boldsymbol{M}^{-1}\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{y}$, the unique sketched least-squares fit, with $\boldsymbol{A} = \boldsymbol{S}^\top\boldsymbol{S}$, $\boldsymbol{M} = \boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}$, and $\operatorname{rank}(\boldsymbol{S}\boldsymbol{X}) = d$. In Lemma B.21, Proposition B.23: When evaluated under $\mathbb{E}_{\text{aux}}$, $\hat{\boldsymbol{\beta}}$ is recomputed by the same fitting construction using the auxiliary response, with $\boldsymbol{\beta}^* = \boldsymbol{0}$ and $\varepsilon \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_n)$.

Let $(\boldsymbol{x}_{\text{new}}, y_{\text{new}})$ be an independent observation from the same model. Our goal is to estimate the prediction risk of the fitted predictor:

$$\begin{aligned} R &:= \mathbb{E}[(y_{\text{new}} - \boldsymbol{x}_{\text{new}}^\top\hat{\boldsymbol{\beta}})^2 \mid \boldsymbol{X}, \boldsymbol{y}, \boldsymbol{S}] \\ &= \sigma^2 + (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*)^\top\boldsymbol{\Sigma}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*). \end{aligned} \tag{1}$$

Here the expectation is over the new observation, while the fitted predictor is held fixed. Thus, R accounts for both noise in the new response and error in the fitted coefficients, measured with respect to the feature distribution. Since $\hat{\boldsymbol{\beta}}$ depends on the observed data and the sketch, R can vary across training samples and sketches. Gaussian design means $\boldsymbol{X} = \boldsymbol{Z}\boldsymbol{\Sigma}^{1/2}$, where $\boldsymbol{Z}$ has independent standard Gaussian entries. *Gaussian data* means Gaussian design and Gaussian errors $\varepsilon \sim \mathcal{N}(\boldsymbol{0}, \sigma^2\boldsymbol{I}_n)$.

Clarification (symbol R):

R is the prediction risk of the realized fitted predictor, integrating only the independent new observation while holding the fitted coefficients fixed; it is a function of the training data and sketch rather than their averaged risk.

We estimate this risk using the residuals on the original observations, $\boldsymbol{u} = \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}$. To describe which parts of these residuals are retained or left out by compression, define

$$\boldsymbol{F} = (\boldsymbol{S}\boldsymbol{S}^\top)^{-1/2}\boldsymbol{S}, \qquad \boldsymbol{P}_{\boldsymbol{S}} = \boldsymbol{F}^\top\boldsymbol{F}, \qquad \boldsymbol{Q}_{\boldsymbol{S}} = \boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{S}}. \tag{2}$$

The row-whitened sketch $\boldsymbol{F} \in \mathbb{R}^{m\times n}$ has orthonormal rows, so $\boldsymbol{F}\boldsymbol{F}^\top = \boldsymbol{I}_m$ and its row space is the same as that of $\boldsymbol{S}$. Consequently, $\boldsymbol{P}_{\boldsymbol{S}}$ projects onto the directions retained by the sketch, while $\boldsymbol{Q}_{\boldsymbol{S}}$ projects onto the directions perpendicular to them. Their ranks are m and $n - m$, respectively, and $\boldsymbol{u} = \boldsymbol{P}_{\boldsymbol{S}}\boldsymbol{u} + \boldsymbol{Q}_{\boldsymbol{S}}\boldsymbol{u}$ gives the corresponding orthogonal decomposition of the residual. This decomposition will be the basis for the risk estimators developed below.

Clarification (symbol u):

Outside an explicit local smooth-field declaration, $\boldsymbol{u} := \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}} \in \mathbb{R}^n$ denotes the residual of the fitted predictor on the original observations. In Theorem 4.1, Theorem 4.2, Theorem 4.3, Theorem 4.4, Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10, Lemma B.12, Lemma B.14, Lemma B.15, Proposition B.18, Proposition 40, Lemma B.20, Lemma B.21, Lemma B.22, Proposition B.23: The fitted coefficient is sketched least squares, so $\boldsymbol{u} = \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}_{\boldsymbol{S}} = \varepsilon - \boldsymbol{Z}\boldsymbol{q}$, with response-dependent quantities recomputed whenever the auxiliary response construction is used. In Lemma B.19, a result cited in the proof of Lemma B.19: In the cited one-vector inequality, $\boldsymbol{u} : \mathbb{R}^{n\times d} \to \mathbb{R}^n$ is a general smooth field satisfying $\|\boldsymbol{u}\| \le 1$, rather than the regression residual; the transposed version interchanges n and d.

3. Risk assessment from the complementary subspace

When $m < n$, the complementary subspace contains $n - m$ directions that are not used for fitting. We use the residuals along these directions to define

$$\widehat{R}_c = \frac{\|\boldsymbol{Q_S u}\|^2}{n-m}.$$

The index c stands for complementary, and the dependence on the sketch is left implicit. To study this estimator, condition on $\mathcal{F}_{\boldsymbol{S}} = \sigma(\boldsymbol{SX}, \boldsymbol{Sy}, \boldsymbol{S})$, which contains the information used for fitting. A $\mathcal{F}_{\boldsymbol{S}}$-measurable coefficient vector therefore depends only on the compressed data. Under Gaussian data, the retained and complementary components are independent conditional on $\boldsymbol{S}$. Conditioning on $\mathcal{F}_{\boldsymbol{S}}$ thus fixes the fitted predictor and its risk R, while the complementary data remain random. We write $\mathcal{L}(\boldsymbol{V} \mid \mathcal{F}_{\boldsymbol{S}})$ for a conditional distribution and χ^2_ν for the sum of squares of ν independent standard Gaussians.

Clarification (symbol $\widehat{R}_c$):

$\widehat{R}_c = \|\boldsymbol{Q_S u}\|^2/(n-m)$ is defined for $m < n$ using the residual of the fitted predictor under consideration. In Theorem 3.1: The residual is $\boldsymbol{u} = \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}$ for any $\mathcal{F}_{\boldsymbol{S}}$-measurable coefficient vector $\hat{\boldsymbol{\beta}}$, where $\mathcal{F}_{\boldsymbol{S}} = \sigma(\boldsymbol{SX}, \boldsymbol{Sy}, \boldsymbol{S})$. In Theorem 4.3, Theorem 4.4: The residual is $\boldsymbol{u} = \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}_{\boldsymbol{S}}$ for the sketched least-squares fit.

Theorem 3.1. *Under the model above with Gaussian data and $m < n$, any $\mathcal{F}_{\boldsymbol{S}}$-measurable $\hat{\boldsymbol{\beta}} \in \mathbb{R}^d$ satisfies*

$$\mathcal{L}(\boldsymbol{Q_S u} \mid \mathcal{F}_{\boldsymbol{S}}) = \mathcal{N}(\boldsymbol{0}, R\boldsymbol{Q_S}),$$
$$\frac{(n-m)\widehat{R}_c}{R} \,\Big|\, \mathcal{F}_{\boldsymbol{S}} \sim \chi^2_{n-m} \quad \text{on } R > 0. \tag{3}$$

Under this regular conditional law, $\mathbb{E}[\widehat{R}_c \mid \mathcal{F}_{\boldsymbol{S}}] = R$ and $\mathrm{Var}(\widehat{R}_c \mid \mathcal{F}_{\boldsymbol{S}}) = 2R^2/(n-m)$, without requiring unconditional moments of R.

Clarification:

On the almost-sure event $\mathrm{rank}(\boldsymbol{S}) = m$, $(\boldsymbol{SS}^\top)^{-1/2}$ denotes the unique symmetric positive definite square root of $(\boldsymbol{SS}^\top)^{-1}$. (the reading of a definition this result uses)

The theorem gives a direct interpretation of $\widehat{R}_c$. Given $\mathcal{F}_{\boldsymbol{S}}$, the $n - m$ residual coordinates in an orthonormal basis of the complementary subspace are independent Gaussians with variance R, so their mean square is an unbiased estimate of R. The chi-square law also gives a confidence interval. If $q_{\nu,p}$ denotes the p-quantile of χ^2_ν and $0 < \alpha < 1$, then

$$\left[\frac{(n-m)\widehat{R}_c}{q_{n-m,1-\alpha/2}}, \frac{(n-m)\widehat{R}_c}{q_{n-m,\alpha/2}}\right] \tag{4}$$

has conditional coverage $1 - \alpha$ when $R > 0$. For $R > 0$, the conditional relative variance of $\widehat{R}_c$ is $2/(n-m)$; fewer complementary directions therefore give less precise assessment. For row sampling, this construction reduces to the familiar use of held-out observations: $\boldsymbol{Q_S u}$ contains the residuals on the unsampled observations, with zeros elsewhere. In this case, conditional unbiasedness requires only the model's second-moment assumptions, while Gaussianity gives the chi-square law. For a dense sketch, individual observations need not be left out, but the complementary directions play the same role. When $m = n$, the complementary subspace disappears and $\widehat{R}_c$ is undefined.

The same conditional law also applies when the compressed responses are fit exactly. When $d > m$ and $\mathrm{rank}(\boldsymbol{SX}) = m$, the minimum-norm fit $\hat{\boldsymbol{\beta}}_{m,\boldsymbol{S}} := \arg\min\{\|\boldsymbol{\beta}\| : \boldsymbol{SX\beta} = \boldsymbol{Sy}\}$ interpolates the compressed equations. Here the index m denotes minimum norm. Since interpolation gives $\boldsymbol{Su} = \boldsymbol{0}$, the residual lies entirely in the complementary subspace, so $\boldsymbol{Q_S u} = \boldsymbol{u}$ and $\widehat{R}_c = \|\boldsymbol{u}\|^2/(n-m)$. For row sampling, this coincides with the single-subsample interpolating formula of Bellec et al. [12].

4. Residual correction for $d < m$

When $d < m$, least squares generally does not fit the compressed responses exactly, so the residual contains both complementary and retained components. The retained component reuses the data used for fitting and therefore requires a correction. Define sketched least squares by

$$\hat{\boldsymbol{\beta}}_{\boldsymbol{S}} := \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{S}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})\|^2 = \boldsymbol{M}^{-1}\boldsymbol{X}^\top \boldsymbol{A}\boldsymbol{y}, \quad \boldsymbol{A} = \boldsymbol{S}^\top \boldsymbol{S}, \quad \boldsymbol{M} = \boldsymbol{X}^\top \boldsymbol{A}\boldsymbol{X}.$$

We assume $\operatorname{rank}(\boldsymbol{S}\boldsymbol{X}) = d$, as holds almost surely under Gaussian design. Here R denotes the prediction risk of $\hat{\boldsymbol{\beta}}_{\boldsymbol{S}}$, $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, and $\boldsymbol{M} \in \mathbb{R}^{d\times d}$. Let $\boldsymbol{u} = \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}_{\boldsymbol{S}}$ and $\boldsymbol{r} = \boldsymbol{A}\boldsymbol{u}$. Adding a multiple of $\boldsymbol{r}$ changes the retained residual while leaving its complementary component unchanged. The correction is determined by

$$b_{\boldsymbol{A}} = \operatorname{Tr}(\boldsymbol{A}) - \operatorname{Tr}(\boldsymbol{A}\boldsymbol{X}\boldsymbol{M}^{-1}\boldsymbol{X}^\top \boldsymbol{A}), \qquad t_{\boldsymbol{\Sigma}} = \operatorname{Tr}(\boldsymbol{\Sigma}\boldsymbol{M}^{-1}),$$

where Tr denotes the sum of the diagonal entries and $b_{\boldsymbol{A}} > 0$ measures the sketch weight remaining after fitting. These quantities lead to two corrected risk estimators,

$$\hat{R}_u = \frac{\|\boldsymbol{u} + t_{\boldsymbol{\Sigma}}\boldsymbol{r}\|^2}{n}, \qquad \hat{R}_o = \frac{\|\boldsymbol{u} + (d/b_{\boldsymbol{A}})\boldsymbol{r}\|^2}{n}.$$

The index u denotes the unobservable version, which depends on the population covariance through $t_{\boldsymbol{\Sigma}}$, while o denotes the observable version based only on the data. When $\boldsymbol{\Sigma}$ is known, $\hat{R}_u$ provides a useful reference for studying the correction. The observable estimator replaces $t_{\boldsymbol{\Sigma}}$ with the trace-calibrated ratio $d/b_{\boldsymbol{A}}$. Both estimators are invariant to an overall rescaling of the sketch. The following result shows that this calibration preserves the accuracy of the population-covariance correction.

Clarification (symbol $\hat{\beta}_S$):

$\hat{\boldsymbol{\beta}}_{\boldsymbol{S}}$ is the unique sketched least-squares coefficient vector on $\{\operatorname{rank}(\boldsymbol{S}\boldsymbol{X}) = d\}$, computed from the response under consideration. In Lemma B.21, Proposition B.23: Under $\mathbb{E}_{\text{aux}}$ the fit is recomputed from the auxiliary response; under the original response law it is the original fit.

Clarification (symbol A):

$\boldsymbol{A} = \boldsymbol{S}^\top \boldsymbol{S}$ is the positive-semidefinite sketch Gram matrix of rank m, using the same sketch representative as $\boldsymbol{S}$. In Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10, Lemma B.11, Lemma B.12, Lemma B.14, Lemma B.15, Lemma B.17, Proposition B.18, Proposition 40, Lemma B.20, Lemma B.21, Lemma B.22, Proposition B.23: $\boldsymbol{A}$ denotes the trace-normalized representative: relative to the original sketch Gram matrix, it is multiplied by $n/\operatorname{Tr}(\boldsymbol{A})$, so the reused symbol satisfies $\operatorname{Tr}(\boldsymbol{A}) = n$. In Theorem 4.1: The statement permits the original sketch scale and imposes its spectral condition on $n\boldsymbol{A}/\operatorname{Tr}(\boldsymbol{A})$; the balanced-sketch proof uses the trace-normalized representative. In Theorem 4.2, Theorem 4.3, Theorem 4.4: $\boldsymbol{A}$ retains the natural sketch scale; in the scaled orthogonal case $\boldsymbol{A} = c_n \boldsymbol{P}_{\boldsymbol{S}}$.

Clarification (symbol M):

$\boldsymbol{M} = \boldsymbol{X}^\top \boldsymbol{A}\boldsymbol{X} = (\boldsymbol{S}\boldsymbol{X})^\top(\boldsymbol{S}\boldsymbol{X})$ is the design-dependent Gram matrix of the current sketch, positive definite on the domain $\operatorname{rank}(\boldsymbol{S}\boldsymbol{X}) = d$. In Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10, Lemma B.12, Lemma B.14, Lemma B.15, Lemma B.17, Proposition B.18, Proposition 40, Lemma B.20, Lemma B.21, Lemma B.22, Proposition B.23: $\boldsymbol{M}$ is formed from the trace-normalized sketch Gram matrix with $\operatorname{Tr}(\boldsymbol{A}) = n$; relative to the natural-scale sketch, it is multiplied by $n/\operatorname{Tr}(\boldsymbol{S}^\top \boldsymbol{S})$.

Clarification (symbol r):

Outside an explicit generic-field declaration, $\boldsymbol{r} := \boldsymbol{A}\boldsymbol{u} \in \mathbb{R}^n$ is the weighted residual at the current sketch scale, recomputed from the current response quantities. In Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10, Lemma B.12, Lemma B.14, Lemma B.15, Lemma B.17, Proposition B.18, Proposition 40, Lemma B.20, Lemma B.21, Lemma B.22, Proposition B.23: The standing weighted residual is $\boldsymbol{r} = \boldsymbol{A}(\boldsymbol{\varepsilon} - \boldsymbol{Z}\boldsymbol{q})$ for the trace-normalized representative, with $\operatorname{Tr}(\boldsymbol{A}) = n$; for the unregularized sketched least-squares fit, $\boldsymbol{Z}^\top \boldsymbol{r} = \boldsymbol{0}$. In Lemma B.13: $\boldsymbol{r} \in \mathbb{R}^n$ is a general random vector field participating in the assumed identity $\partial \boldsymbol{q}/\partial z_{ij} = \boldsymbol{K}\boldsymbol{e}_j r_i - \boldsymbol{L}^\top \boldsymbol{e}_i q_j$; the weighted-residual construction and sketched

normal equations are not imposed. In Theorem 4.3, Theorem 4.4: The weighted residual retains the natural scaled orthogonal sketch scale: $\boldsymbol{r} = c_n \boldsymbol{P}_{\boldsymbol{S}} \boldsymbol{u}$ when $\boldsymbol{A} = c_n \boldsymbol{P}_{\boldsymbol{S}}$.

Clarification (symbol $b_{\boldsymbol{A}}$):

$b_{\boldsymbol{A}}$ is the positive residual sketch mass computed from the current sketch Gram matrix $\boldsymbol{A}$ and design $\boldsymbol{X}$; it depends on $(\boldsymbol{X}, \boldsymbol{S})$ and not on the response. In Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10, Lemma B.12, Lemma B.14, Lemma B.15, Lemma B.17, Proposition B.18, Proposition 40, Lemma B.20, Lemma B.21, Lemma B.22, Proposition B.23: Use the trace-normalized sketch $\overline{\boldsymbol{S}} = \{n/\operatorname{Tr}(\boldsymbol{S}^\top \boldsymbol{S})\}^{1/2} \boldsymbol{S}$ and its Gram matrix; $b_{\boldsymbol{A}}$ denotes the resulting normalized residual sketch mass, equal to $n/\operatorname{Tr}(\boldsymbol{S}^\top \boldsymbol{S})$ times the mass of the natural-scale sketch.

Clarification (symbol $t_{\boldsymbol{\Sigma}}$):

$t_{\boldsymbol{\Sigma}} := \operatorname{Tr}(\boldsymbol{\Sigma} \boldsymbol{M}^{-1})$ depends on the design, sketch and population covariance, is unchanged by replacing only the response law, and scales by c^{-1} when $\boldsymbol{A}$ is replaced by $c\boldsymbol{A}$. In Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10, Lemma B.12, Lemma B.14, Lemma B.15, Lemma B.17, Proposition B.18, Proposition 40, Lemma B.20, Lemma B.21, Lemma B.22, Proposition B.23: The standing balanced-sketch notation evaluates $t_{\boldsymbol{\Sigma}}$ using the trace-normalized sketch, so $\operatorname{Tr}(\boldsymbol{A}) = n$ and $t_{\boldsymbol{\Sigma}} = \operatorname{Tr}((\boldsymbol{Z}^\top \boldsymbol{A} \boldsymbol{Z})^{-1})$. In Theorem 4.3, Theorem 4.4: The scaled orthogonal sketch retains its natural scale $\boldsymbol{A} = c_n \boldsymbol{P}_{\boldsymbol{S}}$; in the rotated whitened coordinates, $c_n t_{\boldsymbol{\Sigma}} = \operatorname{Tr}((\boldsymbol{Z}_1^\top \boldsymbol{Z}_1)^{-1})$.

Clarification (symbol b):

$b := b_{\boldsymbol{A}} = \operatorname{Tr}(\boldsymbol{A}) - \operatorname{Tr}(\boldsymbol{A}\boldsymbol{X}\boldsymbol{M}^{-1}\boldsymbol{X}^\top \boldsymbol{A})$ is the positive residual sketch mass, depending on $(\boldsymbol{X}, \boldsymbol{S})$ and unchanged when only the response law is replaced. In Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10, Lemma B.12, Lemma B.14, Lemma B.15, Lemma B.17, Proposition B.18, Proposition 40, Lemma B.20, Lemma B.21, Lemma B.22, Proposition B.23: The standing balanced-sketch notation uses the trace-normalized representative $\overline{\boldsymbol{A}} = n\boldsymbol{A}/\operatorname{Tr}(\boldsymbol{A})$, reusing b for $b_{\overline{\boldsymbol{A}}} = n b_{\boldsymbol{A}}/\operatorname{Tr}(\boldsymbol{A})$.

Theorem 4.1. *Assume Gaussian design and uniformly sub-Gaussian errors, meaning that their tails admit a uniform Gaussian bound. For all sufficiently large n, suppose almost surely that the positive eigenvalues of $n\boldsymbol{A}/\operatorname{Tr}(\boldsymbol{A})$ lie in $[c_-, c_+]$, with fixed $0 < c_- \le c_+ < \infty$, and $m - d \ge \kappa n$ for a fixed $\kappa > 0$. Under the model conditions of Section 2, a constant C independent of n satisfies*

$$\max\{\mathbb{E}|\hat{R}_o - R|, \mathbb{E}|\hat{R}_u - R|\} \le C n^{-1/2}. \tag{5}$$

Clarification:

There exists a deterministic $K_\varepsilon < \infty$, independent of n, such that $\sup_n \|\varepsilon_1\|_{\psi_2} \le K_\varepsilon$, where $\|W\|_{\psi_2} := \inf\{t > 0 : \mathbb{E}\exp(W^2/t^2) \le 2\}$. (the reading of one of its hypotheses)

Clarification:

There exists a deterministic $n_0 \ge 1$ such that, for every $n \ge n_0$, almost surely every positive eigenvalue of $n\boldsymbol{A}/\operatorname{Tr}(\boldsymbol{A})$ belongs to $[c_-, c_+]$ and $m - d \ge \kappa n$, with c_-, c_+, κ deterministic and independent of n. (the reading of one of its hypotheses)

Clarification (symbol c_-):

c_- is a strictly positive deterministic constant independent of n, with $c_- \le c_+ < \infty$, such that every positive eigenvalue of $n\boldsymbol{A}/\operatorname{Tr}(\boldsymbol{A})$ is at least c_- almost surely for every sufficiently large n.

Clarification (symbol c_+):

c_+ is a finite deterministic constant independent of n, with $0 < c_- \le c_+$, such that every positive eigenvalue of $n\boldsymbol{A}/\operatorname{Tr}(\boldsymbol{A})$ is at most c_+ almost surely for every sufficiently large n.

Clarification (symbol κ):

$\kappa > 0$ is a deterministic constant independent of n such that $m - d \ge \kappa n$ for every sufficiently large n.

Thus, both the population-covariance and observable corrections estimate R with expected absolute error of order $n^{-1/2}$ under the stated conditions. Row sampling and reweightings of orthonormal sketch rows with weights bounded above and away from zero satisfy the spectral condition after normalization.

4.1. Sketch geometry and risk benchmarks

The form of the observable correction becomes especially transparent when the sketch has orthonormal rows. A *scaled orthogonal sketch* satisfies $\boldsymbol{S}\boldsymbol{S}^\top = c_n \boldsymbol{I}_m$ for $c_n > 0$; equivalently, $c_n^{-1/2}\boldsymbol{S}$ has orthonormal rows. In this case, $\boldsymbol{A} = c_n \boldsymbol{P}_{\boldsymbol{S}}$, $b_{\boldsymbol{A}} = c_n(m-d)$, and

$$\hat{R}_o = \frac{\|\boldsymbol{Q}_{\boldsymbol{S}}\boldsymbol{u}\|^2}{n} + \left(\frac{m}{m-d}\right)^2 \frac{\|\boldsymbol{P}_{\boldsymbol{S}}\boldsymbol{u}\|^2}{n}. \tag{6}$$

Thus, $\hat{R}_o$ leaves the complementary residual unchanged and rescales only the retained residual. This distinction is important because the retained component is the part that reuses the fitting data. Although row whitening gives orthonormal rows for any full-row-rank sketch, replacing $\boldsymbol{S}$ by $\boldsymbol{F}$ can change the least-squares weights and hence the fitted predictor; only the interpolation constraints are preserved.

For comparison, GCV treats the residual differently: it rescales the entire full-data residual according to the trace d of the fitted-value map,

$$\hat{R}_{\mathrm{GCV}} = \frac{\|\boldsymbol{u}\|^2/n}{(1-d/n)^2}.$$

We compare scaled orthogonal sketches with dense Gaussian sketches, whose independent entries satisfy $S_{ki} \sim \mathcal{N}(0, 1/m)$. The latter require separate analysis because the preceding theorem's almost-sure spectral bounds do not apply. In the *noninterpolating proportional regime*, $(\gamma_n, \rho_n) := (d/n, m/n) \to (\gamma, \rho)$ with $0 < \gamma < \rho \le 1$. Write $\xrightarrow{\mathrm{p}}$ for convergence in probability.

Clarification (symbol γ_n):

$\gamma_n = d/n$ is the deterministic dimension ratio along the sequence of sample sizes, with d allowed to vary with n. In Theorem 4.2: $(\gamma_n, \rho_n) \to (\gamma, \rho)$ with $0 < \gamma < \rho \le 1$. In Theorem 4.4: $(\gamma_n, \rho_n) \to (\gamma, \rho)$ with $0 < \gamma < \rho < 1$; the additional local-crossover conclusion assumes that γ_n stays in a compact subset of $(0, 1)$.

Clarification (symbol ρ_n):

$\rho_n = m/n$ is the deterministic sketch-size ratio along the sequence of sample sizes, with integer $1 \le m \le n$ for actual sketches. In Theorem 4.2: $(\gamma_n, \rho_n) \to (\gamma, \rho)$ with $0 < \gamma < \rho \le 1$. In Theorem 4.4: $(\gamma_n, \rho_n) \to (\gamma, \rho)$ with $0 < \gamma < \rho < 1$; in defining $\rho_n^{\#}$, ρ_n is continuously extended and $m = n\rho_n$ need not be an integer.

Clarification (symbol γ):

$\gamma = \lim_{n\to\infty} d/n$ is the deterministic limiting feature-to-sample ratio, with $0 < \gamma < 1$. In Theorem 4.2: The joint limiting ratios satisfy $0 < \gamma < \rho \le 1$. In Theorem 4.4: The joint limiting ratios satisfy $0 < \gamma < \rho < 1$.

Clarification (symbol ρ):

$\rho = \lim_{n\to\infty} \rho_n = \lim_{n\to\infty} m/n$ is a limiting sketch proportion satisfying $\gamma < \rho \le 1$. In Theorem 4.4: $\rho < 1$.

Theorem 4.2. *Under Gaussian data with $\sigma^2 > 0$ and this regime, $R, \hat{R}_u, \hat{R}_o \xrightarrow{\mathrm{p}} \mathcal{R}_j$ and $\hat{R}_{GCV} - R \xrightarrow{\mathrm{p}} \Delta_j$, where j identifies the sketch family:*

$$\begin{aligned} \mathcal{R}_{\mathrm{SO}} &= \frac{\sigma^2\rho}{\rho-\gamma}, & \Delta_{\mathrm{SO}} &= \frac{\sigma^2\gamma^2(1-\rho)}{(\rho-\gamma)(1-\gamma)^2}, \\ \mathcal{R}_{\mathrm{G}} &= \sigma^2\left\{\frac{1}{1-\gamma} + \frac{\gamma}{\rho-\gamma}\right\}, & \Delta_{\mathrm{G}} &= \frac{\sigma^2\gamma^2}{(1-\gamma)(\rho-\gamma)}. \end{aligned} \tag{7}$$

Here SO denotes scaled orthogonal sketches and G denotes dense Gaussian sketches with the stated entry law.

The limits $\mathcal{R}_{\mathrm{SO}}$ and $\mathcal{R}_{\mathrm{G}}$ agree with earlier sketch-risk benchmarks [7, 8]. The theorem also shows a difference between the two sketch families that is not visible from sketch size alone. At $\rho = 1$, GCV's limiting gap is zero for scaled orthogonal sketches and positive for dense Gaussian sketches.

4.2. Comparing estimation accuracy

The preceding results establish consistency and characterize large-sample risk, but they do not determine which of the two proposed estimators is more accurate. We therefore compare $\widehat{R}_c$ and $\hat{R}_o$ as estimates of the same fitted predictor's risk. For a risk estimate T, define its mean squared estimation error (MSE) as $\mathrm{MSE}(T) = \mathbb{E}[(T - R)^2]$, where the expectation averages over the training data and sketch randomness. For $d + 3 < m < n$, set $k = m - d$ and $q = n - m$, and define

$$e_{m,d} := \frac{(m-1)(m-3)}{(k-1)(k-3)},$$
$$a_{m,d} := \frac{2m^2}{k^3} + \frac{2d(m-1)}{(k-1)^2(k-3)} + \frac{d^2}{k^2(k-1)^2}.$$

Clarification (symbol k):

k is a deterministic integer parameter bound by its local definition or quantifier. In Theorem 4.3, Theorem 4.4: In the finite-MSE expressions, $k = m - d > 3$ and may vary with n. In Theorem 3.1: In the proof, $k = n - m > 0$ is the rank of $\boldsymbol{Q_S}$ and the complementary chi-square degrees of freedom. In Lemma B.11, Lemma B.12, Lemma B.14, Lemma B.15, Lemma B.20: In moment bounds, $k \geq 1$ is an arbitrary fixed integer moment order, independent of n, rather than a dimension difference. In Theorem 4.1: The proof's invocation of a moment bound at $k = 1$ uses k as moment order. In Lemma B.16: In the proof's rowwise derivative and covariance calculations, k indexes a design row, with $1 \leq k \leq n$ and $k \neq i$ where stated. In another result of the paper: k is a locally bound design-row index, with $1 \leq k \leq n$ and $k \neq i$.

Clarification (symbol $e_{m,d}$):

$e_{m,d}$ is the deterministic dimension-dependent coefficient defined for integer dimensions $1 \leq d$ and $d+3 < m < n$, using $k = m - d$.

Clarification (symbol $a_{m,d}$):

$a_{m,d}$ is the deterministic dimension-dependent coefficient defined for integer dimensions $1 \leq d$ and $d+3 < m < n$, using $k = m - d$.

Clarification (symbol T):

T is local notation for a real-valued random quantity whose interpretation is specified in its surrounding definition. In Theorem 4.3, Theorem 4.4: T denotes a risk estimate of the same fitted predictor's risk R, and $\mathrm{MSE}(T) = \mathbb{E}[(T - R)^2]$ averages over training data and sketch randomness. In Theorem 4.2: In the dense Gaussian proof, $T \sim \chi^2_{n-d}$ is an auxiliary variable independent of J, V, U, B, introduced for marginal distributional representations of residual statistics.

Theorem 4.3. *Under the model of Section 2, suppose Gaussian data, $\sigma^2 > 0$, a scaled orthogonal sketch, and $1 \leq d$ with $d + 3 < m < n$. The mean squared errors are*

$$\mathrm{MSE}(\widehat{R}_c) = \frac{2\sigma^4 e_{m,d}}{q},$$
$$\mathrm{MSE}(\hat{R}_o) = \frac{\sigma^4}{n^2}\{2q e_{m,d} + m^2 a_{m,d}\}. \tag{8}$$

The corrected risk estimate has smaller MSE exactly when

$$q^2 + \frac{m^2 q\, a_{m,d}}{2e_{m,d}} < n^2. \tag{9}$$

Reversing the inequality favors the complementary risk estimate; equality gives equal MSE.

The finite-sample comparison depends on n, m, and d. In the proportional regime, however, it simplifies to a threshold involving only the limiting dimension ratios. Define

$$\rho_*(\gamma) = \frac{3\gamma}{1 + 2\gamma}.$$

Let $\rho_n^{\#}$ denote the local continuous solution of $\mathrm{MSE}(\hat{R}_o) = \mathrm{MSE}(\widehat{R}_c)$ near $\rho_*(\gamma_n)$, using the finite-sample expressions in (8). The finite ratio $\rho_n = m/n$ is treated as continuous for this definition.

Clarification (symbol $\rho_n^{\#}$):

$\rho_n^{\#}$ is a deterministic local root near $\rho_*(d/n)$ of the finite-sample MSE equality after continuously extending $m = n\rho_n$; it need not correspond to an integer sketch size.

Theorem 4.4. *For Gaussian data with $\sigma^2 > 0$, a scaled orthogonal sketch, and $d + 3 < m < n$, suppose $(\gamma_n, \rho_n) \to (\gamma, \rho)$ with $0 < \gamma < \rho < 1$. For sufficiently large n, $\widehat{R}_c$ has smaller MSE if $\rho < \rho_*(\gamma)$, and $\hat{R}_o$ has smaller MSE if $\rho > \rho_*(\gamma)$. At equality, their MSE ratio tends to one. If γ_n stays in a compact subset of $(0, 1)$, the local continuous crossover exists uniquely for sufficiently large n and*

$$\rho_n^{\#} = \rho_*(\gamma_n) - \frac{5}{12n} + O(n^{-2}). \tag{10}$$

Here $O(n^{-2})$ is a remainder bounded by a constant times n^{-2}. The integer-size comparison remains the rule in (9), while the threshold $\rho_*(\gamma)$ gives a simpler interpretation in the proportional regime. When the sketch is relatively small, $\widehat{R}_c$ benefits from having more complementary directions available for assessment; as m increases, those directions become fewer and the corrected estimator $\hat{R}_o$ eventually becomes more accurate. These MSE comparisons hold for every feature covariance allowed by the model, after averaging over training samples and sketches.

5. Simulations

Throughout, the feature vectors are independent $\mathcal{N}_d(\mathbf{0}, \mathbf{\Sigma})$ with $\Sigma_{jk} = 0.5^{|j-k|}$. The first ten coefficients are equal and positive, the others are zero, and $\boldsymbol{\beta}^{*\top}\mathbf{\Sigma}\boldsymbol{\beta}^* = 4$. Errors are independent of the features and sketches and follow either $\mathcal{N}(0, 1)$ or the unscaled uniform law on $[-1, 1]$, whose variance is $1/3$. We compute R from (1) rather than from a test sample; only $\hat{R}_u$ uses the population covariance.

We study three independent sketches. Row sampling selects m indices uniformly without replacement and scales each row by $\sqrt{n/m}$. For a weighted Haar sketch, let $\boldsymbol{F}$ be the transpose of the orthogonal factor in the thin QR decomposition of an $n \times m$ standard Gaussian matrix, with a positive diagonal in the triangular factor, and set $\boldsymbol{S} = c\,\mathbf{diag}(w_1, \ldots, w_m)\boldsymbol{F}$, where $w_j = 1$ for odd j, $w_j = 2$ for even j, and $c^2 = n/\sum_j w_j^2$. These two sketches satisfy $\mathrm{Tr}(\boldsymbol{S}^\top\boldsymbol{S}) = n$. Dense Gaussian sketches have independent $\mathcal{N}(0, 1/m)$ entries and retain their random trace. We fit the compressed equations by least squares when $d < m$ and by minimum norm when $d > m$; weights can affect the fit even when $m = n$. Below, $s(v)$ is the sample standard deviation, and all intervals are pointwise.

5.1. Calibration conditional on a fixed fit

This experiment tests conditional calibration by fixing the compressed data and fitted predictor and resampling only the complementary data. We set $n = 300$, $m = 150$, and $d \in \{100, 300, 500\}$, with row or dense Gaussian sketches and Gaussian errors. For each setting, we generate 100 independent sketches and retained-data realizations, where $\boldsymbol{FX}$ has independent $\mathcal{N}_d(\mathbf{0}, \mathbf{\Sigma})$ rows and $\boldsymbol{F\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_m)$. We fit once using $\boldsymbol{SX} = (\boldsymbol{SF}^\top)(\boldsymbol{FX})$ and $\boldsymbol{Sy} = (\boldsymbol{SF}^\top)\{(\boldsymbol{FX})\boldsymbol{\beta}^* + \boldsymbol{F\varepsilon}\}$, then draw 1,000 independent complementary blocks. With $q = n - m$, each block contains $\boldsymbol{X}_c \in \mathbb{R}^{q\times d}$ with independent $\mathcal{N}_d(\mathbf{0}, \mathbf{\Sigma})$ rows and $\boldsymbol{\varepsilon}_c \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_q)$, giving $\widehat{R}_c = \|\boldsymbol{\varepsilon}_c + \boldsymbol{X}_c(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}})\|^2/q$.

For each fit, we record the mean of $\widehat{R}_c/R$, its sample variance divided by $2/q$, and the coverage of (4). Figure 1 averages these metrics over the 100 fits, with intervals $\bar{v} \pm t_{99,0.975}s(v)/\sqrt{100}$ for the fit-level values v. All six variance intervals contain one. The estimator is centered on the fixed fit's risk, and coverage is close to 95% for both sketches and in both fitting regimes.

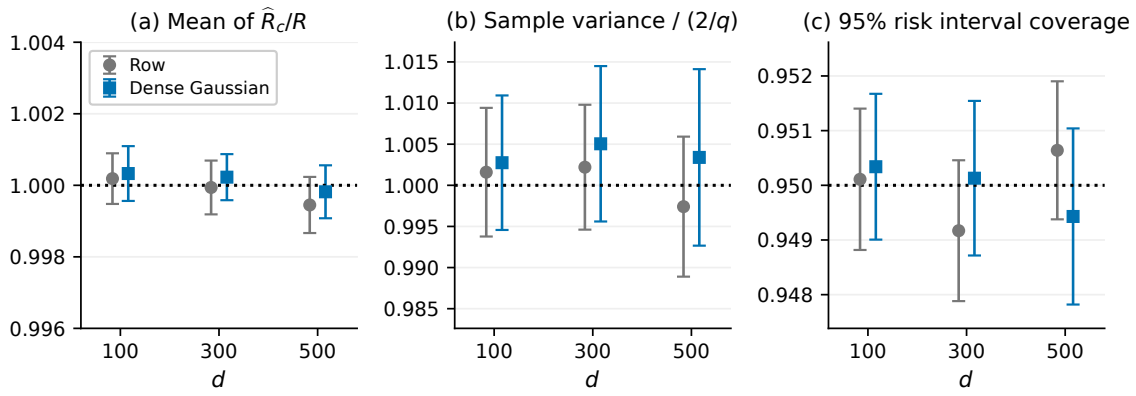


Figure 1: Conditional calibration of $\widehat{R}_c$ at $n = 300$ and $m = 150$ ($q = n - m$). Panels show the mean of $\widehat{R}_c/R$, its conditional sample variance divided by $2/q$, and coverage of the nominal 95% risk interval. Each setting uses 100 fixed fits and 1,000 complementary draws per fit. Error bars are pointwise 95% intervals across fits.

5.2. Risk curves for three sketch families

We next compare mean true risk with mean estimated risk for $n = 100, 150, \ldots, 1000$ and $(d/n, m/n) \in \{(0.30, 0.50), (0.50, 0.70), (0.70, 0.90), (0.80, 1.00)\}$. Each setting uses 1,000 independent datasets. Sketch families share the design and errors within a repetition, while the two error laws share the design and sketch. For each dataset, error law, and sketch realization, all criteria assess the same fitted predictor. Alongside $\hat{R}_o, \hat{R}_u, \hat{R}_{\mathrm{GCV}}, \widehat{R}_c$, we include

$$\widehat{R}_B = \frac{\|\boldsymbol{u}\|^2}{n - 2d + d^2/m}.$$

This row-sampling formula is from Bellec et al. [12, Section 3.1, Remark 2, equation (13)], and its use for weighted Haar and dense Gaussian sketches is an empirical comparison.

Under Gaussian errors (Figure 2), $\hat{R}_o$, $\hat{R}_u$, and $\widehat{R}_c$ track mean true risk more closely than GCV when $m < n$; GCV's upward discrepancy is largest at high d/n and persists as n grows. At larger n, especially for high d/n, the proposed estimates also track mean true risk more closely than $\widehat{R}_B$ for weighted Haar and dense Gaussian sketches, while $\widehat{R}_B$ remains accurate for row sampling. When $m = n$, $\widehat{R}_c$ is undefined. Row sampling then gives full-data OLS and $\hat{R}_o = \hat{R}_{\mathrm{GCV}} = \widehat{R}_B$; for the other sketches, $\widehat{R}_B = \hat{R}_{\mathrm{GCV}}$, while $\hat{R}_o$ and $\hat{R}_u$ are closer to mean true risk.

5.3. Comparing estimation MSE

The final experiment tests the MSE crossover in Theorem 4.3. With Gaussian errors and row sampling, we use $n \in \{100, 300, 500\}$, $d/n \in \{0.30, 0.50, 0.70\}$, and five sketch sizes $m = \mathrm{round}\{n[\rho_*(d/n) + \delta]\}$ for $\delta \in \{-0.08, -0.04, 0, 0.04, 0.08\}$, with half-ties rounded upward. All settings satisfy $d + 3 < m < n$. Each of 1,000 repetitions uses one dataset and one row permutation for all five sketch sizes, taking the first m indices, scaling rows by $\sqrt{n/m}$, and refitting at each m. Define $a_b = (\hat{R}_o^{(b)} - R^{(b)})^2$ and $c_b = (\widehat{R}_c^{(b)} - R^{(b)})^2$. The empirical MSE ratio is $\widehat{\theta} = \bar{a}/\bar{c}$. With $v_b = a_b/\bar{a} - c_b/\bar{c}$, its paired delta-method interval is $\widehat{\theta}\exp\{\pm z_{0.975} s(v)/\sqrt{1000}\}$, where $z_{0.975}$ is the 0.975 standard-normal quantile.

Figure 3 compares empirical and exact MSE ratios. The crossover $\rho_n^{\#}$ satisfies equality in (9) near $\rho_*(d/n)$. Of the 45 intervals, 43 contain the exact ratio. At $\delta = 0$ all contain one; at $\delta = -0.08$ all lie above one, and at $\delta = 0.08$ all lie below one. This confirms lower MSE for $\widehat{R}_c$ below the crossover and for $\hat{R}_o$ above it.

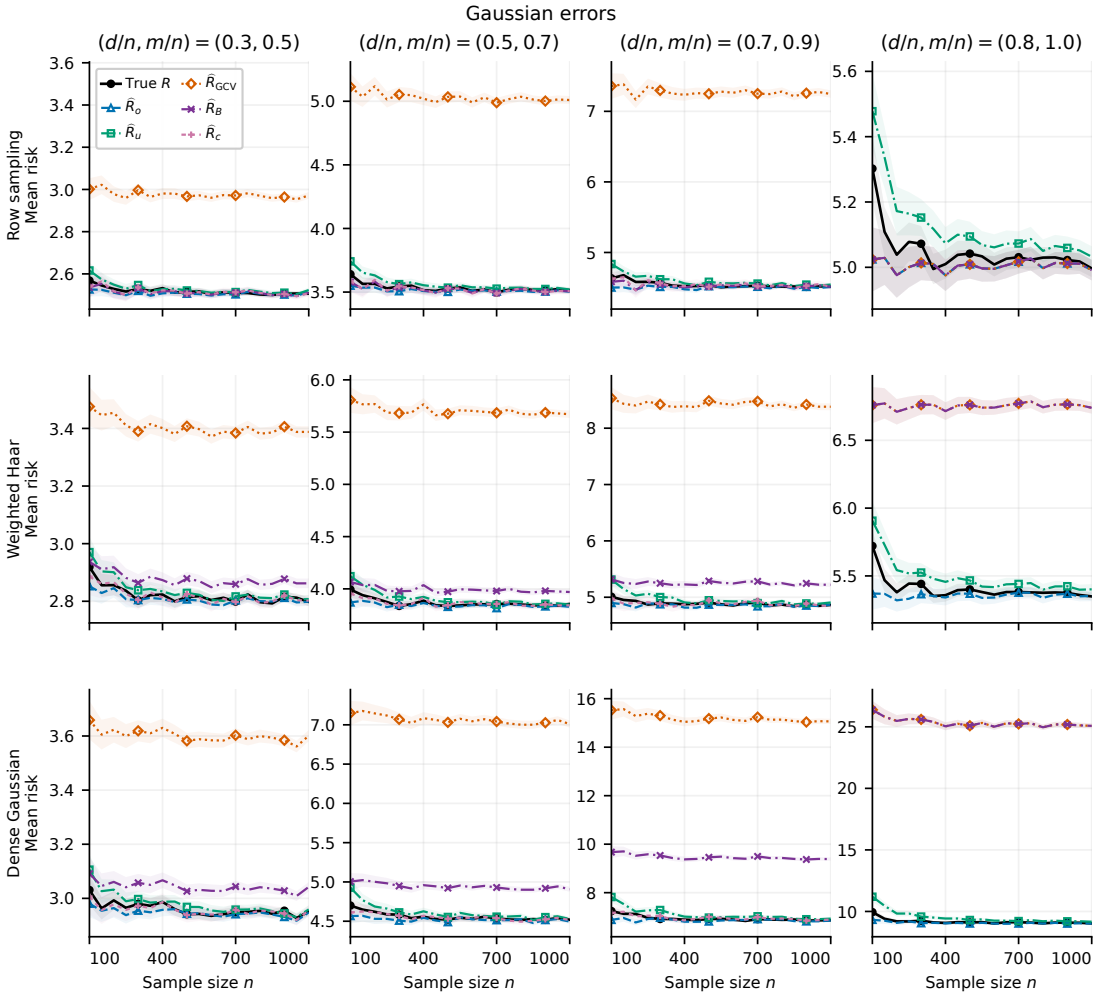


Figure 2: Mean prediction risks and estimates under Gaussian errors. Rows give the three sketch families and columns give $(d/n, m/n)$; each point uses 1,000 repetitions. The black curve is mean true risk, and bands are pointwise 95% Monte Carlo intervals $\bar{T} \pm 1.96 s(T)/\sqrt{1000}$ for each plotted mean. Bellec's formula is applied to every sketch family as an empirical comparison. When $m = n$, $\widehat{R}_c$ is undefined; the observable, GCV, and Bellec curves coincide for row sampling, and Bellec's curve coincides with GCV for the other sketches. Both axes are linear.

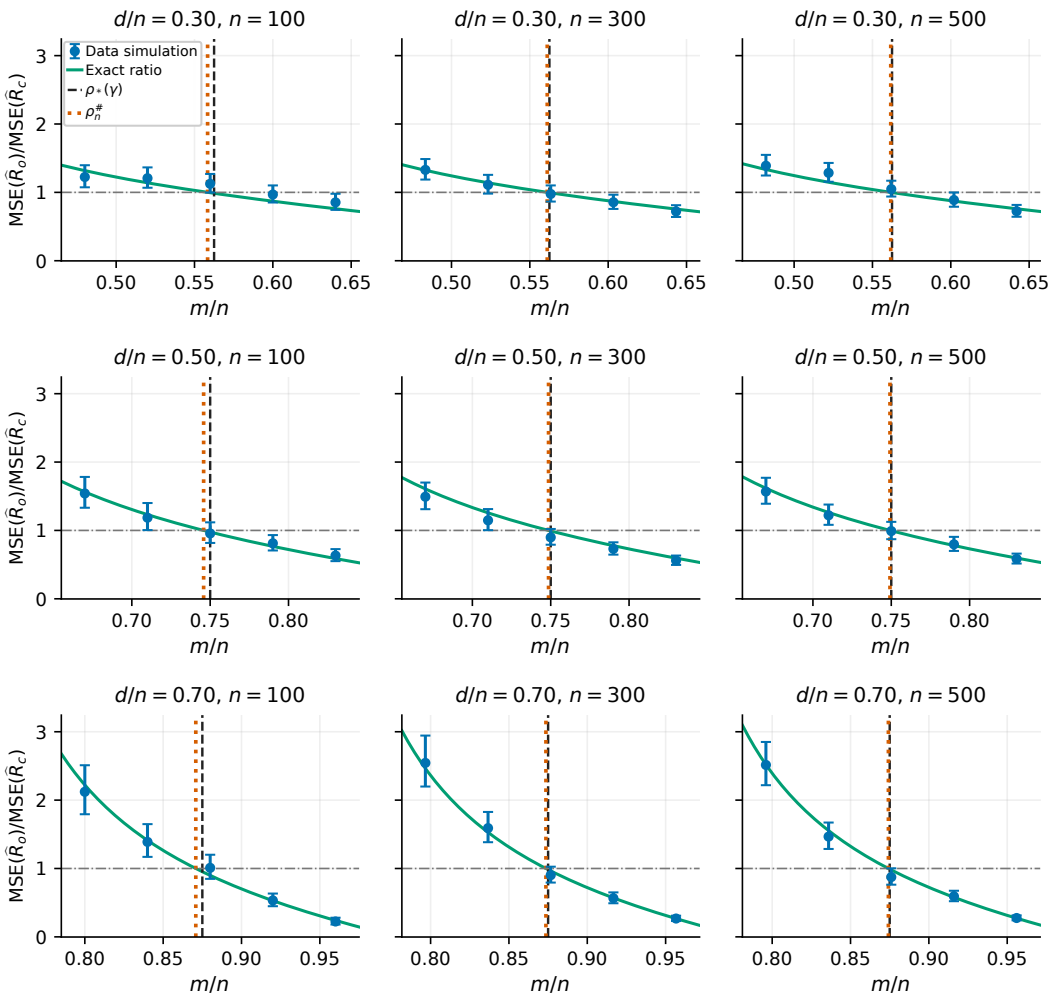


Figure 3: MSE ratio of $\hat{R}_o$ to $\widehat{R}_c$ under Gaussian data and row sampling. Rows give d/n, columns give n, and the horizontal axis gives rounded m/n. Points and pointwise 95% paired intervals use 1,000 repetitions. The green curve is exact; dashed and dotted lines mark $\rho_*(\gamma)$ and $\rho_n^{\#}$. Values above one favor $\widehat{R}_c$.

6. Discussion

Observation compression supports risk assessment without refitting. The complementary law holds for Gaussian data and an independent sketch; row-sampling unbiasedness needs only second moments. The correction bound permits uniformly sub-Gaussian errors but assumes Gaussian features, $m - d$ proportional to n, and a bounded positive spectrum for $n\boldsymbol{A}/\operatorname{Tr}(\boldsymbol{A})$. Extensions to non-Gaussian features and less regular spectra remain open. As m grows, fewer complementary directions remain, and $\widehat{R}_c$ is unavailable at $m = n$. Under Gaussian data, the MSE crossover for scaled orthogonal sketches identifies when $\hat{R}_o$ is more accurate. The weighted Haar and dense Gaussian MSE comparisons remain empirical; broader geometries and m near d require further theory.

Assessment needs the original observations or residual summaries, adding data and computational costs. The simulations use one covariance and signal. Future work should study other designs, structured sketches, end-to-end costs, compact sufficient summaries, and data-dependent or risk-selected sketches that account for design dependence and reused risk.

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A. Additional simulation results

The code for reproducing all simulation results is available at https://github.com/StatResearch-SLAI/RandomSketch.

A.1. Risk assessment across sketch families

Table 1 summarizes mean prediction risks and risk estimates at $n = 1000$ for both error laws; Figure 4 shows the additional curves under uniform errors. These summaries describe calibration across repetitions. To assess estimation accuracy, we compare each estimate with the true risk of the same fitted predictor. For an applicable estimate T and repetition $b = 1, \ldots, 1000$, the paired errors $\Delta_b = T^{(b)} - R^{(b)}$ give bias $\bar{\Delta}$ and root mean squared estimation error (RMSE) $\sqrt{\overline{\Delta^2}}$. Bias measures average signed error, whereas RMSE also reflects variation in the errors across fits. Tables 2 and 3 report these quantities at $n = 1000$. Their estimated Monte Carlo standard errors are $s(\Delta)/\sqrt{1000}$ and $s(\Delta^2)/(2\,\mathrm{RMSE}\,\sqrt{1000})$, respectively; the latter uses the delta method.

The square-sketch setting separates the effects of row space and weighting. At $n = 1000$ and $(d/n, m/n) = (0.80, 1.00)$ with Gaussian errors, the mean true risks are 4.997, 5.349, and 9.029 for row, weighted Haar, and dense Gaussian sketches. The corresponding means of $\hat{R}_o$ are 4.991, 5.352, and 9.012, compared with 4.991, 6.740, and 25.079 for GCV. For the dense Gaussian sketch, the mean unobservable estimate $\hat{R}_u$ is 9.165. All three sketches retain the full observation space, but weighting changes both the fit and its residual calibration. In the same dense square setting under uniform errors, mean risk is 3.021, the mean $\hat{R}_o$ is 3.002, and the mean $\hat{R}_u$ is 3.052. GCV and Bellec's formula both give a mean of 8.345, so the row-sampling expression inherits GCV's mean discrepancy here. Uniform errors have variance $1/3$, compared with variance one for Gaussian errors, so the comparison between Gaussian- and uniform-error settings also changes the noise scale. The uniform-error curves assess calibration empirically.

Across the 24 settings at $n = 1000$, paired mean bias ranges from -0.027 to 0.009 for $\hat{R}_o$ and from -0.006 to 0.136 for $\hat{R}_u$ (Table 2).

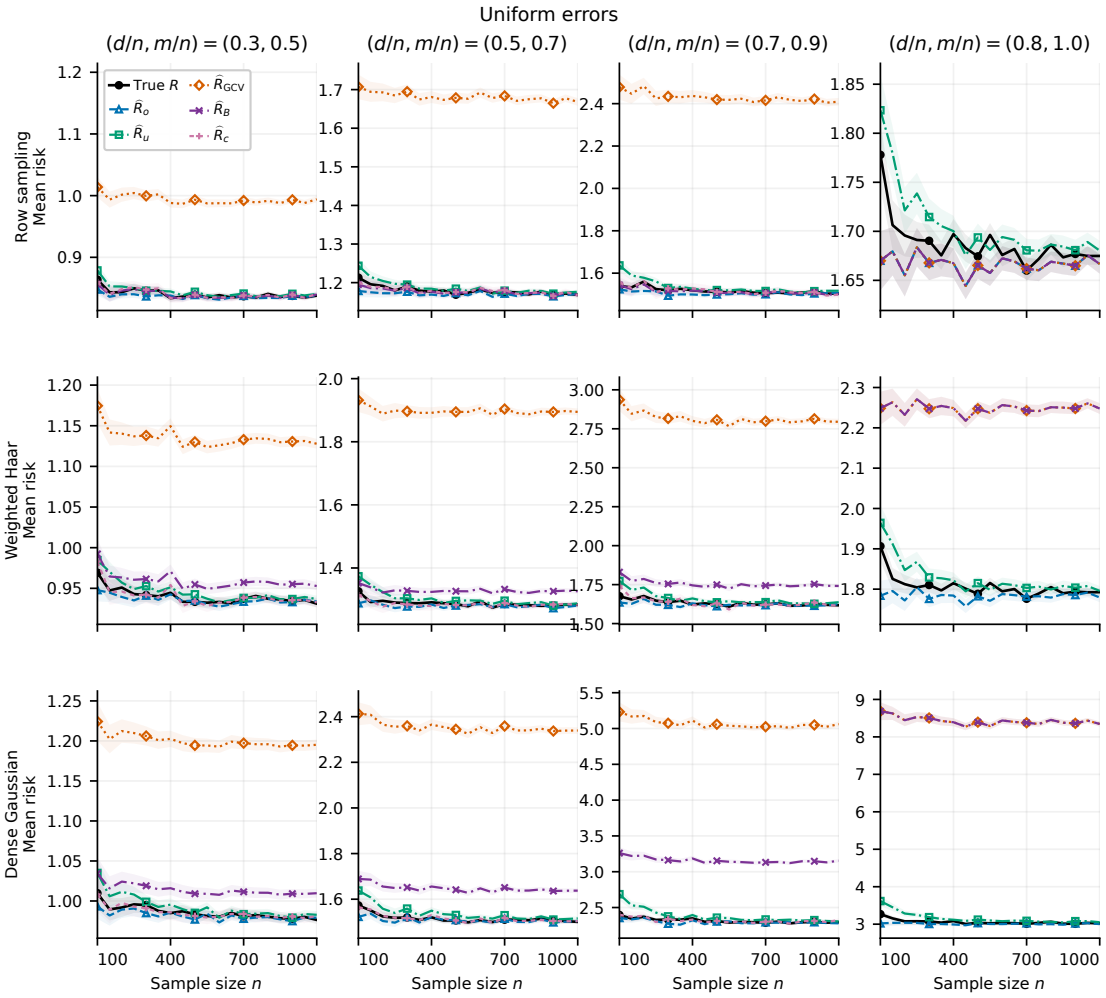


Figure 4: Mean prediction risks and risk estimates under independent uniform errors on $[-1, 1]$, with variance $1/3$. Layout, sample sizes, pairing, and Monte Carlo intervals match Figure 2. Because the errors are uniform, the complementary-risk curves provide empirical calibration rather than an application of the Gaussian chi-square law. The $\widehat{R}_B$ curve uses Bellec's formula for all three sketch families. The complementary estimate is undefined when $m = n$.

Table 1: Mean prediction risks and risk estimates at $n = 1000$, based on 1,000 independent repetitions per setting. Parentheses contain Monte Carlo standard errors. The Bellec column uses the same formula, denoted $\widehat{R}_B$, for all three sketch families. WH and DG denote weighted Haar and dense Gaussian sketches; G and U denote Gaussian and uniform errors. A dash denotes the undefined complementary estimate when $m = n$.

Sketch	Errors	$(d/n, m/n)$	R	$\widehat{R}_o$	$\widehat{R}_u$	$\widehat{R}_{\mathrm{GCV}}$	$\widehat{R}_B$	$\widehat{R}_c$
Row	G	$(0.3, 0.5)$	2.507 (0.006)	2.516 (0.006)	2.523 (0.006)	2.972 (0.008)	2.510 (0.007)	2.509 (0.008)
Row	U	$(0.3, 0.5)$	0.838 (0.002)	0.838 (0.002)	0.841 (0.002)	0.994 (0.003)	0.840 (0.002)	0.841 (0.003)
WH	G	$(0.3, 0.5)$	2.798 (0.008)	2.798 (0.007)	2.807 (0.007)	3.388 (0.010)	2.862 (0.008)	2.802 (0.010)
WH	U	$(0.3, 0.5)$	0.931 (0.002)	0.934 (0.002)	0.937 (0.002)	1.128 (0.003)	0.953 (0.003)	0.933 (0.003)
DG	G	$(0.3, 0.5)$	2.951 (0.008)	2.948 (0.008)	2.959 (0.008)	3.601 (0.011)	3.042 (0.009)	2.956 (0.010)
DG	U	$(0.3, 0.5)$	0.976 (0.003)	0.979 (0.002)	0.983 (0.002)	1.195 (0.003)	1.010 (0.003)	0.981 (0.003)
Row	G	$(0.5, 0.7)$	3.515 (0.010)	3.503 (0.009)	3.520 (0.009)	5.010 (0.016)	3.507 (0.011)	3.508 (0.013)
Row	U	$(0.5, 0.7)$	1.168 (0.003)	1.171 (0.003)	1.176 (0.003)	1.668 (0.005)	1.168 (0.004)	1.167 (0.004)
WH	G	$(0.5, 0.7)$	3.855 (0.011)	3.828 (0.011)	3.849 (0.011)	5.671 (0.018)	3.970 (0.013)	3.842 (0.015)
WH	U	$(0.5, 0.7)$	1.283 (0.003)	1.281 (0.004)	1.288 (0.004)	1.895 (0.006)	1.326 (0.004)	1.282 (0.005)
DG	G	$(0.5, 0.7)$	4.519 (0.014)	4.495 (0.015)	4.528 (0.015)	7.010 (0.023)	4.907 (0.016)	4.514 (0.018)
DG	U	$(0.5, 0.7)$	1.501 (0.004)	1.505 (0.005)	1.516 (0.005)	2.339 (0.007)	1.638 (0.005)	1.504 (0.006)
Row	G	$(0.7, 0.9)$	4.514 (0.013)	4.512 (0.013)	4.545 (0.013)	7.246 (0.027)	4.515 (0.017)	4.516 (0.023)
Row	U	$(0.7, 0.9)$	1.504 (0.004)	1.506 (0.004)	1.517 (0.004)	2.409 (0.009)	1.501 (0.006)	1.499 (0.008)
WH	G	$(0.7, 0.9)$	4.863 (0.014)	4.867 (0.015)	4.907 (0.016)	8.381 (0.032)	5.222 (0.020)	4.838 (0.027)
WH	U	$(0.7, 0.9)$	1.617 (0.005)	1.624 (0.005)	1.637 (0.005)	2.795 (0.010)	1.741 (0.006)	1.614 (0.008)
DG	G	$(0.7, 0.9)$	6.863 (0.023)	6.849 (0.029)	6.929 (0.030)	15.063 (0.056)	9.385 (0.035)	6.892 (0.038)
DG	U	$(0.7, 0.9)$	2.295 (0.007)	2.280 (0.010)	2.306 (0.010)	5.061 (0.019)	3.153 (0.012)	2.325 (0.013)
Row	G	$(0.8, 1.0)$	4.997 (0.014)	4.991 (0.016)	5.032 (0.016)	4.991 (0.016)	4.991 (0.016)	–
Row	U	$(0.8, 1.0)$	1.675 (0.004)	1.667 (0.005)	1.680 (0.005)	1.667 (0.005)	1.667 (0.005)	–
WH	G	$(0.8, 1.0)$	5.349 (0.015)	5.352 (0.020)	5.401 (0.020)	6.740 (0.022)	6.740 (0.022)	–
WH	U	$(0.8, 1.0)$	1.793 (0.005)	1.779 (0.006)	1.796 (0.006)	2.248 (0.007)	2.248 (0.007)	–
DG	G	$(0.8, 1.0)$	9.029 (0.034)	9.012 (0.048)	9.165 (0.050)	25.079 (0.106)	25.079 (0.106)	–
DG	U	$(0.8, 1.0)$	3.021 (0.011)	3.002 (0.015)	3.052 (0.015)	8.345 (0.035)	8.345 (0.035)	–

Table 2: Paired empirical bias at $n = 1000$ in the expanded risk-curve study, with 1,000 repetitions per setting. Parentheses contain Monte Carlo standard errors of the empirical mean biases. Dashes have the same meaning as in Table 1, as do WH, DG, G, and U.

Sketch	Errors	$(d/n, m/n)$	$\widehat{R}_o$	$\widehat{R}_u$	$\widehat{R}_{\mathrm{GCV}}$	$\widehat{R}_B$	$\widehat{R}_c$
Row	G	$(0.3, 0.5)$	0.009 (0.006)	0.017 (0.006)	0.465 (0.005)	0.004 (0.004)	0.002 (0.005)
Row	U	$(0.3, 0.5)$	-0.000 (0.002)	0.003 (0.002)	0.156 (0.002)	0.002 (0.001)	0.003 (0.002)
WH	G	$(0.3, 0.5)$	0.000 (0.007)	0.009 (0.007)	0.590 (0.006)	0.064 (0.005)	0.004 (0.006)
WH	U	$(0.3, 0.5)$	0.003 (0.002)	0.006 (0.002)	0.197 (0.002)	0.022 (0.002)	0.002 (0.002)
DG	G	$(0.3, 0.5)$	-0.004 (0.007)	0.007 (0.007)	0.650 (0.007)	0.091 (0.006)	0.004 (0.006)
DG	U	$(0.3, 0.5)$	0.003 (0.002)	0.006 (0.002)	0.219 (0.002)	0.033 (0.002)	0.005 (0.002)
Row	G	$(0.5, 0.7)$	-0.012 (0.011)	0.005 (0.011)	1.495 (0.011)	-0.008 (0.008)	-0.007 (0.009)
Row	U	$(0.5, 0.7)$	0.002 (0.003)	0.008 (0.003)	0.500 (0.004)	-0.001 (0.003)	-0.001 (0.003)
WH	G	$(0.5, 0.7)$	-0.027 (0.012)	-0.006 (0.012)	1.816 (0.013)	0.114 (0.009)	-0.013 (0.010)
WH	U	$(0.5, 0.7)$	-0.002 (0.004)	0.005 (0.004)	0.611 (0.004)	0.043 (0.003)	-0.001 (0.003)
DG	G	$(0.5, 0.7)$	-0.024 (0.015)	0.009 (0.015)	2.491 (0.016)	0.388 (0.011)	-0.005 (0.012)
DG	U	$(0.5, 0.7)$	0.004 (0.005)	0.015 (0.005)	0.839 (0.005)	0.137 (0.004)	0.003 (0.004)
Row	G	$(0.7, 0.9)$	-0.002 (0.017)	0.031 (0.017)	2.732 (0.024)	0.001 (0.015)	0.002 (0.020)
Row	U	$(0.7, 0.9)$	0.002 (0.006)	0.013 (0.006)	0.905 (0.008)	-0.003 (0.005)	-0.005 (0.007)
WH	G	$(0.7, 0.9)$	0.004 (0.019)	0.043 (0.019)	3.517 (0.026)	0.358 (0.017)	-0.025 (0.022)
WH	U	$(0.7, 0.9)$	0.007 (0.007)	0.020 (0.007)	1.177 (0.008)	0.124 (0.005)	-0.003 (0.007)
DG	G	$(0.7, 0.9)$	-0.013 (0.031)	0.067 (0.031)	8.200 (0.044)	2.523 (0.026)	0.029 (0.030)
DG	U	$(0.7, 0.9)$	-0.015 (0.011)	0.011 (0.011)	2.766 (0.015)	0.858 (0.009)	0.030 (0.010)
Row	G	$(0.8, 1.0)$	-0.005 (0.022)	0.036 (0.022)	-0.005 (0.022)	-0.005 (0.022)	–
Row	U	$(0.8, 1.0)$	-0.008 (0.007)	0.005 (0.007)	-0.008 (0.007)	-0.008 (0.007)	–
WH	G	$(0.8, 1.0)$	0.003 (0.025)	0.052 (0.025)	1.391 (0.026)	1.391 (0.026)	–
WH	U	$(0.8, 1.0)$	-0.013 (0.008)	0.003 (0.008)	0.455 (0.009)	0.455 (0.009)	–
DG	G	$(0.8, 1.0)$	-0.018 (0.051)	0.136 (0.053)	16.049 (0.092)	16.049 (0.092)	–
DG	U	$(0.8, 1.0)$	-0.019 (0.016)	0.031 (0.017)	5.324 (0.031)	5.324 (0.031)	–

Table 3: Empirical RMSE at $n = 1000$ in the expanded risk-curve study, with 1,000 repetitions per setting. Parentheses contain delta-method Monte Carlo standard errors. Dashes have the same meaning as in Table 1, as do WH, DG, G, and U.

Sketch	Errors	$(d/n, m/n)$	$\widehat{R}_o$	$\widehat{R}_u$	$\widehat{R}_{\mathrm{GCV}}$	$\widehat{R}_B$	$\widehat{R}_c$
Row	G	$(0.3, 0.5)$	0.174 (0.004)	0.175 (0.004)	0.493 (0.005)	0.142 (0.003)	0.156 (0.003)
Row	U	$(0.3, 0.5)$	0.060 (0.001)	0.060 (0.001)	0.165 (0.002)	0.046 (0.001)	0.051 (0.001)
WH	G	$(0.3, 0.5)$	0.213 (0.005)	0.214 (0.005)	0.622 (0.006)	0.181 (0.005)	0.185 (0.005)
WH	U	$(0.3, 0.5)$	0.068 (0.002)	0.069 (0.002)	0.206 (0.002)	0.057 (0.001)	0.059 (0.001)
DG	G	$(0.3, 0.5)$	0.216 (0.005)	0.218 (0.005)	0.683 (0.007)	0.200 (0.005)	0.197 (0.005)
DG	U	$(0.3, 0.5)$	0.069 (0.002)	0.070 (0.002)	0.228 (0.002)	0.065 (0.001)	0.061 (0.001)
Row	G	$(0.5, 0.7)$	0.340 (0.008)	0.340 (0.008)	1.535 (0.011)	0.247 (0.006)	0.282 (0.006)
Row	U	$(0.5, 0.7)$	0.110 (0.003)	0.110 (0.003)	0.513 (0.004)	0.079 (0.002)	0.091 (0.002)
WH	G	$(0.5, 0.7)$	0.377 (0.009)	0.377 (0.009)	1.859 (0.013)	0.301 (0.006)	0.311 (0.007)
WH	U	$(0.5, 0.7)$	0.125 (0.003)	0.125 (0.003)	0.625 (0.004)	0.101 (0.002)	0.103 (0.002)
DG	G	$(0.5, 0.7)$	0.487 (0.012)	0.488 (0.012)	2.539 (0.016)	0.519 (0.010)	0.386 (0.009)
DG	U	$(0.5, 0.7)$	0.157 (0.004)	0.159 (0.004)	0.854 (0.005)	0.177 (0.003)	0.127 (0.003)
Row	G	$(0.7, 0.9)$	0.524 (0.012)	0.529 (0.012)	2.831 (0.024)	0.487 (0.011)	0.641 (0.014)
Row	U	$(0.7, 0.9)$	0.184 (0.004)	0.185 (0.004)	0.941 (0.008)	0.164 (0.004)	0.216 (0.005)
WH	G	$(0.7, 0.9)$	0.594 (0.014)	0.598 (0.014)	3.615 (0.027)	0.637 (0.014)	0.699 (0.016)
WH	U	$(0.7, 0.9)$	0.209 (0.005)	0.211 (0.005)	1.207 (0.008)	0.210 (0.005)	0.218 (0.005)
DG	G	$(0.7, 0.9)$	0.979 (0.023)	0.991 (0.023)	8.319 (0.045)	2.653 (0.026)	0.943 (0.022)
DG	U	$(0.7, 0.9)$	0.335 (0.008)	0.339 (0.008)	2.806 (0.015)	0.904 (0.009)	0.327 (0.007)
Row	G	$(0.8, 1.0)$	0.681 (0.015)	0.686 (0.015)	0.681 (0.015)	0.681 (0.015)	–
Row	U	$(0.8, 1.0)$	0.227 (0.005)	0.228 (0.005)	0.227 (0.005)	0.227 (0.005)	–
WH	G	$(0.8, 1.0)$	0.777 (0.018)	0.784 (0.018)	1.623 (0.025)	1.623 (0.025)	–
WH	U	$(0.8, 1.0)$	0.251 (0.006)	0.252 (0.006)	0.531 (0.008)	0.531 (0.008)	–
DG	G	$(0.8, 1.0)$	1.622 (0.038)	1.671 (0.041)	16.311 (0.093)	16.311 (0.093)	–
DG	U	$(0.8, 1.0)$	0.516 (0.012)	0.526 (0.013)	5.413 (0.031)	5.413 (0.031)	–

A.2. Additional MSE comparisons

Figures 5 and 6 extend the MSE comparison to weighted Haar and dense Gaussian sketches. We use the datasets, sketch-size grid, repetitions and paired MSE intervals of Section 5.3, with $\rho_*(d/n)$ as a common grid reference for these empirical comparisons.

For weighted Haar sketches, generate the orthonormal Haar rows at the largest tested size and take the first m rows for each m. Apply alternating weights one and two, and normalize separately at each m so that $\mathrm{Tr}(\boldsymbol{S}^\top \boldsymbol{S}) = n$. For dense Gaussian sketches, generate one standard Gaussian matrix with as many rows as the largest tested size, take its first m rows, and divide by $\sqrt{m}$. Each family uses its own random construction, and the compressed model is refitted at every m.

For both families, all eighteen intervals with $\delta < 0$ lie above one. At the nominal central settings, $\delta = 0$, the weighted Haar ratios range from 0.952 to 1.166; seven intervals contain one and two lie above it. The dense Gaussian ratios range from 1.021 to 1.386; seven intervals lie above one and two contain it. All nine dense Gaussian central point estimates exceed one. At $\delta = 0.04$, six weighted Haar intervals and four dense Gaussian intervals lie below one; the remaining intervals contain one. At $\delta = 0.08$, all nine weighted Haar intervals and eight dense Gaussian intervals lie below one; the remaining dense Gaussian interval contains one. The empirical ratios decrease across the five tested sketch sizes in every panel. For comparison, under row sampling, seventeen of the eighteen intervals with $\delta < 0$ lie above one and seventeen of the eighteen with $\delta > 0$ lie below one; the remaining interval on each side contains one. The central ratios at $\delta = 0$ range from 0.875 to 1.125. Across the row-sampling dimension settings, the exact crossover lies below $\rho_*(d/n)$ by less than $1/n$.

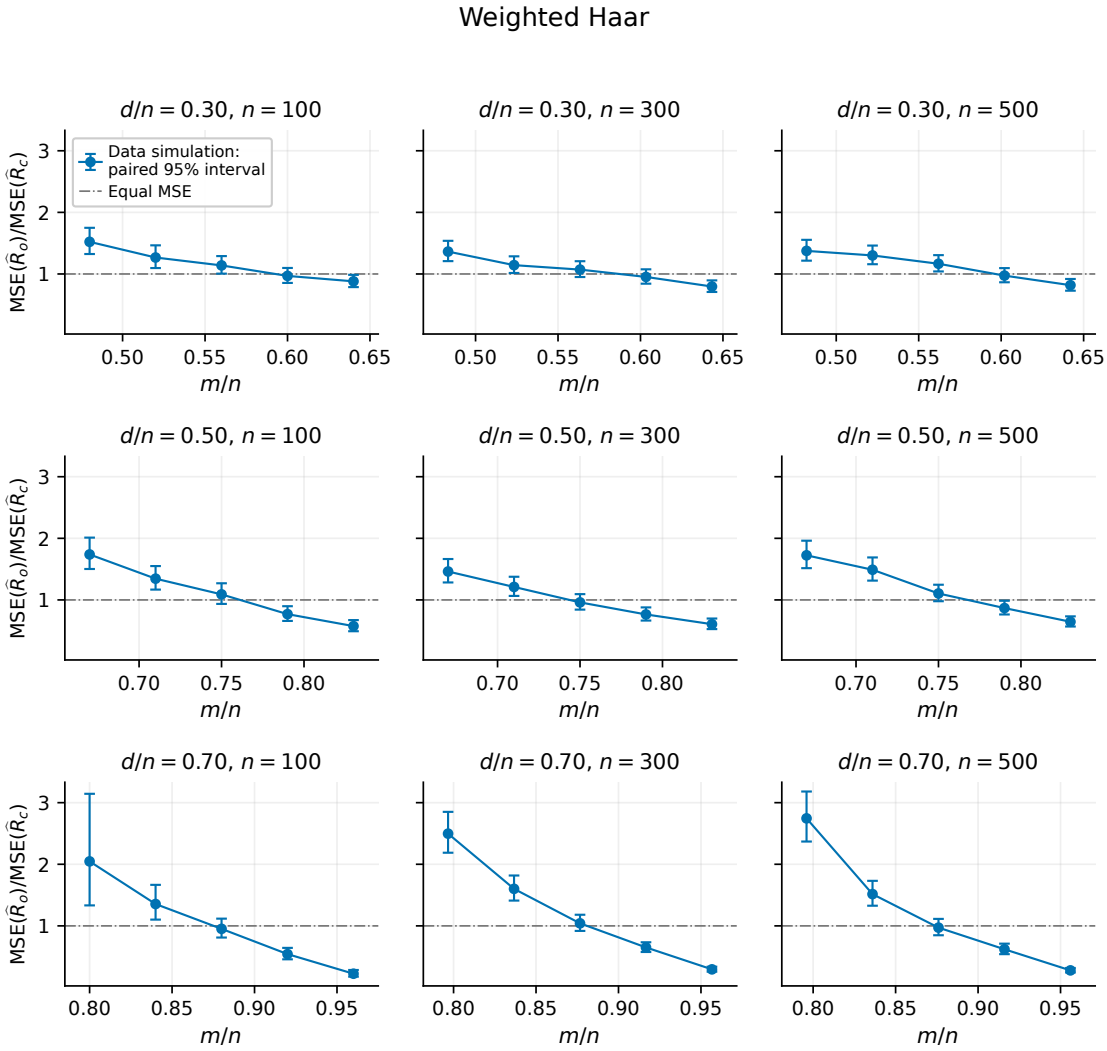


Figure 5: Empirical MSE ratio of the corrected to the complementary risk estimator for weighted Haar sketches under Gaussian data. Rows give $d/n \in \{0.30, 0.50, 0.70\}$ and columns give $n \in \{100, 300, 500\}$; the horizontal axis is the actual m/n. Points and error bars show ratios and pointwise 95% paired Monte Carlo intervals based on 1,000 repetitions per setting. Ratios above one favor the complementary estimate; ratios below one favor the corrected estimate. The horizontal line marks equal MSE, and connecting segments only guide the eye. Only empirical ratios are shown because the scaled orthogonal formula does not apply to the alternating weights.

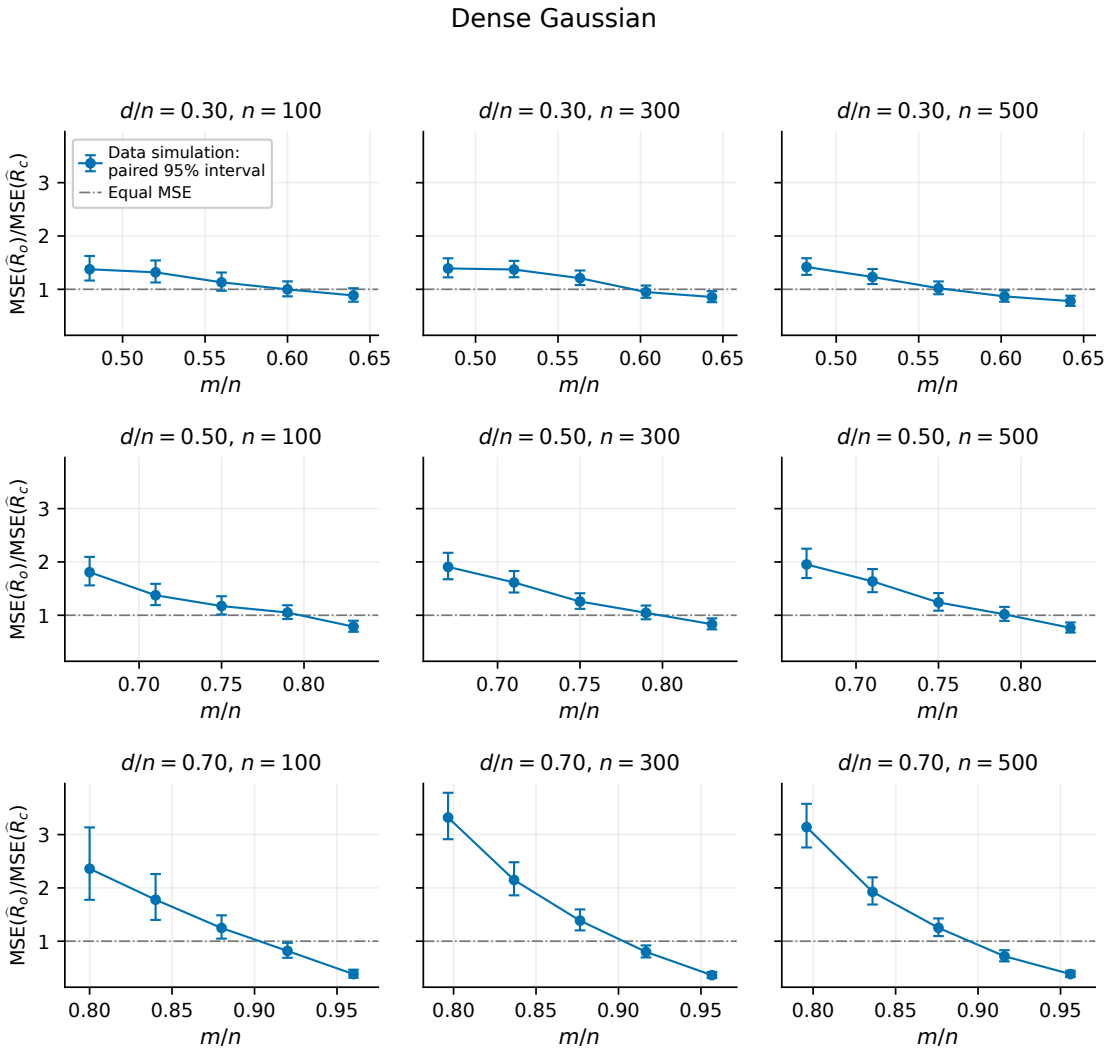


Figure 6: Empirical MSE ratio of the corrected to the complementary risk estimator for raw dense Gaussian sketches under Gaussian data. Layout, dimensions, pairing, repetitions, and intervals match Figure 5. Ratios above one favor the complementary estimate; ratios below one favor the corrected estimate. The horizontal line marks equal MSE, and connecting segments only guide the eye. Only empirical ratios are shown because a dense Gaussian sketch is not a scaled orthogonal sketch.

B. Notation and supporting results

This appendix supplies the notation and auxiliary results used in the proofs. After a compact setup, we develop the moment and trace bounds needed for the corrected estimator. Appendices C and D give the proofs in the order of the main text. For matrices, $\preceq$ denotes the positive-semidefinite order, $\|\cdot\|_{\mathrm{op}}$ the operator norm, and $\|\cdot\|_{\mathrm{F}}$ the Frobenius norm. The table below locates each main proof.

Main result	Proof sections
Theorem 3.1	C.1
Theorem 4.1	D.1.1 and D.1.2
Theorem 4.2	D.2.1 and D.2.2
Theorem 4.3	D.3
Theorem 4.4	D.4

B.1. Proof setup and assumptions

Assumption B.1. The observations, covariance bounds, errors, sketch and independent test pair satisfy the model conditions of Section 2. Write $0 < c_{\boldsymbol{\Sigma}} \le C_{\boldsymbol{\Sigma}} < \infty$ for the fixed constants, independent of n, satisfying $c_{\boldsymbol{\Sigma}} \boldsymbol{I}_d \preceq \boldsymbol{\Sigma} \preceq C_{\boldsymbol{\Sigma}} \boldsymbol{I}_d$.

Clarification (symbol $c_{\boldsymbol{\Sigma}}$):

$c_{\boldsymbol{\Sigma}}$ is a fixed positive constant, independent of n, satisfying $c_{\boldsymbol{\Sigma}} \boldsymbol{I}_d \preceq \boldsymbol{\Sigma}$ and $c_{\boldsymbol{\Sigma}} \le C_{\boldsymbol{\Sigma}}$.

Clarification (symbol $C_{\boldsymbol{\Sigma}}$):

$C_{\boldsymbol{\Sigma}}$ is a fixed finite positive constant, independent of n, satisfying $\boldsymbol{\Sigma} \preceq C_{\boldsymbol{\Sigma}} \boldsymbol{I}_d$.

We use the fit and residual notation of Section 4, and write

$$\boldsymbol{B} := \boldsymbol{S}\boldsymbol{X}, \qquad \boldsymbol{M} = \boldsymbol{B}^\top \boldsymbol{B}, \qquad \boldsymbol{h} := \hat{\boldsymbol{\beta}}_{\boldsymbol{S}} - \boldsymbol{\beta}^*.$$

Assumption B.2. In addition to Assumption B.1, suppose $1 \le d < m \le n$ and $\operatorname{rank}(\boldsymbol{S}\boldsymbol{X}) = d$ almost surely.

Let $\boldsymbol{P}_{\boldsymbol{B}} := \boldsymbol{B}(\boldsymbol{B}^\top \boldsymbol{B})^{-1} \boldsymbol{B}^\top$. The residual sketch mass satisfies

$$b_{\boldsymbol{A}} = \operatorname{Tr}\{(\boldsymbol{I}_m - \boldsymbol{P}_{\boldsymbol{B}}) \boldsymbol{S}\boldsymbol{S}^\top\} > 0,$$

because $\boldsymbol{S}\boldsymbol{S}^\top \succ 0$ and $\boldsymbol{I}_m - \boldsymbol{P}_{\boldsymbol{B}}$ is a projector of rank $m - d > 0$. If $\lambda_1(\boldsymbol{A}) \ge \cdots \ge \lambda_m(\boldsymbol{A}) > 0$ are the positive eigenvalues of $\boldsymbol{A}$, Ky Fan's inequality gives

$$b_{\boldsymbol{A}} \ge \sum_{j=d+1}^{m} \lambda_j(\boldsymbol{A}). \tag{11}$$

Clarification (symbol $\boldsymbol{B}$):

$\boldsymbol{B} \in \mathbb{R}^{m \times d}$ denotes $\boldsymbol{S}\boldsymbol{X}$ except where explicitly rebound locally. In Lemma B.12: In the proof's bound for the sketched least-squares map, $\boldsymbol{B} = \boldsymbol{\Lambda}^{1/2} \boldsymbol{G}_0$ locally, where $\boldsymbol{G}_0 \in \mathbb{R}^{m \times d}$ is the Gaussian matrix used in that proof.

Clarification (symbol $\boldsymbol{h}$):

$\boldsymbol{h} = \hat{\boldsymbol{\beta}}_{\boldsymbol{S}} - \boldsymbol{\beta}^*$ is the coefficient error computed from the response used in the expectation under consideration. In Lemma B.21, Proposition B.23: Under $\mathbb{E}_{\mathrm{aux}}$, recompute $\boldsymbol{h}$ using the auxiliary response and $\boldsymbol{\beta}^* = \boldsymbol{0}$; under the original response law use the original coefficient error.

The fit and both corrected estimators are invariant to the overall sketch scale. Replacing $\boldsymbol{S}$ by $\sqrt{c}\,\boldsymbol{S}$ for any $c > 0$ multiplies $\boldsymbol{A}$, $b_{\boldsymbol{A}}$, and $\boldsymbol{r}$ by c and $t_{\boldsymbol{\Sigma}}$ by c^{-1}, leaving the corrected residuals unchanged. The discrepancy $t_{\boldsymbol{\Sigma}} - d/b_{\boldsymbol{A}}$ scales by c^{-1}. We therefore use

$$\overline{\boldsymbol{S}} := \left\{ \frac{n}{\operatorname{Tr}(\boldsymbol{A})} \right\}^{1/2} \boldsymbol{S}, \qquad \overline{\boldsymbol{A}} := \overline{\boldsymbol{S}}^\top \overline{\boldsymbol{S}}.$$

In the balanced-sketch calculations from Section B.2 through Section B.5.4, we reuse $(\boldsymbol{S}, \boldsymbol{A}, b_{\boldsymbol{A}}, \boldsymbol{r}, t_{\boldsymbol{\Sigma}})$ for the normalized quantities, so $\mathrm{Tr}(\boldsymbol{A}) = n$ almost surely. The sketch-specific calculations in Appendix D retain the natural scalings stated there.

Assumption B.3. In addition to Assumption B.2, suppose $\boldsymbol{X} = \boldsymbol{Z}\boldsymbol{\Sigma}^{1/2}$, where $\boldsymbol{Z}$ has independent standard Gaussian entries and is independent of $(\boldsymbol{S}, \boldsymbol{\varepsilon})$. The errors are uniformly sub-Gaussian: there exists a constant $K_\varepsilon < \infty$, independent of n, such that $\sup_n \|\varepsilon_1\|_{\psi_2} \le K_\varepsilon$, where

$$\|W\|_{\psi_2} := \inf\left\{t > 0 : \mathbb{E}\exp(W^2/t^2) \le 2\right\}.$$

Use the normalization $\mathrm{Tr}(\boldsymbol{A}) = n$, and let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$ be an $\boldsymbol{S}$-measurable spectral decomposition, where $\boldsymbol{U} \in \mathbb{R}^{n\times m}$ satisfies $\boldsymbol{U}^\top\boldsymbol{U} = \boldsymbol{I}_m$ and $\boldsymbol{\Lambda}$ contains the positive eigenvalues of $\boldsymbol{A}$. There exist deterministic constants $n_0 \ge 1$, $0 < c_- \le c_+ < \infty$, and $\kappa > 0$, independent of n, such that, for every $n \ge n_0$, almost surely,

$$c_-\boldsymbol{I}_m \preceq \boldsymbol{\Lambda} \preceq c_+\boldsymbol{I}_m, \qquad m - d \ge \kappa n. \tag{12}$$

Clarification:

In the balanced-sketch calculations, $(\boldsymbol{S}, \boldsymbol{A})$ denote the normalized quantities $(\overline{\boldsymbol{S}}, \overline{\boldsymbol{A}})$ defined in Section B.1, so $\mathrm{Tr}(\boldsymbol{A}) = n$ almost surely.

Clarification (symbol K_ε):

$K_\varepsilon < \infty$ is a deterministic constant independent of n controlling the error ψ_2 norm, using the absolute convention unless the relative convention is explicitly invoked. In Lemma B.11: For the explicitly stated relative version of the noise-energy bound, K_ε instead satisfies $\|\varepsilon_1\|_{\psi_2} \le K_\varepsilon\sigma$; this convention applies to that version only.

Clarification (symbol U):

$\boldsymbol{U}$ is an orthonormal-column matrix, with its column count and defining decomposition specified locally. In Theorem 4.1, Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10, Lemma B.11, Lemma B.12, Lemma B.14, Lemma B.15, Proposition B.18, Proposition 40, Lemma B.20, Lemma B.21, Lemma B.22, Proposition B.23: $\boldsymbol{U} \in \mathbb{R}^{n\times m}$ is an $\boldsymbol{S}$-measurable choice in $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$, with $\boldsymbol{U}^\top\boldsymbol{U} = \boldsymbol{I}_m$ and $\boldsymbol{\Lambda}$ containing the positive eigenvalues; it is held fixed in derivatives with respect to $\boldsymbol{Z}$. In Theorem 4.2: In the dense Gaussian proof, $\boldsymbol{U} = \boldsymbol{Z}(\boldsymbol{Z}^\top\boldsymbol{Z})^{-1/2} \in \mathbb{R}^{n\times d}$ is the thin polar factor of $\boldsymbol{Z}$, with $\boldsymbol{U}^\top\boldsymbol{U} = \boldsymbol{I}_d$; $[\boldsymbol{U}, \boldsymbol{V}]$ is a measurable orthogonal completion.

Clarification (symbol Λ):

$\boldsymbol{\Lambda} \in \mathbb{R}^{m\times m}$ is the diagonal matrix of positive eigenvalues in an $\boldsymbol{S}$-measurable decomposition $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$, where $\boldsymbol{U}^\top\boldsymbol{U} = \boldsymbol{I}_m$ and $\mathrm{Tr}(\boldsymbol{A}) = n$; for all sufficiently large n, $c_-\boldsymbol{I}_m \preceq \boldsymbol{\Lambda} \preceq c_+\boldsymbol{I}_m$ almost surely.

Together with (11), these conditions imply

$$b_{\boldsymbol{A}} \ge c_-(m-d) \ge c_-\kappa n, \qquad \frac{n}{b_{\boldsymbol{A}}} \le \frac{1}{c_-\kappa}.$$

B.2. Coordinates and differential identities

The correction argument in this and the next three subsections uses the trace-normalized representative and Assumption B.3. All fixed-order moment bounds are uniform for sufficiently large n. The threshold may depend on the fixed moment order, and the constants may depend only on the fixed model and sketch parameters. Unless stated otherwise, derivatives are pathwise with respect to $\boldsymbol{Z}$, with $\boldsymbol{S}$, $\boldsymbol{\Sigma}$, $\boldsymbol{\beta}^*$, and the structural noise $\boldsymbol{\varepsilon} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}^*$ held fixed. Write

$$b := b_{\boldsymbol{A}}, \qquad \boldsymbol{q} := \boldsymbol{\Sigma}^{1/2}\boldsymbol{h},$$
$$\widetilde{\boldsymbol{M}} := \boldsymbol{Z}^\top\boldsymbol{A}\boldsymbol{Z}, \qquad \boldsymbol{L} := \boldsymbol{A}\boldsymbol{Z}\widetilde{\boldsymbol{M}}^{-1}, \qquad \boldsymbol{\ell} := \boldsymbol{L}\boldsymbol{q},$$
$$\boldsymbol{v} := \boldsymbol{Z}\boldsymbol{q} - t_{\boldsymbol{\Sigma}}\boldsymbol{r}, \qquad G_{\boldsymbol{\Sigma}} := \|\boldsymbol{q}\|^2 + \frac{1}{n}\|\boldsymbol{r}\|^2 = \|\boldsymbol{\Sigma}^{1/2}\boldsymbol{h}\|^2 + \frac{1}{n}\|\boldsymbol{r}\|^2.$$

In these supporting calculations, the trace coefficient is

$$t_{\boldsymbol{\Sigma}} = \mathrm{Tr}(\boldsymbol{\Sigma}\boldsymbol{M}^{-1}) = \mathrm{Tr}(\widetilde{\boldsymbol{M}}^{-1}).$$

Here $\boldsymbol{q}$ measures coefficient error on the prediction scale, $\boldsymbol{q} = \boldsymbol{L}^\top\boldsymbol{\varepsilon}$ expresses it in terms of training noise, and $\boldsymbol{r} = \boldsymbol{A}(\boldsymbol{\varepsilon} - \boldsymbol{Z}\boldsymbol{q})$ is the weighted residual. Moments of $\boldsymbol{L}$ and $\boldsymbol{r}$ control the derivatives used below. Also write $\boldsymbol{G}_0 := \boldsymbol{U}^\top\boldsymbol{Z} \in \mathbb{R}^{m\times d}$, where $\boldsymbol{U}$ comes from the standing spectral decomposition of $\boldsymbol{A}$. We use $[d] := \{1, \ldots, d\}$ and let $\boldsymbol{e}_j$ denote a standard basis vector, with its ambient dimension determined by context. For any weakly differentiable map $\boldsymbol{f} : \mathbb{R}^{n\times d} \to \mathbb{R}^p$, define its derivative energy by

$$\mathcal{E}_{\boldsymbol{f}} := \sum_{i=1}^{n}\sum_{j=1}^{d} \|\frac{\partial \boldsymbol{f}}{\partial z_{ij}}\|^2.$$

The value of p will be clear from the map under consideration.

Clarification (symbol $\boldsymbol{q}$):

$\boldsymbol{q}$ is an $\mathbb{R}^d$-valued field of the design coordinates. In Theorem 4.1, Theorem 4.2, Theorem 4.3, Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10, Lemma B.12, Lemma B.14, Lemma B.15, Lemma B.17, Proposition B.18, Proposition 40, Lemma B.20, Lemma B.21, Lemma B.22, Proposition B.23: $\boldsymbol{q} = \boldsymbol{\Sigma}^{1/2}(\hat{\boldsymbol{\beta}}_{\boldsymbol{S}} - \boldsymbol{\beta}^*) = \widetilde{\boldsymbol{M}}^{-1}\boldsymbol{Z}^\top\boldsymbol{A}\boldsymbol{\varepsilon} = \boldsymbol{L}^\top\boldsymbol{\varepsilon}$ is the prediction-scale coefficient error; its design derivatives hold $\boldsymbol{S}$, $\boldsymbol{\Sigma}$, $\boldsymbol{\beta}^*$ and $\boldsymbol{\varepsilon}$ fixed. In Lemma B.13: $\boldsymbol{q} \in \mathbb{R}^d$ is a general random vector field satisfying the displayed derivative identities and moment assumptions; it need not be a least-squares coefficient error. In Lemma B.16: Conditional on $(\boldsymbol{S}, \varepsilon)$, $\boldsymbol{q} = \boldsymbol{q}(\boldsymbol{Z}, \boldsymbol{S}, \varepsilon)$ is a general C^∞ field in $\boldsymbol{Z}$; its meaning is not restricted to the coefficient-error specialization. In Lemma B.21, Proposition B.23: In expressions under $\mathbb{E}_{\mathrm{aux}}$, recompute $\boldsymbol{q}$ from the auxiliary response with $\boldsymbol{\beta}^* = \boldsymbol{0}$; expressions under the original response law retain its original coefficient-error field. In another result of the paper: Conditional on $(\boldsymbol{S}, \varepsilon)$, $\boldsymbol{q} = \boldsymbol{q}(\boldsymbol{Z}, \boldsymbol{S}, \varepsilon)$ is a general C^∞ field in $\boldsymbol{Z}$ under the second-order moment premises; it need not be the prediction-scale coefficient error.

Clarification (symbol $\boldsymbol{L}$):

$\boldsymbol{L}$ is an $n \times d$ real matrix with the specialization specified in its scoped uses. In Lemma B.13: $\boldsymbol{L}$ is a generic random matrix in $\mathbb{R}^{n\times d}$ satisfying $\partial\boldsymbol{q}/\partial z_{ij} = \boldsymbol{K}\boldsymbol{e}_j r_i - \boldsymbol{L}^\top\boldsymbol{e}_i q_j$ almost everywhere; it need not equal $\boldsymbol{A}\boldsymbol{Z}\widetilde{\boldsymbol{M}}^{-1}$. In Theorem 4.1, Theorem 4.2, Lemma B.4, Lemma B.5, Lemma B.6, Lemma B.7, Lemma B.8, Lemma B.9, Lemma B.10, Lemma B.12, Lemma B.14, Lemma B.15, Lemma B.17, Proposition B.18, Proposition 40, Lemma B.20, Lemma B.21, Lemma B.22, Proposition B.23: $\boldsymbol{L} = \boldsymbol{A}\boldsymbol{Z}\widetilde{\boldsymbol{M}}^{-1}$ on the domain where $\widetilde{\boldsymbol{M}}$ is invertible.

Clarification (symbol $G_{\boldsymbol{\Sigma}}$):

$G_{\boldsymbol{\Sigma}} := \|\boldsymbol{q}\|^2 + \|\boldsymbol{r}\|^2/n = \|\boldsymbol{\Sigma}^{1/2}\boldsymbol{h}\|^2 + \|\boldsymbol{r}\|^2/n$ is the response energy computed from the current response quantities, and the response-normalized ratios involving this energy are defined to be zero on $\{G_{\boldsymbol{\Sigma}} = 0\}$. In Lemma B.21, Proposition B.23: When the auxiliary response construction is used, $G_{\boldsymbol{\Sigma}}$ is recomputed from its response quantities and is strictly positive almost surely.

Clarification (symbol $\boldsymbol{G}_0$):

$\boldsymbol{G}_0 = \boldsymbol{U}^\top\boldsymbol{Z} \in \mathbb{R}^{m\times d}$ uses the $\boldsymbol{S}$-measurable eigenvector matrix from the standing decomposition $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$, so $\boldsymbol{U}$ is held fixed in differentiation with respect to $\boldsymbol{Z}$.

Clarification (symbol $\boldsymbol{f}$):

$\boldsymbol{f}$ denotes a field on the design-coordinate matrix space, with output dimension determined by the application and derivatives taken in its input coordinates. In Lemma B.19: Conditional on $\mathcal{F}$, $\boldsymbol{f} : \mathbb{R}^{n\times d} \to \mathbb{R}^d$ is C^∞ and may depend on the fixed value of $\mathcal{F}$. In Lemma B.21: For the transposed-design argument, $\boldsymbol{W} = \boldsymbol{Z}^\top \in \mathbb{R}^{d\times n}$ and $\boldsymbol{f}(\boldsymbol{W}) = \boldsymbol{r}/\sqrt{n} \in \mathbb{R}^n$; derivatives are with respect to $w_{ji} = z_{ij}$. In the normalized contraction application, the first field is instead $\boldsymbol{q}_\eta$, followed by its weak-derivative limit $\boldsymbol{q}$.

Clarification (symbol $\mathcal{E}_{\boldsymbol{f}}$):

For a weakly differentiable map $\boldsymbol{f} : \mathbb{R}^{n\times d} \to \mathbb{R}^p$, $\mathcal{E}_{\boldsymbol{f}} = \sum_{i=1}^{n}\sum_{j=1}^{d} \|\frac{\partial \boldsymbol{f}}{\partial z_{ij}}\|^2$ is its pointwise derivative energy, with all other arguments held fixed and p determined by the map.

B.2.1. Pathwise identities

The identities below convert derivatives of the fitted coefficients into residual and coefficient quantities used in the later Stein bounds. We use the notation and derivative convention of Section B.2 on the domain where $\widetilde{\boldsymbol{M}}$ is invertible; the original covariance $\boldsymbol{\Sigma}$ remains arbitrary under the model assumptions.

Lemma B.4. *Recall $\boldsymbol{r} = \boldsymbol{A}(\boldsymbol{\varepsilon} - \boldsymbol{Z}\boldsymbol{q})$. Then $\boldsymbol{Z}^\top \boldsymbol{r} = 0$, and*

$$\frac{\partial \boldsymbol{q}}{\partial z_{ij}} = \widetilde{\boldsymbol{M}}^{-1}\boldsymbol{e}_j r_i - \boldsymbol{L}^\top \boldsymbol{e}_i q_j,$$
$$\sum_{j=1}^{d} \frac{\partial q_j}{\partial z_{ij}} = t_{\boldsymbol{\Sigma}} r_i - \boldsymbol{e}_i^\top \boldsymbol{L}\boldsymbol{q}.$$

Proof of Lemma B.4. The model and normal equations give $\widetilde{\boldsymbol{M}}\boldsymbol{q} = \boldsymbol{Z}^\top \boldsymbol{A}\boldsymbol{\varepsilon}$ and $\boldsymbol{Z}^\top \boldsymbol{r} = 0$. Since $\boldsymbol{A}$ and $\boldsymbol{\varepsilon}$ are fixed under differentiation,

$$\frac{\partial \widetilde{\boldsymbol{M}}}{\partial z_{ij}} = \boldsymbol{e}_j \boldsymbol{e}_i^\top \boldsymbol{A}\boldsymbol{Z} + \boldsymbol{Z}^\top \boldsymbol{A}\boldsymbol{e}_i \boldsymbol{e}_j^\top.$$

Differentiating $\boldsymbol{q} = \widetilde{\boldsymbol{M}}^{-1}\boldsymbol{Z}^\top \boldsymbol{A}\boldsymbol{\varepsilon}$ yields

$$\begin{aligned}\frac{\partial \boldsymbol{q}}{\partial z_{ij}} &= -\widetilde{\boldsymbol{M}}^{-1}\frac{\partial \widetilde{\boldsymbol{M}}}{\partial z_{ij}}\boldsymbol{q} + \widetilde{\boldsymbol{M}}^{-1}\boldsymbol{e}_j \boldsymbol{e}_i^\top \boldsymbol{A}\boldsymbol{\varepsilon} \\ &= \widetilde{\boldsymbol{M}}^{-1}\boldsymbol{e}_j \boldsymbol{e}_i^\top \boldsymbol{A}(\boldsymbol{\varepsilon} - \boldsymbol{Z}\boldsymbol{q}) - \widetilde{\boldsymbol{M}}^{-1}\boldsymbol{Z}^\top \boldsymbol{A}\boldsymbol{e}_i q_j \\ &= \widetilde{\boldsymbol{M}}^{-1}\boldsymbol{e}_j r_i - \boldsymbol{L}^\top \boldsymbol{e}_i q_j.\end{aligned}$$

Taking the jth component and summing over j gives

$$\begin{aligned}\sum_{j=1}^{d} \frac{\partial q_j}{\partial z_{ij}} &= \sum_{j=1}^{d} \boldsymbol{e}_j^\top \widetilde{\boldsymbol{M}}^{-1}\boldsymbol{e}_j r_i - \sum_{j=1}^{d} \boldsymbol{e}_j^\top \boldsymbol{L}^\top \boldsymbol{e}_i q_j \\ &= t_{\boldsymbol{\Sigma}} r_i - \boldsymbol{e}_i^\top \boldsymbol{L}\boldsymbol{q}.\end{aligned}$$

□

Lemma B.5. *For every $j \in [d]$,*

$$\sum_{i=1}^{n} \frac{\partial r_i}{\partial z_{ij}} = -b q_j - \boldsymbol{e}_j^\top \boldsymbol{L}^\top \boldsymbol{r}.$$

Proof of Lemma B.5. Differentiating $\boldsymbol{r} = \boldsymbol{A}(\boldsymbol{\varepsilon} - \boldsymbol{Z}\boldsymbol{q})$ and using Lemma B.4 gives

$$\begin{aligned}\frac{\partial \boldsymbol{r}}{\partial z_{ij}} &= -\boldsymbol{A}\boldsymbol{e}_i q_j - \boldsymbol{A}\boldsymbol{Z}\frac{\partial \boldsymbol{q}}{\partial z_{ij}} \\ &= -\boldsymbol{A}\boldsymbol{e}_i q_j - \boldsymbol{A}\boldsymbol{Z}\widetilde{\boldsymbol{M}}^{-1}\boldsymbol{e}_j r_i + \boldsymbol{A}\boldsymbol{Z}\boldsymbol{L}^\top \boldsymbol{e}_i q_j.\end{aligned}$$

Taking the ith component and summing over i yields

$$\begin{aligned}\sum_{i=1}^{n} \frac{\partial r_i}{\partial z_{ij}} &= -\operatorname{Tr}(\boldsymbol{A})q_j - \boldsymbol{e}_j^\top \boldsymbol{L}^\top \boldsymbol{r} + \operatorname{Tr}(\boldsymbol{A}\boldsymbol{Z}\boldsymbol{L}^\top)q_j \\ &= -b q_j - \boldsymbol{e}_j^\top \boldsymbol{L}^\top \boldsymbol{r}.\end{aligned}$$

Here $\boldsymbol{A}\boldsymbol{Z}\boldsymbol{L}^\top = \boldsymbol{A}\boldsymbol{X}\boldsymbol{M}^{-1}\boldsymbol{X}^\top \boldsymbol{A}$, so the two trace terms combine to $-b q_j$. □

Lemma B.6. *The first product divergence satisfies*

$$\sum_{i=1}^{n}\sum_{j=1}^{d} \frac{\partial r_i q_j}{\partial z_{ij}} = -b\|\boldsymbol{q}\|^2 + t_{\boldsymbol{\Sigma}}\|\boldsymbol{r}\|^2 - 2\boldsymbol{r}^\top \boldsymbol{L}\boldsymbol{q}.$$

Proof of Lemma B.6. The product rule and Lemmas B.4–B.5 give

$$\begin{aligned}\sum_{i=1}^{n}\sum_{j=1}^{d}\frac{\partial r_i q_j}{\partial z_{ij}} &= \sum_{i=1}^{n} r_i \sum_{j=1}^{d}\frac{\partial q_j}{\partial z_{ij}} + \sum_{j=1}^{d} q_j \sum_{i=1}^{n}\frac{\partial r_i}{\partial z_{ij}} \\ &= \sum_{i=1}^{n} r_i(t_{\boldsymbol{\Sigma}} r_i - \boldsymbol{e}_i^\top \boldsymbol{L}\boldsymbol{q}) + \sum_{j=1}^{d} q_j(-bq_j - \boldsymbol{e}_j^\top \boldsymbol{L}^\top \boldsymbol{r}) \\ &= -b\|\boldsymbol{q}\|^2 + t_{\boldsymbol{\Sigma}}\|\boldsymbol{r}\|^2 - 2\boldsymbol{r}^\top \boldsymbol{L}\boldsymbol{q}.\end{aligned}$$

□

Lemma B.7. *The second product divergence satisfies*

$$\sum_{i=1}^{n}\sum_{j=1}^{d}\frac{\partial q_j \boldsymbol{e}_i^\top \boldsymbol{Z}\boldsymbol{q}}{\partial z_{ij}} = t_{\boldsymbol{\Sigma}}\boldsymbol{r}^\top \boldsymbol{Z}\boldsymbol{q} + (n-d-1)\|\boldsymbol{q}\|^2.$$

Proof of Lemma B.7. First, the product rule and Lemma B.4 imply

$$\begin{aligned}\sum_{i=1}^{n}\frac{\partial \boldsymbol{e}_i^\top \boldsymbol{Z}\boldsymbol{q}}{\partial z_{ij}} &= nq_j + \sum_{i=1}^{n}\boldsymbol{e}_i^\top \boldsymbol{Z}\frac{\partial \boldsymbol{q}}{\partial z_{ij}} \\ &= nq_j + \boldsymbol{e}_j^\top \widetilde{\boldsymbol{M}}^{-1}\boldsymbol{Z}^\top \boldsymbol{r} - q_j \operatorname{Tr}(\boldsymbol{Z}\boldsymbol{L}^\top) \\ &= (n-d)q_j,\end{aligned}$$

because $\boldsymbol{Z}^\top \boldsymbol{r} = 0$ and $\boldsymbol{Z}^\top \boldsymbol{L} = \boldsymbol{I}_d$. Applying the product rule once more gives

$$\begin{aligned}\sum_{i=1}^{n}\sum_{j=1}^{d}\frac{\partial q_j \boldsymbol{e}_i^\top \boldsymbol{Z}\boldsymbol{q}}{\partial z_{ij}} &= \sum_{i=1}^{n}\boldsymbol{e}_i^\top \boldsymbol{Z}\boldsymbol{q}\sum_{j=1}^{d}\frac{\partial q_j}{\partial z_{ij}} + \sum_{j=1}^{d} q_j \sum_{i=1}^{n}\frac{\partial \boldsymbol{e}_i^\top \boldsymbol{Z}\boldsymbol{q}}{\partial z_{ij}} \\ &= t_{\boldsymbol{\Sigma}}\boldsymbol{r}^\top \boldsymbol{Z}\boldsymbol{q} - \boldsymbol{q}^\top \boldsymbol{Z}^\top \boldsymbol{L}\boldsymbol{q} + (n-d)\|\boldsymbol{q}\|^2 \\ &= t_{\boldsymbol{\Sigma}}\boldsymbol{r}^\top \boldsymbol{Z}\boldsymbol{q} + (n-d-1)\|\boldsymbol{q}\|^2.\end{aligned}$$

□

Lemma B.8. *The normal equations imply*

$$\sum_{i=1}^{n}\sum_{j=1}^{d}\frac{\partial r_i \boldsymbol{e}_j^\top \boldsymbol{Z}^\top \boldsymbol{r}}{\partial z_{ij}} = 0.$$

Proof of Lemma B.8. The normal equations imply $\boldsymbol{Z}^\top \boldsymbol{r} = 0$ identically on the full-rank domain. Hence every function $r_i \boldsymbol{e}_j^\top \boldsymbol{Z}^\top \boldsymbol{r}$ is identically zero, as is the sum of its derivatives. □

Lemma B.9. *The residual squared-divergence identity is*

$$\sum_{j=1}^{d}\left(\boldsymbol{r}^\top \boldsymbol{Z}\boldsymbol{e}_j - \sum_{i=1}^{n}\frac{\partial r_i}{\partial z_{ij}}\right)^2 = \|b\boldsymbol{q} + \boldsymbol{L}^\top \boldsymbol{r}\|^2.$$

Proof of Lemma B.9. Lemma B.5 and $\boldsymbol{Z}^\top \boldsymbol{r} = 0$ give

$$\boldsymbol{r}^\top \boldsymbol{Z}\boldsymbol{e}_j - \sum_{i=1}^{n}\frac{\partial r_i}{\partial z_{ij}} = bq_j + \boldsymbol{e}_j^\top \boldsymbol{L}^\top \boldsymbol{r}.$$

Squaring and summing over j gives $\|b\boldsymbol{q} + \boldsymbol{L}^\top \boldsymbol{r}\|^2$. □

Lemma B.10. *The coefficient squared-divergence identity is*

$$\sum_{i=1}^{n}\left(\boldsymbol{q}^\top \boldsymbol{Z}^\top \boldsymbol{e}_i - \sum_{j=1}^{d}\frac{\partial q_j}{\partial z_{ij}}\right)^2 = \|\boldsymbol{Z}\boldsymbol{q} - t_{\boldsymbol{\Sigma}}\boldsymbol{r} + \boldsymbol{L}\boldsymbol{q}\|^2.$$

Proof of Lemma B.10. Lemma B.4 gives

$$\boldsymbol{q}^\top \boldsymbol{Z}^\top \boldsymbol{e}_i - \sum_{j=1}^{d}\frac{\partial q_j}{\partial z_{ij}} = \boldsymbol{e}_i^\top(\boldsymbol{Z}\boldsymbol{q} - t_{\boldsymbol{\Sigma}}\boldsymbol{r} + \boldsymbol{L}\boldsymbol{q}).$$

Squaring and summing over i proves the identity. □

B.3. Bounds from the sketch spectrum

The correction proof needs moments of the design and noise, the inverse sketched Gram matrix, the residual, and the fitted coefficient derivatives. The next three subsections establish these bounds in that order.

B.3.1. Design and noise moments

We begin with moments that do not involve matrix inversion.

Lemma B.11. *Suppose Assumption B.3 holds. For every fixed integer $k \geq 1$, there is a constant $C_k < \infty$, independent of n, such that*

$$\mathbb{E}\left(\frac{\|\boldsymbol{Z}\|_{\mathrm{op}}}{\sqrt{n}}\right)^k + \mathbb{E}\left(\frac{\|\boldsymbol{X}\|_{\mathrm{op}}}{\sqrt{n}}\right)^k \leq C_k. \tag{13}$$

More precisely, for every $s \geq 0$,

$$\mathbb{P}\left\{\frac{\|\boldsymbol{X}\|_{\mathrm{op}}}{\sqrt{n}} > \sqrt{C_{\boldsymbol{\Sigma}}}(16+s)\right\} \leq e^{-ns^2/16}. \tag{14}$$

For every fixed integer $k \geq 1$, the error-energy bounds are

$$\mathbb{E}\left(\frac{\|\boldsymbol{\varepsilon}\|^2}{n}\right)^k \leq C_k, \qquad \mathbb{E}\left|\|\boldsymbol{\varepsilon}\|^2 - n\sigma^2\right|^k \leq C_k n^{k/2}. \tag{15}$$

If the sub-Gaussian condition is expressed in the relative form $\|\varepsilon_1\|_{\psi_2} \leq K_\varepsilon \sigma$, the first bound in (15) becomes $\mathbb{E}(\|\boldsymbol{\varepsilon}\|^2/n)^k \leq C(k, K_\varepsilon)(\sigma^2)^k$. Moreover, there are constants $c, C > 0$, independent of n, such that

$$\mathbb{P}\left(\|\boldsymbol{Z}\|_{\mathrm{op}} > C\sqrt{n} \text{ or } \|\boldsymbol{X}\|_{\mathrm{op}} > C\sqrt{n} \text{ or } \|\boldsymbol{\varepsilon}\| > C\sqrt{n}\right) \leq Ce^{-cn}. \tag{16}$$

The constants in the design bounds depend only on the fixed moment order and $C_{\boldsymbol{\Sigma}}$; the noise bounds may additionally depend on K_ε and σ^2. All these bounds also hold conditionally on the independent sketch $\boldsymbol{S}$.

Proof. The spectral bounds in Assumption B.1 and the factorization $\boldsymbol{X} = \boldsymbol{Z}\boldsymbol{\Sigma}^{1/2}$ give

$$\sqrt{c_{\boldsymbol{\Sigma}}}\|\boldsymbol{Z}\|_{\mathrm{op}} \leq \|\boldsymbol{X}\|_{\mathrm{op}} \leq \sqrt{C_{\boldsymbol{\Sigma}}}\|\boldsymbol{Z}\|_{\mathrm{op}}. \tag{17}$$

A finite-net argument gives the Gaussian operator-norm tail. There is a $1/4$-net $\mathcal{N}$ of the unit ball in $\mathbb{R}^d$ with $|\mathcal{N}| \leq 9^d$, and $\|\boldsymbol{Z}\|_{\mathrm{op}} \leq 2\max_{v\in\mathcal{N}}\|\boldsymbol{Z}v\|$. For each fixed $\|v\| \leq 1$, the squared norm $\|\boldsymbol{Z}v\|^2$ has law $\|v\|^2\chi^2_n$. Its exponential moment at $1/4$ is at most $2^{n/2}$. Chernoff's inequality and a union bound therefore give, for $u \geq 0$,

$$\mathbb{P}\{\|\boldsymbol{Z}\|_{\mathrm{op}} \geq u\} \leq 9^d 2^{n/2} e^{-u^2/16}.$$

Since $d \leq n$, setting $u = 16\sqrt{n} + a$ yields

$$\mathbb{P}\{\|\boldsymbol{Z}\|_{\mathrm{op}} \geq 16\sqrt{n} + a\} \leq e^{-a^2/16}, \qquad a \geq 0.$$

Combining this inequality with (17) and taking $a = s\sqrt{n}$ proves (14). Integrating the tail of $\|\boldsymbol{Z}\|_{\mathrm{op}}/\sqrt{n}$ above 16, and using $n \geq 1$, proves (13). Only the covariance upper bound is needed for this step; the lower bound gives the reverse comparison in (17).

For the noise bounds, put $Y_i = \varepsilon_i^2 - \sigma^2$ and define $\|W\|_{\psi_1} := \inf\{t > 0 : \mathbb{E}\exp(|W|/t) \leq 2\}$. A centered sub-Gaussian square is sub-exponential, so $\|Y_i\|_{\psi_1} \leq CK_\varepsilon^2$. Bernstein's inequality therefore gives

$$\mathbb{P}\left(\left|\|\boldsymbol{\varepsilon}\|^2 - n\sigma^2\right| > a\right) \leq 2\exp\left[-c\min\left\{\frac{a^2}{nK_\varepsilon^4}, \frac{a}{K_\varepsilon^2}\right\}\right]. \tag{18}$$

Integrating (18) yields

$$\left(\mathbb{E}\left|\|\boldsymbol{\varepsilon}\|^2 - n\sigma^2\right|^k\right)^{1/k} \leq C_k\sqrt{n}.$$

The triangle inequality in L^k proves the first bound in (15). Taking a sufficiently large fixed threshold in (14) and (18), and using (17), proves (16). Finally, independence of $\boldsymbol{S}$ shows that conditioning on the sketch does not change any of the bounds. □

B.3.2. Inverse moments

The proportional gap between m and d controls the smallest singular value of the Gaussian design restricted to the sketch row space. Together with the lower bound on the positive sketch spectrum, it supplies the inverse moments needed for both the coefficient error and its derivatives. Invertibility itself follows from $\mathrm{rank}(\boldsymbol{S}\boldsymbol{X}) = d$.

Lemma B.12. *Suppose Assumption B.3 holds, and use the trace-normalized representative and spectral decomposition from Assumption B.3, together with the standing notation above. For every fixed integer $k \geq 1$, there are $n_0(k,\kappa) \geq 1$ and $C_k < \infty$, independent of n and uniform over all sketch matrices satisfying (12), such that, for $n \geq n_0(k,\kappa)$,*

$$\mathbb{E}\left[\|\boldsymbol{M}^{-1}\|_{\mathrm{op}}^k \mid \boldsymbol{S}\right] \leq \frac{C_k}{n^k}, \qquad \mathbb{E}\left[\|\boldsymbol{M}^{-1}\boldsymbol{X}^\top\boldsymbol{A}\|_{\mathrm{op}}^k \mid \boldsymbol{S}\right] \leq \frac{C_k}{n^{k/2}}. \tag{19}$$

Moreover,

$$\|\boldsymbol{M}^{-1}\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}\|_{\mathrm{op}} = \|\boldsymbol{I}_d\|_{\mathrm{op}} = 1 \qquad \textit{almost surely}, \tag{20}$$

and the corresponding whitened envelope is

$$\mathbb{E}\left[\left(n\|\widetilde{\boldsymbol{M}}^{-1}\|_{\mathrm{op}}\right)^k + \left(\sqrt{n}\|\boldsymbol{L}\|_{\mathrm{op}}\right)^k + t_{\boldsymbol{\Sigma}}^k + \left\{\mathrm{Tr}(\boldsymbol{M}^{-1})\right\}^k \;\middle|\; \boldsymbol{S}\right] \leq C_k. \tag{21}$$

The same inequalities hold for unconditional expectations. Their constants depend only on k, the fixed constants in (12), and the covariance bounds; in particular, they do not depend on the value of $\boldsymbol{S}$.

Proof. Condition on the sketch and put $\boldsymbol{W} = \boldsymbol{G}_0^\top\boldsymbol{G}_0$. Then $\boldsymbol{G}_0 \in \mathbb{R}^{m\times d}$ has independent standard Gaussian entries and

$$\widetilde{\boldsymbol{M}} = \boldsymbol{G}_0^\top\boldsymbol{\Lambda}\boldsymbol{G}_0 \succeq c_-\boldsymbol{W}. \tag{22}$$

We first prove the required negative moments of $\boldsymbol{W}$ directly. Put $\nu = m - d + 1$. The Schur complement and Gaussian orthogonal invariance show that, for every unit vector $v \in \mathbb{R}^d$, $v^\top\boldsymbol{W}^{-1}v \overset{\mathrm{d}}{=} 1/\chi_\nu^2$. For example, for a coordinate vector, the reciprocal quadratic form is $\boldsymbol{g}_j^\top(\boldsymbol{I}_m - \boldsymbol{P}_{-j})\boldsymbol{g}_j$, where $\boldsymbol{P}_{-j}$ projects onto the other $d-1$ Gaussian columns; this quadratic form has ν Gaussian degrees of freedom. Thus, for every integer $r \geq 1$ with $2r < \nu$,

$$\mathbb{E}[(v^\top\boldsymbol{W}^{-1}v)^r \mid \boldsymbol{S}] = \prod_{j=1}^{r}\frac{1}{\nu - 2j}.$$

For $\|v\| \leq 1$, the same expression is an upper bound.

Choose a $1/4$-net $\mathcal{N}$ of the unit ball with $|\mathcal{N}| \le 9^d$. Positivity of $\boldsymbol{W}^{-1}$ gives $\|\boldsymbol{W}^{-1}\|_{\mathrm{op}} \le 2\max_{v\in\mathcal{N}} v^\top \boldsymbol{W}^{-1} v$. If $4r \le \nu$, summing the directional moment bounds therefore yields

$$\mathbb{E}[(n\|\boldsymbol{W}^{-1}\|_{\mathrm{op}})^r \mid \boldsymbol{S}] \le 9^d \left(\frac{4n}{\nu}\right)^r.$$

Take $r = \lfloor \nu/4 \rfloor$. For all sufficiently large n, $r \ge \nu/8$ and $r \ge k$. Since $\nu \ge \kappa n$ and $d \le n$, the right side is at most D_κ^r, where $D_\kappa = (4/\kappa)9^{8/\kappa}$ is independent of n and the sketch. The elementary inequality $x^k \le 1 + x^r$ for $x \ge 0$ and $r \ge k$, applied to $x = n\|\boldsymbol{W}^{-1}\|_{\mathrm{op}}/D_\kappa$, gives

$$\mathbb{E}[\|\boldsymbol{W}^{-1}\|_{\mathrm{op}}^k \mid \boldsymbol{S}] \le \frac{2D_\kappa^k}{n^k}. \tag{23}$$

This argument also proves integrability, and uses no lower-tail singular-value estimate.

By (22) and order reversal under inversion,

$$\|\widetilde{\boldsymbol{M}}^{-1}\|_{\mathrm{op}} \le c_-^{-1}\|\boldsymbol{W}^{-1}\|_{\mathrm{op}},$$

which proves the first term of (21). For the sketched least-squares map, let $\boldsymbol{B} = \boldsymbol{\Lambda}^{1/2}\boldsymbol{G}_0$ locally in this proof. Since

$$\boldsymbol{L} = \boldsymbol{U}\boldsymbol{\Lambda}^{1/2}\boldsymbol{B}(\boldsymbol{B}^\top\boldsymbol{B})^{-1}, \qquad s_{\min}(\boldsymbol{B}) \ge \sqrt{c_-}\, s_{\min}(\boldsymbol{G}_0),$$

we have

$$\|\boldsymbol{L}\|_{\mathrm{op}} \le \sqrt{\frac{c_+}{c_-}}\, s_{\min}(\boldsymbol{G}_0)^{-1} = \sqrt{\frac{c_+}{c_-}}\|\boldsymbol{W}^{-1}\|_{\mathrm{op}}^{1/2}.$$

Lyapunov's inequality and (23) therefore give $\mathbb{E}[\|\boldsymbol{L}\|_{\mathrm{op}}^k \mid \boldsymbol{S}] \le C_k n^{-k/2}$.

Finally,

$$\boldsymbol{M} = \boldsymbol{\Sigma}^{1/2}\widetilde{\boldsymbol{M}}\boldsymbol{\Sigma}^{1/2}, \qquad \boldsymbol{M}^{-1}\boldsymbol{X}^\top\boldsymbol{A} = \boldsymbol{\Sigma}^{-1/2}\boldsymbol{L}^\top,$$

so the covariance lower bound proves (19). The normal equations prove (20), while

$$t_{\boldsymbol{\Sigma}} \le d\|\widetilde{\boldsymbol{M}}^{-1}\|_{\mathrm{op}}, \qquad \mathrm{Tr}(\boldsymbol{M}^{-1}) \le c_{\boldsymbol{\Sigma}}^{-1} t_{\boldsymbol{\Sigma}}$$

proves the remaining terms in (21). Taking expectations over $\boldsymbol{S}$ gives the unconditional claims. □

B.3.3. Residual contraction and derivative moments

Combining the inverse bounds with the normal equations gives the needed residual and derivative moments. The first lemma isolates a reusable derivative-energy estimate, and the second verifies it for balanced sketches.

Lemma B.13. *Suppose that random matrices $\boldsymbol{K} \in \mathbb{R}^{d\times d}$ and $\boldsymbol{L} \in \mathbb{R}^{n\times d}$ and random vectors $\boldsymbol{q} \in \mathbb{R}^d$ and $\boldsymbol{r} \in \mathbb{R}^n$ satisfy, almost everywhere,*

$$\frac{\partial \boldsymbol{q}}{\partial z_{ij}} = \boldsymbol{K}\boldsymbol{e}_j r_i - \boldsymbol{L}^\top \boldsymbol{e}_i q_j.$$

Then

$$\mathcal{E}_{\boldsymbol{q}} \le 2\|\boldsymbol{K}\|_{\mathrm{F}}^2\|\boldsymbol{r}\|^2 + 2\|\boldsymbol{L}\|_{\mathrm{F}}^2\|\boldsymbol{q}\|^2,$$
$$\mathbb{E}\mathcal{E}_{\boldsymbol{q}}^2 \le 8\mathbb{E}\left[\|\boldsymbol{K}\|_{\mathrm{F}}^4\|\boldsymbol{r}\|^4\right] + 8\mathbb{E}\left[\|\boldsymbol{L}\|_{\mathrm{F}}^4\|\boldsymbol{q}\|^4\right].$$

Consequently, boundedness of the two mixed moments on the right is sufficient for $\mathbb{E}\mathcal{E}_{\boldsymbol{q}}^2 = O(1)$.

Clarification:

The identity $\frac{\partial \boldsymbol{q}}{\partial z_{ij}} = \boldsymbol{K}\boldsymbol{e}_j r_i - \boldsymbol{L}^\top \boldsymbol{e}_i q_j$ holds almost surely under the joint law of the random fields, for every $i = 1, \dots, n$ and $j = 1, \dots, d$. (the reading of one of its hypotheses)

Clarification:

For the derivative versions defining $\mathcal{E}_{\boldsymbol{q}}$, the identity $\frac{\partial \boldsymbol{q}}{\partial z_{ij}} = \boldsymbol{K}\boldsymbol{e}_j r_i - \boldsymbol{L}^\top \boldsymbol{e}_i q_j$ holds almost surely under the law used by $\mathbb{E}$, simultaneously for $1 \le i \le n$ and $1 \le j \le d$. (the reading of one of its hypotheses)

Clarification (symbol $\mathcal{E}_{\boldsymbol{q}}$):

$\mathcal{E}_{\boldsymbol{q}}$ is the pointwise derivative energy of the field $\boldsymbol{q}$ with respect to the coordinates z_{ij}, for $1 \le i \le n$ and $1 \le j \le d$, with its other arguments held fixed.

Clarification (symbol $\boldsymbol{K}$):

$\boldsymbol{K}$ is a real matrix with dimensions and definition specified in its scoped uses. In Lemma B.13: $\boldsymbol{K} \in \mathbb{R}^{d\times d}$ is a generic random matrix satisfying $\partial \boldsymbol{q}/\partial z_{ij} = \boldsymbol{K}\boldsymbol{e}_j r_i - \boldsymbol{L}^\top \boldsymbol{e}_i q_j$ almost everywhere. In Lemma B.14, Lemma B.15, Lemma B.20, Lemma B.21: $\boldsymbol{K} = \widetilde{\boldsymbol{M}}^{-1}$ on the full-rank domain. In Theorem 4.2: In the dense Gaussian proof, $\boldsymbol{K} = (\boldsymbol{G}^\top \boldsymbol{G})^{-1}\boldsymbol{G}^\top \boldsymbol{H} \in \mathbb{R}^{d\times(n-d)}$, where $\sqrt{m}\,\boldsymbol{S}[\boldsymbol{U},\boldsymbol{V}] = [\boldsymbol{G},\boldsymbol{H}]$ and $\boldsymbol{Z} = \boldsymbol{U}\boldsymbol{R}$ is the thin polar decomposition.

Proof. Sum the squared derivative identity over (i,j) and use $\|\boldsymbol{a}-\boldsymbol{b}\|^2 \le 2\|\boldsymbol{a}\|^2 + 2\|\boldsymbol{b}\|^2$ to obtain the pointwise bound. Squaring it proves the mixed-moment bound. Cauchy–Schwarz gives the generic mixed-moment conclusion. □

Clarification (symbol $\boldsymbol{b}$):

$\boldsymbol{b}$ is locally bound vector notation, with its dimension and dependence determined by the surrounding inequality. In Lemma B.13, Lemma B.17: $\boldsymbol{a}$ and $\boldsymbol{b}$ are arbitrary vectors in the same Euclidean space in $\|\boldsymbol{a}-\boldsymbol{b}\|^2 \le 2\|\boldsymbol{a}\|^2 + 2\|\boldsymbol{b}\|^2$. In Lemma B.19, a result cited in the proof of Lemma B.19: In the matrix Gaussian divergence inequality, $\boldsymbol{b} : \mathbb{R}^{n\times d} \to \mathbb{R}^n$ is a smooth field, interpreted conditionally on $\mathcal{F}$ with derivatives taken in $\boldsymbol{Z}$. In Lemma B.19: In the later elementary comparison beginning with $c \ge 1$ and $\|\boldsymbol{u}\| \le 1$, $\boldsymbol{a}$ and $\boldsymbol{b}$ are arbitrary vectors of the same dimension as $\boldsymbol{u}$.

The balanced-sketch moments needed by the Gaussian–Stein tools follow from the design and inverse envelopes in Lemmas B.11 and B.12, together with the derivative-energy inequality in Lemma B.13; it does not use a Gaussian–Stein inequality.

Lemma B.14. *Suppose Assumption B.3 holds, using the trace-normalized representative. Then there are $n_0 \ge 1$ and $C < \infty$, independent of n, such that, for $n \ge n_0$, the map $\boldsymbol{q}$ is weakly differentiable and*

$$\begin{aligned}
\mathbb{E}\|\boldsymbol{q}\|^8 + n^{-4}\mathbb{E}\|\boldsymbol{r}\|^8 + \mathbb{E}\mathcal{E}_{\boldsymbol{q}}^2 &\le C,\\
\mathbb{E}\left[\|\boldsymbol{\varepsilon}\|^2\left(\|\boldsymbol{q}\|^2 + \mathcal{E}_{\boldsymbol{q}}\right)\right] &\le Cn,\\
\mathbb{E}\left[\|\boldsymbol{L}\|_{\mathrm{op}}^2\|\boldsymbol{q}\|^2\right] &\le \frac{C}{n},\\
\mathbb{E}\left[\|\boldsymbol{Z}\|_{\mathrm{op}}^2\|\boldsymbol{q}\|^2 + t_{\boldsymbol{\Sigma}}^2\|\boldsymbol{r}\|^2\right] &\le Cn.
\end{aligned} \tag{24}$$

In particular, Lyapunov's inequality also gives $\mathbb{E}\|\boldsymbol{q}\|^4 \le C$ and $\mathbb{E}\|\boldsymbol{r}\|^4 \le Cn^2$. Thus the direct hypotheses of the remainder and influence lemmas below hold as well.

Clarification:

For the inverse-defined field $\boldsymbol{q}$, the direct remainder hypotheses mean the displayed moment hypotheses, with the smooth-field inequalities applied to $\boldsymbol{q}_\eta$ before taking $\eta \downarrow 0$.

Proof. The design and inverse-moment lemmas apply under Assumption B.3. In the whitened notation above, $\widetilde{\boldsymbol{M}} = \boldsymbol{G}_0^\top \boldsymbol{\Lambda}\boldsymbol{G}_0$. Lemmas B.11 and B.12 imply that, for every fixed integer $k \ge 1$ and all sufficiently large n,

$$\mathbb{E}\left(\frac{\|\boldsymbol{Z}\|_{\mathrm{op}}}{\sqrt{n}}\right)^k + \mathbb{E}\left(n\|\widetilde{\boldsymbol{M}}^{-1}\|_{\mathrm{op}}\right)^k + \mathbb{E}\left(\sqrt{n}\|\boldsymbol{L}\|_{\mathrm{op}}\right)^k + \mathbb{E}t_{\boldsymbol{\Sigma}}^k \le C_k. \tag{25}$$

The model and the normal equations give

$$\boldsymbol{q} = \widetilde{\boldsymbol{M}}^{-1}\boldsymbol{Z}^\top \boldsymbol{A}\boldsymbol{\varepsilon} = \boldsymbol{L}^\top \boldsymbol{\varepsilon}, \qquad \boldsymbol{u} = \boldsymbol{\varepsilon} - \boldsymbol{Z}\boldsymbol{q}, \qquad \boldsymbol{r} = \boldsymbol{A}\boldsymbol{u}. \tag{26}$$

The orthogonal projector onto $\mathrm{col}(\boldsymbol{A}^{1/2}\boldsymbol{Z})$ gives

$$\begin{aligned}\boldsymbol{r} &= (\boldsymbol{A} - \boldsymbol{A}\boldsymbol{Z}\widetilde{\boldsymbol{M}}^{-1}\boldsymbol{Z}^\top\boldsymbol{A})\boldsymbol{\varepsilon},\\ \boldsymbol{0} &\preceq \boldsymbol{A} - \boldsymbol{A}\boldsymbol{Z}\widetilde{\boldsymbol{M}}^{-1}\boldsymbol{Z}^\top\boldsymbol{A} \preceq \boldsymbol{A} \preceq c_+\boldsymbol{I}_n.\end{aligned} \tag{27}$$

Consequently,

$$\|\boldsymbol{q}\| \le \|\boldsymbol{L}\|_{\mathrm{op}}\|\boldsymbol{\varepsilon}\|, \qquad \|\boldsymbol{r}\| \le c_+\|\boldsymbol{\varepsilon}\|.$$

Because $\boldsymbol{L}$ is measurable with respect to $(\boldsymbol{Z},\boldsymbol{S})$ and $\boldsymbol{\varepsilon}$ is independent of $(\boldsymbol{Z},\boldsymbol{S})$, the eighth-order cases of (25) and Lemma B.11 yield

$$\mathbb{E}\|\boldsymbol{q}\|^8 \le C, \qquad \mathbb{E}\|\boldsymbol{r}\|^8 \le Cn^4. \tag{28}$$

Off the Gaussian-null rank-deficient set, differentiation of (26), with $\boldsymbol{K} := \widetilde{\boldsymbol{M}}^{-1}$, gives

$$\frac{\partial \boldsymbol{q}}{\partial z_{ij}} = \boldsymbol{K}\boldsymbol{e}_j r_i - \boldsymbol{L}^\top \boldsymbol{e}_i q_j, \tag{29}$$

and hence

$$\mathcal{E}_{\boldsymbol{q}} \le 2\|\boldsymbol{K}\|_{\mathrm{F}}^2\|\boldsymbol{r}\|^2 + 2\|\boldsymbol{L}\|_{\mathrm{F}}^2\|\boldsymbol{q}\|^2. \tag{30}$$

This is the pointwise conclusion of Lemma B.13, applied with $\boldsymbol{K} = \widetilde{\boldsymbol{M}}^{-1}$ and the present $\boldsymbol{L}$. Since $d \le n$, (25) implies

$$\mathbb{E}\|\boldsymbol{K}\|_{\mathrm{F}}^8 \le d^4\mathbb{E}\|\boldsymbol{K}\|_{\mathrm{op}}^8 \le Cn^{-4}, \qquad \mathbb{E}\|\boldsymbol{L}\|_{\mathrm{F}}^8 \le d^4\mathbb{E}\|\boldsymbol{L}\|_{\mathrm{op}}^8 \le C.$$

The mixed-moment conclusion of Lemma B.13, followed by Cauchy–Schwarz and (28), therefore proves

$$\mathbb{E}\mathcal{E}_{\boldsymbol{q}}^2 \le 8\left(\mathbb{E}\|\boldsymbol{K}\|_{\mathrm{F}}^8\right)^{1/2}\left(\mathbb{E}\|\boldsymbol{r}\|^8\right)^{1/2} + 8\left(\mathbb{E}\|\boldsymbol{L}\|_{\mathrm{F}}^8\right)^{1/2}\left(\mathbb{E}\|\boldsymbol{q}\|^8\right)^{1/2} \le C.$$

Lemma B.11 gives $\mathbb{E}\|\boldsymbol{\varepsilon}\|^4 \le Cn^2$. Thus Cauchy–Schwarz and Lyapunov's inequality give

$$\mathbb{E}\left[\|\boldsymbol{\varepsilon}\|^2\left(\|\boldsymbol{q}\|^2 + \mathcal{E}_{\boldsymbol{q}}\right)\right] \le \left(\mathbb{E}\|\boldsymbol{\varepsilon}\|^4\right)^{1/2}\left\{2\mathbb{E}\|\boldsymbol{q}\|^4 + 2\mathbb{E}\mathcal{E}_{\boldsymbol{q}}^2\right\}^{1/2} \le Cn.$$

Finally, the fourth-order cases of the two envelope lemmas give

$$\mathbb{E}\|\boldsymbol{Z}\|_{\mathrm{op}}^4 \le Cn^2, \qquad \mathbb{E}\|\boldsymbol{L}\|_{\mathrm{op}}^4 \le Cn^{-2}, \qquad \mathbb{E}t_{\boldsymbol{\Sigma}}^4 \le C.$$

Cauchy–Schwarz, together with the fourth moments implied by (28), now proves the last two lines of (24).

We use weak derivatives through Gaussian integration by parts: for each scalar component f and every smooth, compactly supported test function φ, conditional on $(\boldsymbol{A},\boldsymbol{\varepsilon})$,

$$\mathbb{E}_{\boldsymbol{Z}}\left[\frac{\partial f}{\partial z_{ij}}\varphi\right] = \mathbb{E}_{\boldsymbol{Z}}\left[f\left(z_{ij}\varphi - \frac{\partial \varphi}{\partial z_{ij}}\right)\right].$$

The required identity is obtained explicitly by ridge approximation. For $\eta > 0$, put

$$\begin{aligned}\boldsymbol{K}_\eta &= (\widetilde{\boldsymbol{M}} + \eta\boldsymbol{I}_d)^{-1}, & \boldsymbol{q}_\eta &= \boldsymbol{K}_\eta\boldsymbol{Z}^\top\boldsymbol{A}\boldsymbol{\varepsilon},\\ \boldsymbol{L}_\eta &= \boldsymbol{A}\boldsymbol{Z}\boldsymbol{K}_\eta, & \boldsymbol{r}_\eta &= \boldsymbol{A}(\boldsymbol{\varepsilon} - \boldsymbol{Z}\boldsymbol{q}_\eta).\end{aligned}$$

These maps are smooth and satisfy

$$\frac{\partial \boldsymbol{q}_\eta}{\partial z_{ij}} = \boldsymbol{K}_\eta\boldsymbol{e}_j(\boldsymbol{r}_\eta)_i - \boldsymbol{L}_\eta^\top\boldsymbol{e}_i(\boldsymbol{q}_\eta)_j.$$

The order relations $\boldsymbol{K}_\eta \preceq \widetilde{\boldsymbol{M}}^{-1}$ and $\boldsymbol{K}_\eta\widetilde{\boldsymbol{M}}\boldsymbol{K}_\eta \preceq \boldsymbol{K}_\eta$, together with $\boldsymbol{A}^2 \preceq c_+\boldsymbol{A}$, give

$$\|\boldsymbol{K}_\eta\|_{\mathrm{op}} \le \|\widetilde{\boldsymbol{M}}^{-1}\|_{\mathrm{op}}, \qquad \|\boldsymbol{L}_\eta\|_{\mathrm{op}}^2 \le c_+\|\widetilde{\boldsymbol{M}}^{-1}\|_{\mathrm{op}}.$$

Also $\boldsymbol{q}_\eta = \boldsymbol{L}_\eta^\top \boldsymbol{\varepsilon}$ and $\mathbf{0} \preceq \boldsymbol{A} - \boldsymbol{A}\boldsymbol{Z}\boldsymbol{K}_\eta\boldsymbol{Z}^\top\boldsymbol{A} \preceq \boldsymbol{A}$, so $\|\boldsymbol{r}_\eta\| \le c_+\|\boldsymbol{\varepsilon}\|$. In particular, uniformly over $\eta > 0$,

$$\|\boldsymbol{q}_\eta\|^2 \le c_+\|\widetilde{\boldsymbol{M}}^{-1}\|_{\mathrm{op}}\|\boldsymbol{\varepsilon}\|^2, \qquad \mathcal{E}_{\boldsymbol{q}_\eta} \le 4dc_+^2\|\widetilde{\boldsymbol{M}}^{-1}\|_{\mathrm{op}}^2\|\boldsymbol{\varepsilon}\|^2.$$

For each fixed $(\boldsymbol{S}, \boldsymbol{\varepsilon})$, the first two inverse moments make these integrable dominators under the Gaussian design law. On the full-rank event, the fields and their displayed derivatives converge pointwise as $\eta \downarrow 0$. Dominated convergence therefore gives convergence in Gaussian $W^{1,2}$ and passes the integration-by-parts identity to the limit. Thus $\boldsymbol{q}$ has weak derivative (29). □

Clarification (symbol $\boldsymbol{q}_\eta$):

For each $\eta > 0$, $\boldsymbol{q}_\eta = (\widetilde{\boldsymbol{M}} + \eta\boldsymbol{I}_d)^{-1}\boldsymbol{Z}^\top\boldsymbol{A}\boldsymbol{\varepsilon}$ is a smooth field on all of $\mathbb{R}^{n\times d}$ in the design coordinates $\boldsymbol{Z}$, with $(\boldsymbol{A}, \boldsymbol{\varepsilon})$ held fixed.

Clarification (symbol $\boldsymbol{r}_\eta$):

For each $\eta > 0$, $\boldsymbol{r}_\eta = \boldsymbol{A}(\boldsymbol{\varepsilon} - \boldsymbol{Z}\boldsymbol{q}_\eta)$ is a smooth field on all of $\mathbb{R}^{n\times d}$ in the design coordinates $\boldsymbol{Z}$, with $(\boldsymbol{A}, \boldsymbol{\varepsilon})$ held fixed.

The representation $\boldsymbol{q} = \boldsymbol{L}^\top\boldsymbol{\varepsilon}$ and the residual contraction give moments of the coefficient error and weighted residual. Combined with inverse moments, these control the derivative energies and Gaussian fluctuations used in risk calibration.

Lemma B.15. *Under Assumption B.3, the maps $\boldsymbol{q}$ and $\boldsymbol{r}$ are weakly differentiable with respect to $\boldsymbol{Z}$ for all sufficiently large n. For every fixed integer $k \ge 1$, there are $n_0(k) \ge 1$ and $C_k < \infty$, independent of n, such that, for $n \ge n_0(k)$,*

$$\begin{aligned} \mathbb{E}\|\boldsymbol{q}\|^{2k} &\le C_k, & \mathbb{E}\|\boldsymbol{r}\|^{2k} &\le C_k n^k, \\ \mathbb{E}\mathcal{E}_{\boldsymbol{q}}^k &\le C_k, & \mathbb{E}\mathcal{E}_{\boldsymbol{r}}^k &\le C_k n^k. \end{aligned} \tag{31}$$

If also $\|\boldsymbol{\Sigma}^{1/2}\boldsymbol{\beta}^*\| \le B$, then

$$\mathbb{E}\|\boldsymbol{\Sigma}^{1/2}\hat{\boldsymbol{\beta}}\|^{2k} + \mathbb{E}\|\hat{\boldsymbol{\beta}}\|^{2k} \le C_k(B).$$

Moreover, the uniform lower spectral bound in Assumption B.1 implies

$$\mathbb{E}\|\boldsymbol{h}\|^{2k} \le C_k.$$

Clarification (symbol B):

B is a nonnegative real scalar whose meaning is local to its scoped use. In Lemma B.15, Lemma B.20, Lemma B.21: $B < \infty$ is a fixed bound, independent of n, in the optional assumption $\|\boldsymbol{\Sigma}^{1/2}\beta^*\| \le B$; the resulting coefficient moment constants may depend on B. In Theorem 4.2: In the dense Gaussian proof, $B \sim \chi^2_{n-d-1}$ is an auxiliary variable independent of T, J, V, U, where $T \sim \chi^2_{n-d}$, $J \sim \chi^2_d$, $V \sim \chi^2_{m-d+1}$, and $U \sim \chi^2_{m-d}$; these variables give marginal distributional representations, without asserting joint equality of the represented statistics.

Proof of Lemma B.15. The inverse envelope and the exact representations of $(\boldsymbol{q}, \boldsymbol{r})$ are given in (25)–(26). The $\boldsymbol{q}$-derivative bound is (30). Differentiating $\boldsymbol{r} = \boldsymbol{A}(\boldsymbol{\varepsilon} - \boldsymbol{Z}\boldsymbol{q})$ additionally gives

$$\begin{aligned} \frac{\partial \boldsymbol{r}}{\partial z_{ij}} &= -\boldsymbol{A}\boldsymbol{e}_i q_j - \boldsymbol{A}\boldsymbol{Z}\frac{\partial \boldsymbol{q}}{\partial z_{ij}}, \\ \mathcal{E}_{\boldsymbol{r}} &\le 2\|\boldsymbol{A}\|_{\mathrm{F}}^2\|\boldsymbol{q}\|^2 + 2\|\boldsymbol{A}\boldsymbol{Z}\|_{\mathrm{op}}^2\mathcal{E}_{\boldsymbol{q}}. \end{aligned} \tag{32}$$

Equations (26) and (27) give

$$\|\boldsymbol{q}\| \le \|\boldsymbol{L}\|_{\mathrm{op}}\|\boldsymbol{\varepsilon}\|, \qquad \|\boldsymbol{r}\| \le c_+\|\boldsymbol{\varepsilon}\|.$$

Since $\boldsymbol{L}$ is measurable with respect to $(\boldsymbol{Z}, \boldsymbol{S})$ and $\boldsymbol{\varepsilon}$ is independent of $(\boldsymbol{Z}, \boldsymbol{S})$, the inverse envelope (25) and the noise-energy moments in Lemma B.11 imply, for every fixed integer $k \ge 1$,

$$\begin{aligned} \mathbb{E}\|\boldsymbol{q}\|^{2k} &\le \mathbb{E}\|\boldsymbol{L}\|_{\mathrm{op}}^{2k}\,\mathbb{E}\|\boldsymbol{\varepsilon}\|^{2k} \le C_k n^{-k}\, C_k n^k \le C_k, \\ \mathbb{E}\|\boldsymbol{r}\|^{2k} &\le c_+^{2k}\mathbb{E}\|\boldsymbol{\varepsilon}\|^{2k} \le C_k n^k. \end{aligned} \tag{33}$$

Thus the response moments are established before, and independently of, any derivative bound.

Put $\boldsymbol{K} = \widetilde{\boldsymbol{M}}^{-1}$. Since $d \le n$, the inverse envelope gives

$$\begin{aligned}\left(\mathbb{E}\|\boldsymbol{K}\|_{\mathrm{F}}^{4k}\right)^{1/2} &\le d^k \left(\mathbb{E}\|\boldsymbol{K}\|_{\mathrm{op}}^{4k}\right)^{1/2} \le C_k n^{-k},\\ \left(\mathbb{E}\|\boldsymbol{L}\|_{\mathrm{F}}^{4k}\right)^{1/2} &\le d^k \left(\mathbb{E}\|\boldsymbol{L}\|_{\mathrm{op}}^{4k}\right)^{1/2} \le C_k. \end{aligned} \tag{34}$$

Raising (30) to the kth power and applying Cauchy–Schwarz, (33), and (34), we obtain

$$\begin{aligned}\mathbb{E}\mathcal{E}_{\boldsymbol{q}}^k \le{}& C_k \left(\mathbb{E}\|\boldsymbol{K}\|_{\mathrm{F}}^{4k}\right)^{1/2} \left(\mathbb{E}\|\boldsymbol{r}\|^{4k}\right)^{1/2}\\ &+ C_k \left(\mathbb{E}\|\boldsymbol{L}\|_{\mathrm{F}}^{4k}\right)^{1/2} \left(\mathbb{E}\|\boldsymbol{q}\|^{4k}\right)^{1/2} \le C_k. \end{aligned} \tag{35}$$

Because $\mathrm{Tr}(\boldsymbol{A}) = n$ and $\boldsymbol{0} \preceq \boldsymbol{A} \preceq c_+ \boldsymbol{I}_n$,

$$\|\boldsymbol{A}\|_{\mathrm{F}}^2 = \mathrm{Tr}(\boldsymbol{A}^2) \le c_+ n, \qquad \|\boldsymbol{A}\boldsymbol{Z}\|_{\mathrm{op}} \le c_+ \|\boldsymbol{Z}\|_{\mathrm{op}}.$$

Consequently, (32) implies

$$\frac{\mathcal{E}_{\boldsymbol{r}}}{n} \le 2c_+ \|\boldsymbol{q}\|^2 + 2c_+^2 \frac{\|\boldsymbol{Z}\|_{\mathrm{op}}^2}{n} \mathcal{E}_{\boldsymbol{q}}.$$

Another application of Cauchy–Schwarz, using Lemma B.11 and (35) at order $2k$, yields

$$\mathbb{E}\left(\frac{\mathcal{E}_{\boldsymbol{r}}}{n}\right)^k \le C_k \mathbb{E}\|\boldsymbol{q}\|^{2k} + C_k \left\{\mathbb{E}\left(\frac{\|\boldsymbol{Z}\|_{\mathrm{op}}}{\sqrt{n}}\right)^{4k}\right\}^{1/2} \left(\mathbb{E}\mathcal{E}_{\boldsymbol{q}}^{2k}\right)^{1/2} \le C_k. \tag{36}$$

Equations (33), (35), and (36) prove (31).

Finally, $\boldsymbol{\Sigma}^{1/2}\hat{\boldsymbol{\beta}} = \boldsymbol{\Sigma}^{1/2}\boldsymbol{\beta}^* + \boldsymbol{q}$. Hence, when $\|\boldsymbol{\Sigma}^{1/2}\boldsymbol{\beta}^*\| \le B$,

$$\mathbb{E}\|\boldsymbol{\Sigma}^{1/2}\hat{\boldsymbol{\beta}}\|^{2k} \le 2^{2k-1}\left(B^{2k} + \mathbb{E}\|\boldsymbol{q}\|^{2k}\right) \le C_k(B).$$

The covariance lower bound also gives

$$\begin{aligned}\mathbb{E}\|\boldsymbol{h}\|^{2k} &\le c_{\boldsymbol{\Sigma}}^{-k}\mathbb{E}\|\boldsymbol{q}\|^{2k} \le C_k,\\ \mathbb{E}\|\hat{\boldsymbol{\beta}}\|^{2k} &\le c_{\boldsymbol{\Sigma}}^{-k}\mathbb{E}\|\boldsymbol{\Sigma}^{1/2}\hat{\boldsymbol{\beta}}\|^{2k} \le C_k(B),\end{aligned}$$

which proves the asserted Euclidean moment bounds.

The ridge approximation constructed in the proof of Lemma B.14, now combined with the all-order bounds just proved, also converges for $\boldsymbol{r}$ and its derivative. Indeed,

$$\frac{\partial \boldsymbol{r}_\eta}{\partial z_{ij}} = -\boldsymbol{A}\boldsymbol{e}_i(q_\eta)_j - \boldsymbol{A}\boldsymbol{Z}\frac{\partial \boldsymbol{q}_\eta}{\partial z_{ij}}.$$

The same bound as (32), with ridge quantities in place of $(\boldsymbol{q}, \boldsymbol{r})$, has uniformly bounded second moments after division by n: use $\|\boldsymbol{A}\|_{\mathrm{F}}^2 \le c_+ n$, $\|\boldsymbol{A}\boldsymbol{Z}\|_{\mathrm{op}} \le c_+\|\boldsymbol{Z}\|_{\mathrm{op}}$, and the fixed-order moments already established. Pointwise convergence and uniform integrability of the squared derivatives therefore give convergence in Gaussian $W^{1,2}$. This proves the asserted weak differentiability of both $\boldsymbol{q}$ and $\boldsymbol{r}$. □

B.4. Stein bounds and influence controls

With the moment bounds in place, we can control the Stein remainders and influence vectors that enter the risk identity.

B.4.1. Direct Stein and influence bounds

Write $\boldsymbol{z}_i^\top$ for the ith row of $\boldsymbol{Z}$. Since $\boldsymbol{u} = \boldsymbol{\varepsilon} - \boldsymbol{Z}\boldsymbol{q}$, Lemma B.4 expresses the divergence of the coefficient error through the weighted residual:

$$\left(\sum_{j=1}^{d} \frac{\partial q_j}{\partial z_{ij}}\right)_{i=1}^{n} = t_{\boldsymbol{\Sigma}}\boldsymbol{r} - \boldsymbol{\ell}. \tag{37}$$

The first- and second-order Stein fluctuations are

$$\Xi_1 := \boldsymbol{\varepsilon}^\top \boldsymbol{Z}\boldsymbol{q} - \sum_{i=1}^{n}\sum_{j=1}^{d} \frac{\partial(\varepsilon_i q_j)}{\partial z_{ij}},$$

$$\Xi_2 := n\|\boldsymbol{q}\|^2 - \sum_{i=1}^{n}\left(\boldsymbol{z}_i^\top \boldsymbol{q} - \sum_{j=1}^{d} \frac{\partial q_j}{\partial z_{ij}}\right)^2.$$

Clarification (symbol Ξ_2):

Ξ_2 is the displayed scalar functional of the field $\boldsymbol{q}$ and design $\boldsymbol{Z}$, with derivatives taken in the design coordinates while $\boldsymbol{S}$ and the structural noise $\boldsymbol{\varepsilon}$ are held fixed. In Lemma B.16: The definition is applied to a general field $\boldsymbol{q}(\boldsymbol{Z}, \boldsymbol{S}, \boldsymbol{\varepsilon})$ that is C^∞ in $\boldsymbol{Z}$ conditional on $(\boldsymbol{S}, \boldsymbol{\varepsilon})$.

Clarification (symbol $\boldsymbol{z}_i$):

For $i \in \{1, \dots, n\}$, $\boldsymbol{z}_i = (z_{i1}, \dots, z_{id})^\top \in \mathbb{R}^d$ is the column vector whose transpose is row i of $\boldsymbol{Z}$. In Lemma B.16: In the rowwise proof, $\mathbb{E}_i$ integrates $\boldsymbol{z}_i$ conditional on $(\boldsymbol{Z}_{-i}, \boldsymbol{S}, \boldsymbol{\varepsilon})$; consequently $\boldsymbol{a}_i = \mathbb{E}_i[\boldsymbol{q}]$ does not depend on the integrated row.

The preceding moment bounds verify the hypotheses of the next two lemmas for balanced sketches.

Lemma B.16. *Suppose that $\boldsymbol{Z}$ has i.i.d. standard Gaussian entries and is independent of $(\boldsymbol{S}, \boldsymbol{\varepsilon})$. Conditional on $(\boldsymbol{S}, \boldsymbol{\varepsilon})$, let $\boldsymbol{q} = \boldsymbol{q}(\boldsymbol{Z}, \boldsymbol{S}, \boldsymbol{\varepsilon})$ be C^∞ in $\boldsymbol{Z}$. If, for a constant $C_1 < \infty$ independent of n,*

$$\mathbb{E}\left[\|\boldsymbol{\varepsilon}\|^2\left(\|\boldsymbol{q}\|^2 + \mathcal{E}_{\boldsymbol{q}}\right)\right] \le C_1 n,$$

then

$$\mathbb{E}\Xi_1^2 \le C_1 n.$$

Independently, if for constants $C_q, C_D < \infty$ independent of n,

$$\mathbb{E}\|\boldsymbol{q}\|^4 \le C_q, \qquad \mathbb{E}\mathcal{E}_{\boldsymbol{q}}^2 \le C_D,$$

then $\mathbb{E}|\Xi_2| \le C(C_q, C_D)\sqrt{n}$.

Clarification:

The expectation in $\mathbb{E}[\|\boldsymbol{\varepsilon}\|^2(\|\boldsymbol{q}\|^2 + \mathcal{E}_{\boldsymbol{q}})] \le C_1 n$ is unconditional over the joint law of $(\boldsymbol{Z}, \boldsymbol{S}, \boldsymbol{\varepsilon})$. (the reading of one of its hypotheses)

Proof. Condition on $(\boldsymbol{S}, \boldsymbol{\varepsilon})$ and vectorize $\boldsymbol{Z}$. The first-order Gaussian divergence inequality applied to the vector field $F_{ij} = \varepsilon_i q_j$ gives

$$\mathbb{E}\Xi_1^2 \le \mathbb{E}\left[\|\boldsymbol{\varepsilon}\|^2\left(\|\boldsymbol{q}\|^2 + \mathcal{E}_{\boldsymbol{q}}\right)\right].$$

The first displayed assumption proves the claim for Ξ_1.

For the second claim, let $\mathbb{E}_i$ denote expectation over the row $\boldsymbol{z}_i$, conditional on $(\boldsymbol{Z}_{-i}, \boldsymbol{S}, \boldsymbol{\varepsilon})$, and set

$$\boldsymbol{a}_i := \mathbb{E}_i[\boldsymbol{q}], \qquad \eta_i := \boldsymbol{z}_i^\top(\boldsymbol{q} - \boldsymbol{a}_i) - \sum_{j=1}^{d} \frac{\partial q_j}{\partial z_{ij}},$$

$$\chi_i := (\boldsymbol{z}_i^\top \boldsymbol{a}_i)^2 - \|\boldsymbol{a}_i\|^2.$$

The vector $\boldsymbol{a}_i$ depends only on the other design rows and is therefore independent of $\boldsymbol{z}_i$ under $\mathbb{E}_i$. Gaussian integration by parts gives $\mathbb{E}_i\eta_i = 0$, while the Gaussian second moment gives $\mathbb{E}_i\chi_i = 0$. The terms below isolate the effect of row i from the dependence of $\boldsymbol{q}$ on that row. Writing $Q = \|\boldsymbol{q}\|^2$, direct expansion gives

$$\sum_{i=1}^{n}\left(\boldsymbol{z}_i^\top\boldsymbol{q} - \sum_{j=1}^{d}\frac{\partial q_j}{\partial z_{ij}}\right)^2 - nQ$$
$$= \sum_i \eta_i^2 + 2\sum_i \eta_i \boldsymbol{z}_i^\top\boldsymbol{a}_i + \sum_i \chi_i + \sum_i(\|\boldsymbol{a}_i\|^2 - Q).$$

The Gaussian Poincaré inequality with constant at most 2, together with the first-order divergence inequality, applied conditionally row by row, gives

$$\mathbb{E}\sum_i \eta_i^2 \le 3\mathbb{E}\mathcal{E}_{\boldsymbol{q}}, \qquad \mathbb{E}\left|2\sum_i \eta_i \boldsymbol{z}_i^\top\boldsymbol{a}_i\right| \le 2\sqrt{3n\mathbb{E}Q\,\mathbb{E}\mathcal{E}_{\boldsymbol{q}}},$$
$$\mathbb{E}\left|\sum_i(\|\boldsymbol{a}_i\|^2 - Q)\right| \le 2\sqrt{2n\mathbb{E}Q\,\mathbb{E}\mathcal{E}_{\boldsymbol{q}}}.$$

To verify the derivative term, let $D_k\boldsymbol{a}_i$ be the Jacobian of $\boldsymbol{a}_i$ with respect to the row $\boldsymbol{z}_k$ and put $b_{ik} = \mathbb{E}[\|D_k\boldsymbol{a}_i\|_{\mathrm{F}}^2\|\boldsymbol{a}_i\|^2]$. The rowwise Gaussian covariance inequality for centered quadratic forms gives

$$|\mathbb{E}[\chi_i\chi_k]| \le 4\sqrt{b_{ik}b_{ki}}, \qquad i \ne k.$$

For this step, condition first on $(\boldsymbol{Z}_{-\{i,k\}}, \boldsymbol{S}, \boldsymbol{\varepsilon})$ so that only the two Gaussian rows remain random. The displayed covariance inequality is Lemma 7.1, the covariance step in the second-order Gaussian inequalities of Bellec [10]. Differentiation under conditional expectation gives $D_i\boldsymbol{a}_i = 0$ and $D_k\boldsymbol{a}_i = \mathbb{E}_i[D_k\boldsymbol{q}]$ for $k \ne i$. Jensen's inequality gives $\|\boldsymbol{a}_i\|^2 \le \mathbb{E}_i Q$ and $\sum_k\|D_k\boldsymbol{a}_i\|_{\mathrm{F}}^2 \le \mathbb{E}_i\mathcal{E}_{\boldsymbol{q}}$. Apply Cauchy–Schwarz after summing over i, k. Also $\mathbb{E}_i\chi_i^2 = 2\|\boldsymbol{a}_i\|^4$ by the Gaussian fourth moment. These observations give

$$\sum_{i,k} b_{ik} \le n\{\mathbb{E}Q^2\,\mathbb{E}\mathcal{E}_{\boldsymbol{q}}^2\}^{1/2}, \qquad \sum_i \mathbb{E}\chi_i^2 \le 2n\mathbb{E}Q^2.$$

Consequently,

$$\mathbb{E}\left|\sum_i \chi_i\right| \le \left\{2n\mathbb{E}Q^2 + 4n(\mathbb{E}Q^2\,\mathbb{E}\mathcal{E}_{\boldsymbol{q}}^2)^{1/2}\right\}^{1/2}.$$

Combining the preceding displays gives

$$\mathbb{E}|\Xi_2| \le 3\mathbb{E}\mathcal{E}_{\boldsymbol{q}} + (2\sqrt{3} + 2\sqrt{2})\sqrt{n\mathbb{E}Q\,\mathbb{E}\mathcal{E}_{\boldsymbol{q}}}$$
$$+ \left\{2n\mathbb{E}Q^2 + 4n(\mathbb{E}Q^2\,\mathbb{E}\mathcal{E}_{\boldsymbol{q}}^2)^{1/2}\right\}^{1/2}.$$

Cauchy–Schwarz and the two moment assumptions now give the asserted $O(\sqrt{n})$ rate. The calculations can first be made with smooth compact cutoffs; dominated convergence for the moment bounds and Fatou's lemma remove the cutoffs. □

Clarification (symbol a):

$\boldsymbol{a}$ is locally bound vector notation, with its dimension and dependence determined by the surrounding definition or inequality. In Lemma B.13, Lemma B.17: $\boldsymbol{a}$ and $\boldsymbol{b}$ are arbitrary vectors in the same Euclidean space in $\|\boldsymbol{a} - \boldsymbol{b}\|^2 \le 2\|\boldsymbol{a}\|^2 + 2\|\boldsymbol{b}\|^2$. In Lemma B.19, a result cited in the proof of Lemma B.19: In the matrix Gaussian divergence inequality, $\boldsymbol{a} : \mathbb{R}^{n\times d} \to \mathbb{R}^d$ is a smooth field, interpreted conditionally on $\mathcal{F}$ with derivatives taken in $\boldsymbol{Z}$. In Lemma B.19: In the later elementary comparison beginning with $c \ge 1$ and $\|\boldsymbol{u}\| \le 1$, $\boldsymbol{a}$ and $\boldsymbol{b}$ are arbitrary vectors of the same dimension as $\boldsymbol{u}$. In Theorem 4.2: In the dense Gaussian

proof, $\boldsymbol{a} = \boldsymbol{U}^\top \boldsymbol{\varepsilon} \in \mathbb{R}^d$, where $\boldsymbol{U} = \boldsymbol{Z}(\boldsymbol{Z}^\top \boldsymbol{Z})^{-1/2}$. Conditional on $\boldsymbol{Z}$, $\boldsymbol{a}$ is centered Gaussian with covariance $\sigma^2 \boldsymbol{I}_d$ and is independent of $\boldsymbol{G}, \boldsymbol{H}, \boldsymbol{c}$; the joint conditional law of these blocks does not depend on $\boldsymbol{Z}$.

Clarification (symbol $\boldsymbol{a}_i$):

For $i \in \{1, \dots, n\}$, $\boldsymbol{a}_i = \mathbb{E}[\boldsymbol{q} \mid \boldsymbol{Z}_{-i}, \boldsymbol{S}, \varepsilon]$ is measurable with respect to $(\boldsymbol{Z}_{-i}, \boldsymbol{S}, \varepsilon)$ and is held fixed when integrating over $\boldsymbol{z}_i$ under $\mathbb{E}_i$.

Clarification (symbol η_i):

For $i \in \{1, \dots, n\}$, $\eta_i = \boldsymbol{z}_i^\top (\boldsymbol{q} - \boldsymbol{a}_i) - \sum_{j=1}^d \frac{\partial q_j}{\partial z_{ij}}$, where $\boldsymbol{a}_i = \mathbb{E}_i[\boldsymbol{q}]$ and $\mathbb{E}_i$ integrates row $\boldsymbol{z}_i$ conditional on $(\boldsymbol{Z}_{-i}, \boldsymbol{S}, \varepsilon)$, so $\mathbb{E}_i \eta_i = 0$.

Clarification (symbol χ_i):

For $i \in \{1, \dots, n\}$, $\chi_i = (\boldsymbol{z}_i^\top \boldsymbol{a}_i)^2 - \|\boldsymbol{a}_i\|^2$, where $\boldsymbol{a}_i = \mathbb{E}_i[\boldsymbol{q}]$ and $\mathbb{E}_i$ integrates row $\boldsymbol{z}_i$ conditional on $(\boldsymbol{Z}_{-i}, \boldsymbol{S}, \varepsilon)$, so $\mathbb{E}_i \chi_i = 0$.

Clarification (symbol $D_k \boldsymbol{a}_i$):

For $i, k \in \{1, \dots, n\}$, $D_k \boldsymbol{a}_i$ differentiates $\boldsymbol{a}_i$ with respect to $\boldsymbol{z}_k$, holding the other design rows, $\boldsymbol{S}$ and ε fixed, with $D_i \boldsymbol{a}_i = 0$ and $D_k \boldsymbol{a}_i = \mathbb{E}_i[D_k \boldsymbol{q}]$ for $k \neq i$.

Clarification (symbol b_{ik}):

For $i, k \in \{1, \dots, n\}$, $b_{ik} = \mathbb{E}[\|D_k \boldsymbol{a}_i\|_{\mathrm{F}}^2 \|\boldsymbol{a}_i\|^2]$ is an unconditional expectation under the original law, rather than an expectation conditional on $(\boldsymbol{Z}_{-\{i,k\}}, \boldsymbol{S}, \varepsilon)$; in particular, $b_{ii} = 0$.

For the inverse-defined coefficient error used here, apply these inequalities to the smooth ridge fits and let the ridge parameter tend to zero. The uniform moment bounds and almost-sure convergence of the fits and their first derivatives justify Fatou's lemma. Weak differentiability of the limit is established separately in Lemma B.14.

Lemma B.17. *For the influence vectors $\boldsymbol{\ell}$ and $\boldsymbol{v}$ in the standing notation, the direct bounds are*

$$\mathbb{E}\|\boldsymbol{\ell}\|^2 \le \mathbb{E}\left[\|\boldsymbol{L}\|_{\mathrm{op}}^2 \|\boldsymbol{q}\|^2\right],$$
$$\mathbb{E}\|\boldsymbol{v}\|^2 \le 2\mathbb{E}\left[\|\boldsymbol{Z}\|_{\mathrm{op}}^2 \|\boldsymbol{q}\|^2\right] + 2\mathbb{E}\left[t_{\boldsymbol{\Sigma}}^2 \|\boldsymbol{r}\|^2\right].$$

If $\boldsymbol{Z}^\top \boldsymbol{r} = 0$, the second inequality sharpens to

$$\mathbb{E}\|\boldsymbol{v}\|^2 = \mathbb{E}\|\boldsymbol{Z}\boldsymbol{q}\|^2 + \mathbb{E}[t_{\boldsymbol{\Sigma}}^2 \|\boldsymbol{r}\|^2].$$

Thus each influence vector inherits the rate of its corresponding mixed moment. In particular, $O(1)$ and $O(n)$ mixed-moment bounds give the same respective rates for $(\boldsymbol{\ell}, \boldsymbol{v})$; balanced sketches satisfy the stronger rate $\mathbb{E}\|\boldsymbol{\ell}\|^2 = O(n^{-1})$.

Proof. The first two inequalities follow from $\|\boldsymbol{L}\boldsymbol{q}\| \le \|\boldsymbol{L}\|_{\mathrm{op}} \|\boldsymbol{q}\|$ and $\|\boldsymbol{a} - \boldsymbol{b}\|^2 \le 2\|\boldsymbol{a}\|^2 + 2\|\boldsymbol{b}\|^2$. If $\boldsymbol{Z}^\top \boldsymbol{r} = 0$, the cross term is zero. Cauchy–Schwarz gives the claims under the corresponding mixed-moment conditions. □

B.4.2. Balanced-sketch risk controls

Proposition B.18. *Under Assumption B.3, for all sufficiently large n, the map $\boldsymbol{q} = \boldsymbol{q}(\boldsymbol{Z}, \boldsymbol{S}, \varepsilon)$ is weakly differentiable with respect to $\boldsymbol{Z}$. There are constants $C_\Xi, C_\ell, C_v < \infty$, independent of n, such that, for all sufficiently large n,*

$$\mathbb{E}[\Xi_1^2] \le C_\Xi n, \qquad \mathbb{E}|\Xi_2| \le C_\Xi \sqrt{n}, \tag{38}$$
$$\mathbb{E}\|\boldsymbol{\ell}\|^2 \le \frac{C_\ell}{n}, \qquad \mathbb{E}\|\boldsymbol{v}\|^2 \le C_v n. \tag{39}$$

Proof. Conditional on the full-row-rank sketch, the columns of $\boldsymbol{S}\boldsymbol{Z}$ are independent Gaussian vectors with positive-definite covariance $\boldsymbol{S}\boldsymbol{S}^\top$. Since $d < m$ and $\boldsymbol{\Sigma}$ is invertible, $\mathrm{rank}(\boldsymbol{S}\boldsymbol{X}) = d$ almost

surely, so all displayed quantities are well defined. Lemma B.14 gives weak differentiability and all direct moment bounds required below. Apply Lemma B.16 first to the smooth positive-ridge coefficient fields. The uniform inverse and derivative moment bounds from Lemma B.14 hold along this approximation. The fields and their coordinate derivatives converge almost surely as the ridge parameter decreases to zero, so Fatou's lemma gives (38). Lemma B.17 gives the bound for $\boldsymbol{v}$, while the direct estimate

$$\mathbb{E}\|\boldsymbol{\ell}\|^2 \le \mathbb{E}[\|\boldsymbol{L}\|_{\mathrm{op}}^2\|\boldsymbol{q}\|^2] \le Cn^{-1}$$

gives the stronger balanced-sketch rate in (39). □

B.5. Trace identities for residual sketch mass

The observable correction replaces $t_{\boldsymbol{\Sigma}}$ by d/b. To control this substitution, we express $d-bt_{\boldsymbol{\Sigma}}$ through five quantities that admit Stein bounds.

B.5.1. Trace identity and response weighting

Define the five trace-calibration quantities

$$\begin{aligned}
Q_{1,\boldsymbol{\Sigma}} &:= \boldsymbol{r}^\top \boldsymbol{X}\boldsymbol{h} + b\|\boldsymbol{\Sigma}^{1/2}\boldsymbol{h}\|^2 - t_{\boldsymbol{\Sigma}}\|\boldsymbol{r}\|^2,\\
Q_{2,\boldsymbol{\Sigma}} &:= \boldsymbol{h}^\top \boldsymbol{X}^\top \boldsymbol{X}\boldsymbol{h} - t_{\boldsymbol{\Sigma}}\boldsymbol{r}^\top \boldsymbol{X}\boldsymbol{h} - (n-d)\|\boldsymbol{\Sigma}^{1/2}\boldsymbol{h}\|^2,\\
Q_{3,\boldsymbol{\Sigma}} &:= \frac{1}{n}\boldsymbol{r}^\top \boldsymbol{X}\boldsymbol{\Sigma}^{-1}\boldsymbol{X}^\top \boldsymbol{r} + \frac{b}{n}\boldsymbol{h}^\top \boldsymbol{X}^\top \boldsymbol{r},\\
Q_{4,\boldsymbol{\Sigma}} &:= \frac{d}{n}\|\boldsymbol{r}\|^2 - \frac{1}{n}\|\boldsymbol{\Sigma}^{-1/2}\boldsymbol{X}^\top \boldsymbol{r} + b\boldsymbol{\Sigma}^{1/2}\boldsymbol{h}\|^2,\\
Q_{5,\boldsymbol{\Sigma}} &:= n\|\boldsymbol{\Sigma}^{1/2}\boldsymbol{h}\|^2 - \|\boldsymbol{X}\boldsymbol{h} - t_{\boldsymbol{\Sigma}}\boldsymbol{r}\|^2.
\end{aligned}$$

A direct whitening and expansion give

$$\left(\frac{b}{n} - t_{\boldsymbol{\Sigma}}\right) Q_{1,\boldsymbol{\Sigma}} + Q_{2,\boldsymbol{\Sigma}} + Q_{3,\boldsymbol{\Sigma}} + Q_{4,\boldsymbol{\Sigma}} + Q_{5,\boldsymbol{\Sigma}} = (d - bt_{\boldsymbol{\Sigma}})\left(\|\boldsymbol{\Sigma}^{1/2}\boldsymbol{h}\|^2 + \frac{\|\boldsymbol{r}\|^2}{n}\right). \tag{40}$$

Since $\boldsymbol{Z}\boldsymbol{q} = \boldsymbol{X}\boldsymbol{h}$ and $\boldsymbol{Z}^\top \boldsymbol{r} = \boldsymbol{\Sigma}^{-1/2}\boldsymbol{X}^\top \boldsymbol{r}$, the trace-calibration quantities $Q_{k,\boldsymbol{\Sigma}}$ have the same values in both coordinate systems. Thus (40) becomes

$$\left(\frac{b}{n} - t_{\boldsymbol{\Sigma}}\right) Q_{1,\boldsymbol{\Sigma}} + \sum_{k=2}^{5} Q_{k,\boldsymbol{\Sigma}} = (d - bt_{\boldsymbol{\Sigma}})G_{\boldsymbol{\Sigma}}. \tag{41}$$

The normal equations give $\boldsymbol{Z}^\top \boldsymbol{r} = \boldsymbol{0}$; after whitening, $Q_{3,\boldsymbol{\Sigma}} = \{\|\boldsymbol{Z}^\top \boldsymbol{r}\|^2 + b\boldsymbol{q}^\top \boldsymbol{Z}^\top \boldsymbol{r}\}/n = 0$.

Proof of the trace identity (40). Using $\boldsymbol{Z}\boldsymbol{q} = \boldsymbol{X}\boldsymbol{h}$, $\boldsymbol{Z}^\top \boldsymbol{r} = \boldsymbol{\Sigma}^{-1/2}\boldsymbol{X}^\top \boldsymbol{r}$, and $\|\boldsymbol{q}\|^2 = \|\boldsymbol{\Sigma}^{1/2}\boldsymbol{h}\|^2$, direct expansion gives

$$\sum_{k=2}^{5} Q_{k,\boldsymbol{\Sigma}} = \left(t_{\boldsymbol{\Sigma}} - \frac{b}{n}\right)\boldsymbol{r}^\top \boldsymbol{Z}\boldsymbol{q} + \left(d - \frac{b^2}{n}\right)\|\boldsymbol{q}\|^2 + \left(\frac{d}{n} - t_{\boldsymbol{\Sigma}}^2\right)\|\boldsymbol{r}\|^2.$$

Adding $(b/n - t_{\boldsymbol{\Sigma}})Q_{1,\boldsymbol{\Sigma}}$ cancels the mixed term and yields (41), which is (40) in whitened coordinates. □

For trace calibration, introduce an auxiliary probability law under which $\boldsymbol{\beta}^* = \boldsymbol{0}$ and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_n)$ independently of $(\boldsymbol{X}, \boldsymbol{S})$, while leaving the joint law of $(\boldsymbol{X}, \boldsymbol{S})$ unchanged. Write $\mathbb{E}_{\mathrm{aux}}$ for expectation under this law. Recompute all response-dependent quantities $(\hat{\boldsymbol{\beta}}, \boldsymbol{h}, \boldsymbol{u}, \boldsymbol{r}, \boldsymbol{q}, Q_{1,\boldsymbol{\Sigma}}, \ldots, Q_{5,\boldsymbol{\Sigma}})$ under the auxiliary law, whereas b and $t_{\boldsymbol{\Sigma}}$ are unchanged. Set

$$\begin{aligned}
B_{\boldsymbol{A}} &:= \frac{2|\boldsymbol{u}^\top \boldsymbol{r}|}{n} + \left(|t_{\boldsymbol{\Sigma}}| + \frac{d}{b}\right)\frac{\|\boldsymbol{r}\|^2}{n},\\
\mathcal{T}_{\boldsymbol{A}} &:= \frac{n}{b}\left\{\left|\frac{b}{n} - t_{\boldsymbol{\Sigma}}\right|\frac{|Q_{1,\boldsymbol{\Sigma}}|}{G_{\boldsymbol{\Sigma}}} + \sum_{k=2}^{5}\frac{|Q_{k,\boldsymbol{\Sigma}}|}{G_{\boldsymbol{\Sigma}}}\right\}.
\end{aligned}$$

Under the auxiliary law $G_{\boldsymbol{\Sigma}} > 0$ almost surely: conditional on the rank-m sketch, a nondegenerate Gaussian $\boldsymbol{\varepsilon}$ lies in $\ker(\boldsymbol{A})$ with probability zero. Under the original law, every ratio in $\mathcal{T}_{\boldsymbol{A}}$ is defined to be zero on $\{G_{\boldsymbol{\Sigma}} = 0\}$; on that event $\boldsymbol{q} = \boldsymbol{r} = \mathbf{0}$ and hence $B_{\boldsymbol{A}} = 0$ as well.

Clarification (symbol $\mathcal{T}_A$):

$\mathcal{T}_{\boldsymbol{A}} := \frac{n}{b}\left\{\left|\frac{b}{n} - t_{\boldsymbol{\Sigma}}\right| \frac{|Q_{1,\boldsymbol{\Sigma}}|}{G_{\boldsymbol{\Sigma}}} + \sum_{k=2}^{5} \frac{|Q_{k,\boldsymbol{\Sigma}}|}{G_{\boldsymbol{\Sigma}}}\right\}$ is evaluated using the response quantities of the law under consideration, with every displayed ratio defined to be zero on $\{G_{\boldsymbol{\Sigma}} = 0\}$.

B.5.2. Normalized Gaussian–Stein inequalities

Because trace calibration divides by response energy, we need normalized inequalities. We condition on the sketch and noise and integrate over the Gaussian design.

For maps $\boldsymbol{f} : \mathbb{R}^{n\times d} \to \mathbb{R}^d$ and $\boldsymbol{\psi} : \mathbb{R}^{n\times d} \to \mathbb{R}^n$, write

$$G_{\boldsymbol{f},\boldsymbol{\psi}} := \|\boldsymbol{f}\|^2 + \frac{\|\boldsymbol{\psi}\|^2}{n}, \qquad \mathcal{E}_{\boldsymbol{f},\boldsymbol{\psi}} := \mathcal{E}_{\boldsymbol{f}} + \frac{1}{n}\mathcal{E}_{\boldsymbol{\psi}},$$

and define the five Gaussian contractions below. The first three are integration-by-parts errors controlled in L^2; the last two compare Gaussian quadratic forms and are controlled in L^1:

$$\Gamma_1 := \boldsymbol{\psi}^\top \boldsymbol{Z}\boldsymbol{f} - \sum_{i=1}^{n}\sum_{j=1}^{d} \frac{\partial(\psi_i f_j)}{\partial z_{ij}},$$

$$\Gamma_2 := \|\boldsymbol{Z}\boldsymbol{f}\|^2 - \sum_{i=1}^{n}\sum_{j=1}^{d} \frac{\partial\{f_j \boldsymbol{e}_i^\top \boldsymbol{Z}\boldsymbol{f}\}}{\partial z_{ij}},$$

$$\Gamma_3 := \frac{1}{n}\|\boldsymbol{Z}^\top\boldsymbol{\psi}\|^2 - \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{d} \frac{\partial\{\psi_i \boldsymbol{e}_j^\top \boldsymbol{Z}^\top\boldsymbol{\psi}\}}{\partial z_{ij}},$$

$$\Gamma_4 := \frac{d}{n}\|\boldsymbol{\psi}\|^2 - \frac{1}{n}\sum_{j=1}^{d}\left(\boldsymbol{\psi}^\top \boldsymbol{Z}\boldsymbol{e}_j - \sum_{i=1}^{n}\frac{\partial\psi_i}{\partial z_{ij}}\right)^2,$$

$$\Gamma_5 := n\|\boldsymbol{f}\|^2 - \sum_{i=1}^{n}\left(\boldsymbol{f}^\top \boldsymbol{Z}^\top \boldsymbol{e}_i - \sum_{j=1}^{d}\frac{\partial f_j}{\partial z_{ij}}\right)^2.$$

Thus $\mathcal{E}_{\boldsymbol{f},\boldsymbol{\psi}}$ is the raw derivative energy for the normalized stacked map $(\boldsymbol{f}, \boldsymbol{\psi}/\sqrt{n})$. For a sigma-field $\mathcal{F}$, also write

$$\Theta_{\mathcal{F}} := \mathbb{E}\left[\frac{\mathcal{E}_{\boldsymbol{f},\boldsymbol{\psi}}}{G_{\boldsymbol{f},\boldsymbol{\psi}}} \,\middle|\, \mathcal{F}\right],$$

where the ratio is interpreted as zero when $G_{\boldsymbol{f},\boldsymbol{\psi}} = 0$.

Clarification (symbol ψ):

$\boldsymbol{\psi} : \mathbb{R}^{n\times d} \to \mathbb{R}^n$ is the second design-coordinate field in the normalized energies and Gaussian contractions. In Lemma B.19: Conditional on $\mathcal{F}$, $\boldsymbol{\psi} = \boldsymbol{\psi}(\boldsymbol{Z})$ is C^∞ and may depend on the fixed value of $\mathcal{F}$. In Theorem 4.1, Lemma B.21, Lemma B.22, Proposition B.23: In the specialization defining ζ_s, $\boldsymbol{\psi} = \boldsymbol{r}$; the smooth approximation uses $\boldsymbol{r}_\eta$ before passing to the weak-derivative limit.

Clarification (symbol $\mathcal{F}$):

$\mathcal{F}$ is the conditioning sigma-field independent of $\boldsymbol{Z}$, and expectations conditional on it integrate the Gaussian design while holding the conditioning information fixed. In Lemma B.21: $\mathcal{F} = \sigma(\boldsymbol{A}, \boldsymbol{\varepsilon})$.

Clarification (symbol $G_{f,\psi}$):

$G_{\boldsymbol{f},\boldsymbol{\psi}}$ is the pointwise squared norm of the stacked field $(\boldsymbol{f}, \boldsymbol{\psi}/\sqrt{n})$, and ratios with this energy as denominator are assigned zero on $\{G_{\boldsymbol{f},\boldsymbol{\psi}} = 0\}$. In Lemma B.21: The fields are $(\boldsymbol{q}_\eta, \boldsymbol{r}_\eta)$ during ridge approximation and $(\boldsymbol{q}, \boldsymbol{r})$ in the limit; $G_{\boldsymbol{q},\boldsymbol{r}} = G_{\boldsymbol{\Sigma}}$.

Clarification (symbol $\mathcal{E}_{f,\psi}$):

$\mathcal{E}_{f,\psi}$ is the pointwise raw derivative energy of $(f, \psi/\sqrt{n})$ with respect to the design coordinates z_{ij}, holding any conditioning information fixed. In Lemma B.21: The fields are (q_η, r_η) during ridge approximation and (q, r) in the limit, and differentiation holds (A, ε) fixed.

Clarification (symbol $\Theta_{\mathcal{F}}$):

$\Theta_{\mathcal{F}}$ is the $\mathcal{F}$-measurable conditional expectation of the normalized derivative energy, with the zero-denominator convention in its definition. In Lemma B.21: Here $\mathcal{F} = \sigma(A, \varepsilon)$; the fields are (q_η, r_η) for $\eta > 0$ during the ridge approximation and (q, r) in the limit, where the conditional energy is written $\Theta_{A,\varepsilon} = \mathbb{E}_Z[\mathcal{E}_{q,r}/G_\Sigma]$.

Lemma B.19. *Let $\mathcal{F}$ be a sigma-field independent of $Z \in \mathbb{R}^{n\times d}$, whose entries are i.i.d. standard Gaussian. Conditional on $\mathcal{F}$, let $f = f(Z) \in \mathbb{R}^d$ and $\psi = \psi(Z) \in \mathbb{R}^n$ be C^∞. All ratios below are defined to be zero on $\{G_{f,\psi} = 0\}$. Assume, almost surely,*

$$\mathbb{E}\left[\left(1 + \frac{\|Z\|_{\mathrm{op}}^2}{n}\right)\frac{\mathcal{E}_{f,\psi}}{G_{f,\psi}} \;\middle|\; \mathcal{F}\right] < \infty.$$

Then, for an absolute constant $C < \infty$,

$$\mathbb{E}\left[\frac{\Gamma_1^2 + \Gamma_2^2 + \Gamma_3^2}{G_{f,\psi}^2} \;\middle|\; \mathcal{F}\right] \le C\mathbb{E}\left[n + d + \|Z\|_{\mathrm{op}}^2 + (n+d)\left(1 + \frac{\|Z\|_{\mathrm{op}}^2}{n}\right)\frac{\mathcal{E}_{f,\psi}}{G_{f,\psi}} \;\middle|\; \mathcal{F}\right]. \tag{42}$$

Moreover,

$$\mathbb{E}\left[\frac{|\Gamma_4| + |\Gamma_5|}{G_{f,\psi}} \;\middle|\; \mathcal{F}\right] \le C\left\{\sqrt{n+d}\left(1 + \Theta_{\mathcal{F}}^{1/2}\right) + \Theta_{\mathcal{F}}\right\}. \tag{43}$$

Proof. Work conditionally on $\mathcal{F}$ wherever the conditional expectations are finite. For a smooth scalar function, a zero at which a specified coordinate derivative is nonzero is isolated along that coordinate line. Such zeros are countable on each line. Since Gaussian coordinates have no atoms, Fubini's theorem implies that each coordinate derivative vanishes almost everywhere on the function's zero set. Apply this fact to the finitely many components of both fields. Their first derivatives therefore vanish almost everywhere on $\{G_{f,\psi} = 0\}$, and so do all five contractions.

For $\delta > 0$, write

$$a_\delta := \left(G_{f,\psi} + \delta^2\right)^{1/2}, \qquad \eta_\delta := \frac{f}{a_\delta}, \qquad \upsilon_\delta := \frac{\psi}{\sqrt{n}\, a_\delta}, \qquad \omega := (\eta_\delta^\top, \upsilon_\delta^\top)^\top.$$

The positive δ makes these normalized fields smooth on the whole matrix space. Differentiation of their stacked vector gives

$$\sum_{i=1}^{n}\sum_{j=1}^{d}\left\{\|\frac{\partial \eta_\delta}{\partial z_{ij}}\|^2 + \|\frac{\partial \upsilon_\delta}{\partial z_{ij}}\|^2\right\} \le \frac{\mathcal{E}_{f,\psi}}{G_{f,\psi} + \delta^2}.$$

Indeed, the derivative of a normalized vector is its original derivative multiplied by the symmetric contraction $a_\delta^{-1}(I_{d+n} - \omega\omega^\top)$.

We use the first-order matrix Gaussian divergence inequality appearing in the proof of Proposition 14 of Bellec and Shen [14]: for smooth fields $a : \mathbb{R}^{n\times d} \to \mathbb{R}^d$ and $b : \mathbb{R}^{n\times d} \to \mathbb{R}^n$,

$$\mathbb{E}\left[\left\{b^\top Z a - \sum_{i,j}\frac{\partial(b_i a_j)}{\partial z_{ij}}\right\}^2 \;\middle|\; \mathcal{F}\right] \le \mathbb{E}\left[\|b\|^2\|a\|^2 + 2\sum_{i,j}\left\{\|a\|^2\|\frac{\partial b}{\partial z_{ij}}\|^2 + \|b\|^2\|\frac{\partial a}{\partial z_{ij}}\|^2\right\} \;\middle|\; \mathcal{F}\right]. \tag{44}$$

This inequality follows directly from Gaussian integration by parts. We derive the required scale-invariant form from (44), rather than relying on the displayed normalization in Proposition 14. Apply it successively to the normalized pairs

$$(\boldsymbol{a}, \boldsymbol{b}) = (\boldsymbol{\eta}_\delta, \boldsymbol{v}_\delta), \qquad (\boldsymbol{\eta}_\delta, \boldsymbol{Z}\boldsymbol{\eta}_\delta), \qquad (\boldsymbol{Z}^\top \boldsymbol{v}_\delta, \boldsymbol{v}_\delta).$$

The first divergence is multiplied by $\sqrt{n}$, so its second-moment bound is multiplied by n. The product rule separates the three resulting divergences as follows:

$$\sqrt{n}\left\{\boldsymbol{v}_\delta^\top \boldsymbol{Z}\boldsymbol{\eta}_\delta - \sum_{i,j} \frac{\partial(v_{\delta,i}\eta_{\delta,j})}{\partial z_{ij}}\right\} = \frac{\Gamma_1}{G_{\boldsymbol{f},\boldsymbol{\psi}} + \delta^2} + \mathcal{R}_{1,\delta},$$

$$\|\boldsymbol{Z}\boldsymbol{\eta}_\delta\|^2 - \sum_{i,j} \frac{\partial\{\eta_{\delta,j}\boldsymbol{e}_i^\top \boldsymbol{Z}\boldsymbol{\eta}_\delta\}}{\partial z_{ij}} = \frac{\Gamma_2}{G_{\boldsymbol{f},\boldsymbol{\psi}} + \delta^2} + \mathcal{R}_{2,\delta},$$

$$\|\boldsymbol{Z}^\top \boldsymbol{v}_\delta\|^2 - \sum_{i,j} \frac{\partial\{v_{\delta,i}\boldsymbol{e}_j^\top \boldsymbol{Z}^\top \boldsymbol{v}_\delta\}}{\partial z_{ij}} = \frac{\Gamma_3}{G_{\boldsymbol{f},\boldsymbol{\psi}} + \delta^2} + \mathcal{R}_{3,\delta}.$$

For the second and third lines, the extra direct derivative terms come from $\partial(\boldsymbol{Z}\boldsymbol{\eta}_\delta)/\partial z_{ij} = \boldsymbol{e}_i\eta_{\delta,j} + \boldsymbol{Z}\,\partial\boldsymbol{\eta}_\delta/\partial z_{ij}$ and the analogous identity for $\boldsymbol{Z}^\top\boldsymbol{v}_\delta$. This is where the $n+d+\|\boldsymbol{Z}\|_{\mathrm{op}}^2$ term in the bound below originates. The normalizer remainders are explicitly

$$\mathcal{R}_{1,\delta} := -\sum_{i,j} \psi_i f_j \frac{\partial a_\delta^{-2}}{\partial z_{ij}},$$

$$\mathcal{R}_{2,\delta} := -\sum_{i,j} f_j(\boldsymbol{e}_i^\top \boldsymbol{Z}\boldsymbol{f}) \frac{\partial a_\delta^{-2}}{\partial z_{ij}},$$

$$\mathcal{R}_{3,\delta} := -\frac{1}{n}\sum_{i,j} \psi_i(\boldsymbol{e}_j^\top \boldsymbol{Z}^\top\boldsymbol{\psi}) \frac{\partial a_\delta^{-2}}{\partial z_{ij}}.$$

In particular,

$$\sum_{i,j}\left(\frac{\partial G_{\boldsymbol{f},\boldsymbol{\psi}}}{\partial z_{ij}}\right)^2 \le 4G_{\boldsymbol{f},\boldsymbol{\psi}}\mathcal{E}_{\boldsymbol{f},\boldsymbol{\psi}}.$$

Using also $\|\boldsymbol{f}\|^2 \le G_{\boldsymbol{f},\boldsymbol{\psi}} + \delta^2$, $\|\boldsymbol{\psi}\|^2 \le n(G_{\boldsymbol{f},\boldsymbol{\psi}} + \delta^2)$, and the operator-norm bounds for their images under $\boldsymbol{Z}$, Cauchy–Schwarz gives

$$|\mathcal{R}_{1,\delta}|^2 \le 4n\frac{\mathcal{E}_{\boldsymbol{f},\boldsymbol{\psi}}}{G_{\boldsymbol{f},\boldsymbol{\psi}} + \delta^2},$$

$$|\mathcal{R}_{2,\delta}|^2 + |\mathcal{R}_{3,\delta}|^2 \le C(n+d)\left(1 + \frac{\|\boldsymbol{Z}\|_{\mathrm{op}}^2}{n}\right)\frac{\mathcal{E}_{\boldsymbol{f},\boldsymbol{\psi}}}{G_{\boldsymbol{f},\boldsymbol{\psi}} + \delta^2}.$$

Combining these estimates with (44) proves (42) with $G_{\boldsymbol{f},\boldsymbol{\psi}} + \delta^2$ in the intermediate denominators. Let $\delta \downarrow 0$. The derivative ratios are dominated by $\mathcal{E}_{\boldsymbol{f},\boldsymbol{\psi}}/G_{\boldsymbol{f},\boldsymbol{\psi}}$ under the zero convention. Dominated convergence on the right and Fatou's lemma on the left prove (42). This is the scale-invariant normalization obtained by the argument in the proof of the cited proposition.

For the second-order terms, we use a conditional corollary of Proposition 16 of Bellec and Shen [14]. The one-vector inequality used here states that, for a smooth $\boldsymbol{u} : \mathbb{R}^{n\times d} \to \mathbb{R}^n$ with $\|\boldsymbol{u}\| \le 1$,

$$\mathbb{E}\left[\left|d\|\boldsymbol{u}\|^2 - \sum_{j=1}^d\left(\boldsymbol{u}^\top \boldsymbol{Z}\boldsymbol{e}_j - \sum_{i=1}^n \frac{\partial u_i}{\partial z_{ij}}\right)^2\right| \,\middle|\, \mathcal{F}\right]$$
$$\le C\sqrt{d}\left\{1 + \mathbb{E}\left[\sum_{i,j}\|\frac{\partial \boldsymbol{u}}{\partial z_{ij}}\|^2 \,\middle|\, \mathcal{F}\right]^{1/2}\right\} + C\mathbb{E}\left[\sum_{i,j}\|\frac{\partial \boldsymbol{u}}{\partial z_{ij}}\|^2 \,\middle|\, \mathcal{F}\right]. \tag{45}$$

The transposed version holds with n and d interchanged. Conditional on $\mathcal{F}$, (45) is the deterministic-field inequality in the proof of the cited proposition.

Apply (45) to $\boldsymbol{\upsilon}_\delta$ and its transposed version to $\boldsymbol{\eta}_\delta$. To identify the target contractions, define

$$a_{4,\delta,j} := \frac{\boldsymbol{\psi}^\top \boldsymbol{Z}\boldsymbol{e}_j - \sum_i \partial\psi_i/\partial z_{ij}}{\sqrt{n}\, a_\delta}, \qquad b_{4,\delta,j} := \boldsymbol{\upsilon}_\delta^\top \boldsymbol{Z}\boldsymbol{e}_j - \sum_i \frac{\partial \upsilon_{\delta,i}}{\partial z_{ij}},$$

$$a_{5,\delta,i} := \frac{\boldsymbol{f}^\top \boldsymbol{Z}^\top \boldsymbol{e}_i - \sum_j \partial f_j/\partial z_{ij}}{a_\delta}, \qquad b_{5,\delta,i} := \boldsymbol{\eta}_\delta^\top \boldsymbol{Z}^\top \boldsymbol{e}_i - \sum_j \frac{\partial \eta_{\delta,j}}{\partial z_{ij}}.$$

Writing $\boldsymbol{a}_{s,\delta}$ and $\boldsymbol{b}_{s,\delta}$ for the corresponding vectors, the definitions give the exact identities

$$\frac{\Gamma_4}{G_{\boldsymbol{f},\boldsymbol{\psi}} + \delta^2} = d\|\boldsymbol{\upsilon}_\delta\|^2 - \|\boldsymbol{a}_{4,\delta}\|^2,$$
$$\frac{\Gamma_5}{G_{\boldsymbol{f},\boldsymbol{\psi}} + \delta^2} = n\|\boldsymbol{\eta}_\delta\|^2 - \|\boldsymbol{a}_{5,\delta}\|^2.$$

The product rule and Cauchy–Schwarz yield

$$\|\boldsymbol{a}_{4,\delta} - \boldsymbol{b}_{4,\delta}\|^2 + \|\boldsymbol{a}_{5,\delta} - \boldsymbol{b}_{5,\delta}\|^2 \le G_{\boldsymbol{f},\boldsymbol{\psi}} \sum_{i,j} \left(\frac{\partial a_\delta^{-1}}{\partial z_{ij}}\right)^2 \le \frac{\mathcal{E}_{\boldsymbol{f},\boldsymbol{\psi}}}{G_{\boldsymbol{f},\boldsymbol{\psi}} + \delta^2}.$$

For $c \ge 1$, $\|\boldsymbol{u}\| \le 1$, and vectors $\boldsymbol{a}, \boldsymbol{b}$, the following elementary comparison holds:

$$\left|c\|\boldsymbol{u}\|^2 - \|\boldsymbol{a}\|^2\right| \le 2\left|c\|\boldsymbol{u}\|^2 - \|\boldsymbol{b}\|^2\right| + 3\|\boldsymbol{a} - \boldsymbol{b}\|^2 + 2\sqrt{c}\|\boldsymbol{a} - \boldsymbol{b}\|.$$

This comparison follows by writing $|\|\boldsymbol{a}\|^2 - \|\boldsymbol{b}\|^2| \le \|\boldsymbol{a} - \boldsymbol{b}\|^2 + 2\|\boldsymbol{b}\|\|\boldsymbol{a} - \boldsymbol{b}\|$ and then bounding $\|\boldsymbol{b}\|$ using $|c\|\boldsymbol{u}\|^2 - \|\boldsymbol{b}\|^2|$. It therefore converts the two applications of (45) into

$$\mathbb{E}\left[\frac{|\Gamma_4| + |\Gamma_5|}{G_{\boldsymbol{f},\boldsymbol{\psi}} + \delta^2} \;\middle|\; \mathcal{F}\right] \le C\left\{\sqrt{n+d}\left(1 + \Theta_{\delta,\mathcal{F}}^{1/2}\right) + \Theta_{\delta,\mathcal{F}}\right\},$$

where

$$\Theta_{\delta,\mathcal{F}} := \mathbb{E}\left[\frac{\mathcal{E}_{\boldsymbol{f},\boldsymbol{\psi}}}{G_{\boldsymbol{f},\boldsymbol{\psi}} + \delta^2} \;\middle|\; \mathcal{F}\right] \le \Theta_{\mathcal{F}}.$$

Fatou's lemma as $\delta \downarrow 0$ proves (43). This also shows directly that Proposition 16 is applied to the two normalized fields separately; no result for an undifferentiated stacked map is being assumed. □

B.5.3. Remainders and response-weighted fluctuations

Define

$$\Delta_1 := -2\boldsymbol{r}^\top \boldsymbol{L}\boldsymbol{q}, \qquad \Delta_2 := -\|\boldsymbol{q}\|^2,$$
$$\Delta_4 := \|\boldsymbol{L}^\top \boldsymbol{r}\|^2 + 2b\boldsymbol{q}^\top \boldsymbol{L}^\top \boldsymbol{r}, \qquad \Delta_5 := \|\boldsymbol{L}\boldsymbol{q}\|^2 + 2\|\boldsymbol{q}\|^2 - 2t_{\boldsymbol{\Sigma}} \boldsymbol{r}^\top \boldsymbol{L}\boldsymbol{q}.$$

Lemma B.20. *Under Assumption B.3, for every fixed integer $k \ge 1$, there are $n_0(k) \ge 1$ and $C_k < \infty$ such that, for $n \ge n_0(k)$,*

$$\mathbb{E}\left(|\Delta_1|^k + |\Delta_2|^k + |\Delta_5|^k + n^{-k}|\Delta_4|^k\right) \le C_k. \tag{46}$$

Define every normalized ratio below to be zero on $\{G_{\boldsymbol{\Sigma}} = 0\}$. Then the stronger response-normalized bounds

$$\mathbb{E}\left[\sum_{s\in\{1,2,5\}} \left|\frac{\Delta_s}{G_{\boldsymbol{\Sigma}}}\right|^k + \left|\frac{\Delta_4}{nG_{\boldsymbol{\Sigma}}}\right|^k\right] \le C_k \tag{47}$$

hold. In particular, at $k = 2$,

$$\mathbb{E}\left[\sum_{s\in\{1,2,5\}} \frac{\Delta_s^2}{nG_{\boldsymbol{\Sigma}}^2} + \frac{\Delta_4^2}{n^3 G_{\boldsymbol{\Sigma}}^2}\right] \le \frac{C_2}{n}.$$

Thus the normalized estimate also yields the displayed $O(n^{-1})$ scaled L^2 bound. No third remainder is needed because $Q_{3,\boldsymbol{\Sigma}} = 0$ by the sketched normal equations.

Specialize the Gaussian contractions from Lemma B.19 to the coefficient error and residual:

$$\zeta_s := \Gamma_s|_{(\boldsymbol{f},\psi)=(\boldsymbol{q},\boldsymbol{r})}, \qquad s = 1, \dots, 5. \tag{48}$$

Clarification (symbol ζ_s):

The contractions ζ_s, $s \in \{1, \dots, 5\}$, use the trace-normalized residual field $\boldsymbol{r}$ and derivatives with respect to $\boldsymbol{Z}$, holding $\boldsymbol{S}$, $\boldsymbol{\Sigma}$, $\boldsymbol{\beta}^*$ and the structural noise $\boldsymbol{\varepsilon}$ fixed, with weak derivatives on the full-rank domain.

Lemma B.21. *Under Assumption B.3, there are $n_0 \geq 1$ and $C < \infty$, independent of n, such that, for $n \geq n_0$,*

$$\begin{aligned} &\mathbb{E}\zeta_1^2 \leq Cn, && \mathbb{E}\zeta_2^2 \leq Cn, && \zeta_3 = 0, \\ &\mathbb{E}|\zeta_4| \leq C\sqrt{n}, && \mathbb{E}|\zeta_5| \leq C\sqrt{n}. \end{aligned} \tag{49}$$

Every ratio below is defined to be zero on $\{G_{\boldsymbol{\Sigma}} = 0\}$. Moreover, if $\mathbb{E}_\star$ denotes either expectation under the auxiliary response law defined above or expectation under the original response law, then

$$\mathbb{E}_\star\left[\frac{\zeta_1^2 + \zeta_2^2}{nG_{\boldsymbol{\Sigma}}^2} + \frac{|\zeta_4| + |\zeta_5|}{\sqrt{n}\, G_{\boldsymbol{\Sigma}}}\right] \leq C. \tag{50}$$

The exact decompositions are

$$\begin{aligned} &Q_{1,\boldsymbol{\Sigma}} = \zeta_1 + \Delta_1, && Q_{2,\boldsymbol{\Sigma}} = \zeta_2 + \Delta_2, \\ &Q_{3,\boldsymbol{\Sigma}} = \zeta_3 = 0, && Q_{4,\boldsymbol{\Sigma}} = \zeta_4 + \frac{\Delta_4}{n}, && Q_{5,\boldsymbol{\Sigma}} = \zeta_5 + \Delta_5. \end{aligned} \tag{51}$$

Together with Lemma B.20, these estimates imply the trace bounds (62)– (63).

The $O(\sqrt{n})$ rate for ζ_4 in (49) reflects fluctuations present even for a scaled orthogonal sketch. Indeed, with $\boldsymbol{A} = \boldsymbol{I}_n$, nondegenerate Gaussian errors, and $d/n \to \gamma \in (0, 1)$, the normal equations and Lemma B.5 give

$$\zeta_4 = \frac{d}{n}\|\boldsymbol{r}\|^2 - \frac{(n-d)^2}{n}\|\boldsymbol{q}\|^2,$$

whose Gaussian projection fluctuations are of order $\sqrt{n}$. Hence a uniform $O(1)$ fourth-fluctuation bound is not valid for general random sketches. The normalized bound (50) is the quantity needed for trace calibration.

For a constant $C_0 > 0$, define

$$\Omega_n := \left\{n\|(\boldsymbol{G}_0^\top \boldsymbol{G}_0)^{-1}\|_{\mathrm{op}} \leq C_0, \quad \|\boldsymbol{Z}\|_{\mathrm{op}} \leq C_0\sqrt{n}, \quad \|\boldsymbol{\varepsilon}\| \leq C_0\sqrt{n}\right\}.$$

Clarification (symbol C_0):

$C_0 > 0$ is a deterministic threshold independent of n and of the realized sketch, chosen using only the fixed model and balanced-sketch parameters. In Lemma B.20, Lemma B.21, Lemma B.22: The threshold is chosen sufficiently large to control all three constraints defining Ω_n, including $C_0 \geq eD_J$ in the inverse-norm argument.

We defer the proofs of the preceding two lemmas until after the following auxiliary result, which is needed for the response-weighted trace bound.

Lemma B.22. *Suppose Assumption B.3 holds. The constant C_0 can be chosen sufficiently large so that there are constants $c, C > 0$, uniform over all balanced sketch matrices, such that, for all sufficiently large n,*

$$\mathbb{P}(\Omega_n^c \mid \boldsymbol{S}) \leq Ce^{-cn} \quad \text{almost surely}, \qquad \mathbb{P}(\Omega_n^c) \leq Ce^{-cn}. \tag{52}$$

On the whole sample space, using the zero convention for the ratios in $\mathcal{T}_{\boldsymbol{A}}$,

$$\begin{aligned} \frac{B_{\boldsymbol{A}}\mathcal{T}_{\boldsymbol{A}}}{n} \leq 8\frac{n}{b}\left(1 + \frac{\|\boldsymbol{Z}\|_{\mathrm{op}}^2}{n}\right)(1 + t_{\boldsymbol{\Sigma}})^2 \\ \times \left\{\frac{2(\|\boldsymbol{\varepsilon}\| + \|\boldsymbol{Z}\|_{\mathrm{op}}\|\boldsymbol{q}\|)\|\boldsymbol{r}\|}{n} + \left(t_{\boldsymbol{\Sigma}} + \frac{n}{b}\right)\frac{\|\boldsymbol{r}\|^2}{n}\right\}. \end{aligned} \tag{53}$$

Consequently, for every fixed $N > 0$, there is $C_N < \infty$ such that, for all sufficiently large n,

$$\mathbb{E}\left[B_{\boldsymbol{A}}\mathcal{T}_{\boldsymbol{A}}\mathbf{1}_{\Omega_n^c}\right] \leq C_N n^{-N}. \tag{54}$$

Proof. Conditional on the sketch, $\boldsymbol{G}_0$ is an $m \times d$ standard Gaussian matrix. Put $\nu = m - d + 1$ and choose an integer $J \geq 1$ with $J\kappa \geq 1$. Then $n \leq J\nu$. The growing-moment argument in the proof of Lemma B.12, with $r = \lfloor \nu/4 \rfloor$ and $D_J = 4J\, 9^{8J}$, gives, for all sufficiently large n,

$$\mathbb{E}\left[\left\{n\|(\boldsymbol{G}_0^\top \boldsymbol{G}_0)^{-1}\|_{\mathrm{op}}\right\}^r \;\middle|\; \boldsymbol{S}\right] \leq 2D_J^r, \qquad r \geq \frac{n}{8J}.$$

Markov's inequality therefore implies

$$\mathbb{P}\left\{n\|(\boldsymbol{G}_0^\top \boldsymbol{G}_0)^{-1}\|_{\mathrm{op}} \geq eD_J \;\middle|\; \boldsymbol{S}\right\} \leq 2e^{-r} \leq 2e^{-n/(8J)}.$$

Choose $C_0 \geq eD_J$ and enlarge it as needed for the Gaussian operator-norm and sub-Gaussian noise-norm bounds in (16). A union bound proves (52), uniformly in the sketch.

The definition of $\mathcal{T}_{\boldsymbol{A}}$, Cauchy–Schwarz, and $0 < b \leq n$, $d \leq n$, $t_{\boldsymbol{\Sigma}} \geq 0$ give

$$\mathcal{T}_{\boldsymbol{A}} \leq 8n\frac{n}{b}\left(1 + \frac{\|\boldsymbol{Z}\|_{\mathrm{op}}^2}{n}\right)(1 + t_{\boldsymbol{\Sigma}})^2. \tag{55}$$

For example, $\|\boldsymbol{q}\|^2 \leq G_{\boldsymbol{\Sigma}}$, $\|\boldsymbol{r}\|^2 \leq nG_{\boldsymbol{\Sigma}}$, and $|\boldsymbol{r}^\top \boldsymbol{Z}\boldsymbol{q}| \leq (n + \|\boldsymbol{Z}\|_{\mathrm{op}}^2)G_{\boldsymbol{\Sigma}}$ bound each term in the definition. Expanding their coefficients gives (55); the same bound holds at zero response energy by the zero convention. Since $\boldsymbol{u} = \boldsymbol{\varepsilon} - \boldsymbol{Z}\boldsymbol{q}$, the definition of $B_{\boldsymbol{A}}$ gives

$$B_{\boldsymbol{A}} \leq \frac{2(\|\boldsymbol{\varepsilon}\| + \|\boldsymbol{Z}\|_{\mathrm{op}}\|\boldsymbol{q}\|)\|\boldsymbol{r}\|}{n} + \left(t_{\boldsymbol{\Sigma}} + \frac{n}{b}\right)\frac{\|\boldsymbol{r}\|^2}{n}.$$

Multiplication proves (53).

Every fixed moment of n/b, $t_{\boldsymbol{\Sigma}}$, $\|\boldsymbol{Z}\|_{\mathrm{op}}/\sqrt{n}$, $\|\boldsymbol{q}\|$, $\|\boldsymbol{r}\|/\sqrt{n}$, and $\|\boldsymbol{\varepsilon}\|/\sqrt{n}$ is uniformly bounded, by the preceding inverse, response, and noise moment bounds. Repeated Hölder inequalities applied to the polynomial in (53) therefore give $\mathbb{E}(B_{\boldsymbol{A}}\mathcal{T}_{\boldsymbol{A}})^2 \leq Cn^2$. Cauchy–Schwarz and (52) now yield

$$\begin{aligned}\mathbb{E}\left[B_{\boldsymbol{A}}\mathcal{T}_{\boldsymbol{A}}\mathbf{1}_{\Omega_n^c}\right] &\leq \{\mathbb{E}(B_{\boldsymbol{A}}\mathcal{T}_{\boldsymbol{A}})^2\}^{1/2}\mathbb{P}(\Omega_n^c)^{1/2} \\ &\leq Cne^{-cn/2} \leq C_N n^{-N}.\end{aligned}$$

This proves (54). □

We prove Lemmas B.20 and B.21 in three stages. We first obtain unnormalized fluctuation bounds, then apply Lemma B.19 through a ridge approximation, and finally combine its contractions with the algebraic remainder bounds.

Proofs of Lemmas B.20 and B.21. Retain the notation and bounds from the proof of Lemmas B.14 and B.15. Conditional on the sketch, $\boldsymbol{G}_0$ is an $m \times d$ standard Gaussian matrix. The Ky Fan lower bound (11) gives

$$b \geq c_-(m - d) \geq c_-\kappa n, \qquad \frac{n}{b} \leq \frac{1}{c_-\kappa}. \tag{56}$$

For the response-normalized bounds, no negative response moment is needed. With $G_{\boldsymbol{\Sigma}} = \|\boldsymbol{q}\|^2 + \|\boldsymbol{r}\|^2/n$, define every ratio as zero when $G_{\boldsymbol{\Sigma}} = 0$. Equations (30)–(32) give pointwise

$$\frac{1}{G_{\boldsymbol{\Sigma}}}\sum_{i,j}\left\{\|\frac{\partial \boldsymbol{q}}{\partial z_{ij}}\|^2 + \frac{1}{n}\|\frac{\partial \boldsymbol{r}}{\partial z_{ij}}\|^2\right\} \leq P(H_n), \tag{57}$$

where P is a fixed polynomial with nonnegative coefficients and all fixed moments of

$$H_n := 1 + \frac{\|\boldsymbol{Z}\|_{\mathrm{op}}}{\sqrt{n}} + n\|\widetilde{\boldsymbol{M}}^{-1}\|_{\mathrm{op}} + \sqrt{n}\|\boldsymbol{L}\|_{\mathrm{op}}$$

are uniformly bounded.

For the first stage, we establish the unnormalized fluctuation bounds. In the raw Stein bounds below, apply the smooth-field inequalities first to the positive-ridge fields $\boldsymbol{q}_\eta$ and $\boldsymbol{r}_\eta$. The uniform moment bounds established above control their fourth moments and derivative energies. Almost-sure convergence of the fields and their derivatives, followed by Fatou's lemma, gives the displayed bounds at $\eta = 0$. Conditional on $(\boldsymbol{A}, \boldsymbol{\varepsilon})$, the first-order Gaussian divergence inequality applied to the matrix field $\boldsymbol{r}\boldsymbol{q}^\top$ gives

$$\begin{aligned}\mathbb{E}\zeta_1^2 &\le C\mathbb{E}\left[\|\boldsymbol{r}\|^2\|\boldsymbol{q}\|^2 + \sum_{i,j}\|\frac{\partial(\boldsymbol{r}\boldsymbol{q}^\top)}{\partial z_{ij}}\|_{\mathrm{F}}^2\right] \\ &\le C\mathbb{E}\left[\|\boldsymbol{r}\|^2\|\boldsymbol{q}\|^2 + \|\boldsymbol{q}\|^2\mathcal{E}_{\boldsymbol{r}} + \|\boldsymbol{r}\|^2\mathcal{E}_{\boldsymbol{q}}\right] \le Cn,\end{aligned}$$

where the last step uses (31) and Cauchy–Schwarz. Applying the same inequality to $(\boldsymbol{Z}\boldsymbol{q})\boldsymbol{q}^\top$ and using the product rule yields

$$\sum_{i,j}\|\frac{\partial\{(\boldsymbol{Z}\boldsymbol{q})\boldsymbol{q}^\top\}}{\partial z_{ij}}\|_{\mathrm{F}}^2 \le C\left\{n\|\boldsymbol{q}\|^4 + \|\boldsymbol{Z}\|_{\mathrm{op}}^2\|\boldsymbol{q}\|^2\mathcal{E}_{\boldsymbol{q}}\right\}.$$

Together with $\|(\boldsymbol{Z}\boldsymbol{q})\boldsymbol{q}^\top\|_{\mathrm{F}}^2 \le \|\boldsymbol{Z}\|_{\mathrm{op}}^2\|\boldsymbol{q}\|^4$, Hölder's inequality, (25), and (31) prove $\mathbb{E}\zeta_2^2 \le Cn$. The normal equations give $\boldsymbol{Z}^\top\boldsymbol{r} = 0$, and Lemma B.8 makes the divergence in the definition of ζ_3 zero. Hence $\zeta_3 = 0$ pathwise.

Apply the second claim of Lemma B.16 to $\boldsymbol{q}(\boldsymbol{Z})$. Its proof uses only the fourth-moment and derivative-energy assumptions and gives $\mathbb{E}|\zeta_5| \le C\sqrt{n}$. Apply the same argument after transposing the design: with $\boldsymbol{W} = \boldsymbol{Z}^\top \in \mathbb{R}^{d\times n}$ and $\boldsymbol{f}(\boldsymbol{W}) = \boldsymbol{r}/\sqrt{n}$, its second-order fluctuation is exactly

$$d\|\boldsymbol{f}\|^2 - \sum_{j=1}^{d}\left(\boldsymbol{e}_j^\top\boldsymbol{W}\boldsymbol{f} - \sum_{i=1}^{n}\frac{\partial f_i}{\partial w_{ji}}\right)^2 = \zeta_4.$$

Here $w_{ji} = z_{ij}$ and $\partial f_i/\partial w_{ji} = n^{-1/2}\partial r_i/\partial z_{ij}$, so transposition changes only the orientation of the design indices. The required fourth moments of $\boldsymbol{f}$ and its derivative energy are bounded by (31). Hence $\mathbb{E}|\zeta_4| \le C\sqrt{d} \le C\sqrt{n}$, proving the unnormalized second-order bounds.

For the response-normalized statements, condition on $(\boldsymbol{A}, \boldsymbol{\varepsilon})$, let $\mathbb{E}_{\boldsymbol{Z}}$ denote expectation over $\boldsymbol{Z}$, and set

$$\Theta_{\boldsymbol{A},\boldsymbol{\varepsilon}} := \mathbb{E}_{\boldsymbol{Z}}\left[\frac{\mathcal{E}_{\boldsymbol{q},\boldsymbol{r}}}{G_{\boldsymbol{\Sigma}}}\right].$$

Here we use the derivative energy from Lemma B.19 and the same zero convention.

The ridge approximations verify the regularity requirements of Lemma B.19. For $\eta > 0$, use the ridge approximations from the proof of Lemma B.14. Conditional on $(\boldsymbol{A}, \boldsymbol{\varepsilon})$, these maps are smooth on all of $\mathbb{R}^{n\times d}$. Their behavior at zero energy is particularly simple: if $\boldsymbol{q}_\eta = \boldsymbol{r}_\eta = \boldsymbol{0}$ at one design, then $\boldsymbol{A}\boldsymbol{\varepsilon} = \boldsymbol{r}_\eta + \boldsymbol{A}\boldsymbol{Z}\boldsymbol{q}_\eta = \boldsymbol{0}$. The coefficient and residual fields are consequently identically zero as functions of $\boldsymbol{Z}$, and so are their derivatives. Use the quantities $G_{\boldsymbol{q}_\eta,\boldsymbol{r}_\eta}$ and $\mathcal{E}_{\boldsymbol{q}_\eta,\boldsymbol{r}_\eta}$ from Lemma B.19, again using the zero convention. The derivative formulas are

$$\begin{aligned}\frac{\partial\boldsymbol{q}_\eta}{\partial z_{ij}} &= \boldsymbol{K}_\eta\boldsymbol{e}_j(\boldsymbol{r}_\eta)_i - \boldsymbol{L}_\eta^\top\boldsymbol{e}_i(\boldsymbol{q}_\eta)_j, \\ \frac{\partial\boldsymbol{r}_\eta}{\partial z_{ij}} &= -\boldsymbol{A}\boldsymbol{e}_i(\boldsymbol{q}_\eta)_j - \boldsymbol{A}\boldsymbol{Z}\frac{\partial\boldsymbol{q}_\eta}{\partial z_{ij}}.\end{aligned}$$

Therefore, with $\mathcal{E}_{\boldsymbol{q}_\eta}$ and $\mathcal{E}_{\boldsymbol{r}_\eta}$ denoting the two derivative energies, use $G_{\boldsymbol{q}_\eta,\boldsymbol{r}_\eta} \ge \|\boldsymbol{q}_\eta\|^2$ and $G_{\boldsymbol{q}_\eta,\boldsymbol{r}_\eta} \ge \|\boldsymbol{r}_\eta\|^2/n$ to obtain

$$\begin{aligned}\frac{\mathcal{E}_{\boldsymbol{q}_\eta}}{G_{\boldsymbol{q}_\eta,\boldsymbol{r}_\eta}} &\le 2n\|\boldsymbol{K}_\eta\|_{\mathrm{F}}^2 + 2\|\boldsymbol{L}_\eta\|_{\mathrm{F}}^2, \\ \frac{\mathcal{E}_{\boldsymbol{r}_\eta}}{nG_{\boldsymbol{q}_\eta,\boldsymbol{r}_\eta}} &\le \frac{2\|\boldsymbol{A}\|_{\mathrm{F}}^2}{n} + \frac{2\|\boldsymbol{A}\boldsymbol{Z}\|_{\mathrm{op}}^2}{n}\frac{\mathcal{E}_{\boldsymbol{q}_\eta}}{G_{\boldsymbol{q}_\eta,\boldsymbol{r}_\eta}}.\end{aligned}$$

The singular-value comparison in Lemma B.12 gives, on the full-rank event,

$$\|\boldsymbol{K}_\eta\|_{\mathrm{op}} \le \|\widetilde{\boldsymbol{M}}^{-1}\|_{\mathrm{op}}, \qquad \|\boldsymbol{L}_\eta\|_{\mathrm{op}} \le \|\boldsymbol{L}\|_{\mathrm{op}}.$$

Indeed,

$$\boldsymbol{L}_\eta = \boldsymbol{L}\left\{\widetilde{\boldsymbol{M}}(\widetilde{\boldsymbol{M}}+\eta\boldsymbol{I}_d)^{-1}\right\}, \qquad \|\widetilde{\boldsymbol{M}}(\widetilde{\boldsymbol{M}}+\eta\boldsymbol{I}_d)^{-1}\|_{\mathrm{op}} \le 1.$$

Thus both bounds hold uniformly in η, and in particular $\sqrt{n}\|\boldsymbol{L}_\eta\|_{\mathrm{op}} \le H_n$. Together with $d \le n$, $\|\boldsymbol{A}\|_{\mathrm{F}}^2 = \mathrm{Tr}(\boldsymbol{A}^2) \le c_+ \mathrm{Tr}(\boldsymbol{A}) = c_+ n$, and $\|\boldsymbol{A}\boldsymbol{Z}\|_{\mathrm{op}} \le c_+\|\boldsymbol{Z}\|_{\mathrm{op}}$, the preceding two displays yield, after enlarging the fixed polynomial P if necessary,

$$\frac{\mathcal{E}_{\boldsymbol{q}_\eta,\boldsymbol{r}_\eta}}{G_{\boldsymbol{q}_\eta,\boldsymbol{r}_\eta}} \le P(H_n) \qquad (\eta > 0). \tag{58}$$

The zero-denominator event is also explicit. Indeed, $\boldsymbol{q} = \widetilde{\boldsymbol{M}}^{-1}\boldsymbol{Z}^\top\boldsymbol{A}\boldsymbol{\varepsilon}$ and $\boldsymbol{r} = \boldsymbol{A}(\boldsymbol{\varepsilon} - \boldsymbol{Z}\boldsymbol{q})$ show that both $\boldsymbol{q}$ and $\boldsymbol{r}$ vanish exactly when $\boldsymbol{A}\boldsymbol{\varepsilon} = 0$. The same reasoning applies to the ridge versions. Thus

$$G_{\boldsymbol{\Sigma}} = 0 \iff \boldsymbol{A}\boldsymbol{\varepsilon} = \boldsymbol{0} \iff G_{\boldsymbol{q}_\eta,\boldsymbol{r}_\eta} = 0 \text{ for every } \eta > 0.$$

On this event, $\boldsymbol{q}_\eta$, $\boldsymbol{r}_\eta$, their first derivatives, and all five Γ-contractions in Lemma B.19 vanish. On its complement, all the denominators are positive. On the Gaussian full-rank event, as $\eta \downarrow 0$, $\boldsymbol{q}_\eta$, $\boldsymbol{r}_\eta$, and their displayed derivatives converge pointwise to $\boldsymbol{q}$, $\boldsymbol{r}$, and their weak derivatives. By (48), the five ridge-normalized contractions converge to their corresponding $\zeta_s/G_{\boldsymbol{\Sigma}}$ limits.

Apply that lemma to $(\boldsymbol{q}_\eta, \boldsymbol{r}_\eta)$ with $\mathcal{F}$ the sigma-field generated by $(\boldsymbol{A}, \boldsymbol{\varepsilon})$. Conditional on this sigma-field, $\boldsymbol{Z}$ still has i.i.d. standard Gaussian entries. The bound (58), the conditional inverse envelope (21), and the conditional Gaussian operator-norm bound in Lemma B.11 give an integrable dominating variable for both right-hand sides, uniformly over balanced sketch matrices. This bound is independent of the fixed value of $\boldsymbol{\varepsilon}$. The same statement holds when conditioning on $\boldsymbol{A}$ rather than on $\boldsymbol{S}$ by the tower property. Dominated convergence therefore applies to those right-hand sides and to $\mathbb{E}_{\boldsymbol{Z}}[\mathcal{E}_{\boldsymbol{q}_\eta,\boldsymbol{r}_\eta}/G_{\boldsymbol{q}_\eta,\boldsymbol{r}_\eta}]$; Fatou's lemma applies to the nonnegative left-hand sides. Passing to the limit in Lemma B.19 gives

$$\mathbb{E}_{\boldsymbol{Z}}\left[\frac{\zeta_1^2+\zeta_2^2+\zeta_3^2}{G_{\boldsymbol{\Sigma}}^2}\right] \le C\mathbb{E}_{\boldsymbol{Z}}\left[n+d+\|\boldsymbol{Z}\|_{\mathrm{op}}^2 + (n+d)\left(1+\frac{\|\boldsymbol{Z}\|_{\mathrm{op}}^2}{n}\right)\frac{\mathcal{E}_{\boldsymbol{q},\boldsymbol{r}}}{G_{\boldsymbol{\Sigma}}}\right], \tag{59}$$

$$\mathbb{E}_{\boldsymbol{Z}}\left[\frac{|\zeta_4|+|\zeta_5|}{G_{\boldsymbol{\Sigma}}}\right] \le C\left\{\sqrt{n+d}\left(1+\Theta_{\boldsymbol{A},\boldsymbol{\varepsilon}}^{1/2}\right)+\Theta_{\boldsymbol{A},\boldsymbol{\varepsilon}}\right\}. \tag{60}$$

The same domination proves uniform integrability of the normalized derivative energies along the ridge approximation. No negative moment of $G_{\boldsymbol{\Sigma}}$ is used.

By (57), $\mathcal{E}_{\boldsymbol{q},\boldsymbol{r}}/G_{\boldsymbol{\Sigma}} \le P(H_n)$ pointwise. Since $d = O(n)$ and all fixed moments of H_n are uniformly bounded, integration over $(\boldsymbol{A}, \boldsymbol{\varepsilon})$ in (59) gives an $O(n)$ right-hand side. Moreover,

$$\mathbb{E}_\star\Theta_{\boldsymbol{A},\boldsymbol{\varepsilon}} \le C, \qquad \mathbb{E}_\star\Theta_{\boldsymbol{A},\boldsymbol{\varepsilon}}^{1/2} \le C,$$

so (60) is $O(\sqrt{n})$ after integration. The pointwise envelope is independent of the response scale, and therefore these conclusions hold under both response laws denoted by $\mathbb{E}_\star$. Dividing the two resulting estimates by n and $\sqrt{n}$, respectively, proves (50); together with the preceding argument, this also completes (49).

The algebraic remainders follow from elementary Cauchy–Schwarz bounds. Since $0 < b/n \le 1$, these bounds give pointwise

$$\frac{|\Delta_1|}{G_{\boldsymbol{\Sigma}}} \le \sqrt{n}\|\boldsymbol{L}\|_{\mathrm{op}}, \qquad \frac{|\Delta_2|}{G_{\boldsymbol{\Sigma}}} \le 1,$$

$$\frac{|\Delta_4|}{nG_{\boldsymbol{\Sigma}}} \le \|\boldsymbol{L}\|_{\mathrm{op}}^2 + \frac{b}{n}\sqrt{n}\|\boldsymbol{L}\|_{\mathrm{op}}, \qquad \frac{|\Delta_5|}{G_{\boldsymbol{\Sigma}}} \le \|\boldsymbol{L}\|_{\mathrm{op}}^2 + 2 + t_{\boldsymbol{\Sigma}}\sqrt{n}\|\boldsymbol{L}\|_{\mathrm{op}}.$$

The inverse envelope (25) proves (47) at every fixed order. The same envelope, (31), and $b \le n$ prove (46). This establishes Lemma B.20.

Finally, Lemmas B.6–B.10, together with the definitions in (48), give the exact decompositions (51). To obtain the rate for $\mathcal{T}_{\boldsymbol{A}}$, first note that Equation (56) gives $n/b = O(1)$. By the normalized zeta and remainder bounds, $Q_{1,\boldsymbol{\Sigma}}/G_{\boldsymbol{\Sigma}}$ and $Q_{2,\boldsymbol{\Sigma}}/G_{\boldsymbol{\Sigma}}$ have L^2 size $O(\sqrt{n})$, whereas $Q_{4,\boldsymbol{\Sigma}}/G_{\boldsymbol{\Sigma}}$ and $Q_{5,\boldsymbol{\Sigma}}/G_{\boldsymbol{\Sigma}}$ have L^1 size $O(\sqrt{n})$; $Q_{3,\boldsymbol{\Sigma}} = 0$. The normalized remainders have bounded fixed-order moments. Hölder's inequality combines the L^2 bound for $Q_1/G_{\boldsymbol{\Sigma}}$ with the bounded second moment of $b/n - t_{\boldsymbol{\Sigma}}$ from (25). Substitution in the definition of $\mathcal{T}_{\boldsymbol{A}}$ gives $\mathbb{E}_{\mathrm{aux}}\mathcal{T}_{\boldsymbol{A}} \le C\sqrt{n}$. For the response-weighted version, positive spectral mass on the range of $\boldsymbol{A}$ gives

$$0 \le \boldsymbol{u}^\top \boldsymbol{r} = \boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{u} \le c_-^{-1}\|\boldsymbol{A}\boldsymbol{u}\|^2 = c_-^{-1}\|\boldsymbol{r}\|^2.$$

Consequently,

$$\frac{B_{\boldsymbol{A}}}{G_{\boldsymbol{\Sigma}}} \le \frac{2}{c_-} + t_{\boldsymbol{\Sigma}} + \frac{d}{b}. \tag{61}$$

The right side has bounded fixed moments by (25)–(56). Applying the normalized inequalities under the original response law gives the unweighted bound $\mathbb{E}\mathcal{T}_{\boldsymbol{A}} \le C\sqrt{n}$ as well. To justify the response weight without invoking an unproved L^2 second-order inequality, use the event Ω_n from Lemma B.22. On Ω_n, the inverse-norm bound in its definition and the balanced spectrum imply

$$n\|\widetilde{\boldsymbol{M}}^{-1}\|_{\mathrm{op}} + t_{\boldsymbol{\Sigma}} + \sqrt{n}\|\boldsymbol{L}\|_{\mathrm{op}} \le C.$$

Together with $\|\boldsymbol{\varepsilon}\| \le C_0\sqrt{n}$, equations (26) and (27) give $\|\boldsymbol{q}\| \le \|\boldsymbol{L}\|_{\mathrm{op}}\|\boldsymbol{\varepsilon}\| = O(1)$ and $\|\boldsymbol{r}\|/\sqrt{n} \le c_+\|\boldsymbol{\varepsilon}\|/\sqrt{n} = O(1)$. Thus $G_{\boldsymbol{\Sigma}} \le C$ on Ω_n. Hence (61) gives $B_{\boldsymbol{A}} \le C$, and therefore $\mathbb{E}[B_{\boldsymbol{A}}\mathcal{T}_{\boldsymbol{A}}\mathbf{1}_{\Omega_n}] \le C\sqrt{n}$. Lemma B.22 controls the complementary event, so, for example with $N = 1$, $\mathbb{E}[B_{\boldsymbol{A}}\mathcal{T}_{\boldsymbol{A}}] \le C\sqrt{n} + C_1 n^{-1} \le C'\sqrt{n}$. Thus both trace bounds (62) and (63) hold. This completes the proofs of Lemmas B.20 and B.21. □

B.5.4. Trace calibration and the observable risk criterion

For the trace-normalized representative, the balanced-sketch lemmas in Appendix B.5.3 give two bounds with distinct roles. There are constants $C_Q, C_{QB} < \infty$, independent of n, such that

$$\mathbb{E}_{\mathrm{aux}}[\mathcal{T}_{\boldsymbol{A}}] \le C_Q\sqrt{n}, \tag{62}$$
$$\mathbb{E}[B_{\boldsymbol{A}}\mathcal{T}_{\boldsymbol{A}}] \le C_{QB}\sqrt{n}, \tag{63}$$

where the second expectation is under the original response law. The first bound controls trace calibration, and the second controls the data-driven substitution. The four nonzero Q-terms are Stein deviations plus the explicit influence remainders in Lemmas B.6–B.10. The normalized second-order inequality controls $\zeta_4/G_{\boldsymbol{\Sigma}}$ and $\zeta_5/G_{\boldsymbol{\Sigma}}$ in L^1, which is sufficient for the argument.

Proposition B.23. *Under Assumption B.3, there is a constant $C < \infty$, independent of n, such that, for all sufficiently large n, $\mathbb{E}_{\boldsymbol{X},\boldsymbol{S}}|t_{\boldsymbol{\Sigma}} - d/b| \le C/\sqrt{n}$. Here $\mathbb{E}_{\boldsymbol{X},\boldsymbol{S}}$ is expectation under the original marginal law of $(\boldsymbol{X}, \boldsymbol{S})$. In particular, $t_{\boldsymbol{\Sigma}} - d/b = O_{\mathbb{P}}(n^{-1/2})$.*

Proof. Divide the exact identity (40) by $bG_{\boldsymbol{\Sigma}}$ and use the triangle inequality. Under the auxiliary law, $|t_{\boldsymbol{\Sigma}} - d/b| \le \mathcal{T}_{\boldsymbol{A}}/n$. The required estimate $\mathbb{E}_{\mathrm{aux}}\mathcal{T}_{\boldsymbol{A}} \le C\sqrt{n}$ is (62), proved under the standing balanced assumptions by Lemmas B.20 and B.21. Taking $\mathbb{E}_{\mathrm{aux}}$ now proves the claim. The left side is a function of $(\boldsymbol{X}, \boldsymbol{S})$ alone, whose law is unchanged by the auxiliary construction. □

When $b \asymp_{\mathbb{P}} n$, this rate corresponds only to $d - bt_{\boldsymbol{\Sigma}} = O_{\mathbb{P}}(\sqrt{n})$; the unnormalized discrepancy need not converge to zero. The balance assumption is also substantive: Lemmas B.12 and B.14 use both the proportional gap and the bounded positive sketch spectrum.

C. Complementary assessment

This appendix proves the conditional law for a general predictor fitted from compressed data.

C.1. Proof of Theorem 3.1

For an $\mathcal{F}_{\boldsymbol{S}}$-measurable coefficient vector $\widetilde{\boldsymbol{\beta}}$, write $\widetilde{\boldsymbol{u}} = \boldsymbol{y} - \boldsymbol{X}\widetilde{\boldsymbol{\beta}}$ and let $R(\widetilde{\boldsymbol{\beta}})$ denote the prediction risk in (1) evaluated at this predictor. Similarly, $\widehat{R}_c(\widetilde{\boldsymbol{\beta}})$ denotes the complementary risk estimate evaluated at this predictor. This risk is finite almost surely without unconditional moment assumptions; its defining conditional expectation is understood in the extended nonnegative sense.

Proof. Put $k = n - m > 0$ and $\boldsymbol{Q} = \boldsymbol{Q}_{\boldsymbol{S}}$. The full-row-rank assumption makes $\boldsymbol{S}\boldsymbol{S}^\top$ invertible. Hence $\boldsymbol{Q}$ is a Borel function of $\boldsymbol{S}$, and direct matrix multiplication gives

$$\boldsymbol{Q}^\top = \boldsymbol{Q}, \qquad \boldsymbol{Q}^2 = \boldsymbol{Q}, \qquad \boldsymbol{S}\boldsymbol{Q} = \boldsymbol{0}, \qquad \operatorname{rank}(\boldsymbol{Q}) = k.$$

The last identity follows because $\operatorname{range}(\boldsymbol{Q}) = \ker(\boldsymbol{S})$. Furthermore, $\mathcal{F}_{\boldsymbol{S}} = \sigma(\boldsymbol{S}, \boldsymbol{S}\boldsymbol{X}, \boldsymbol{S}\boldsymbol{\varepsilon})$ because $\boldsymbol{S}\boldsymbol{y} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{\beta}^* + \boldsymbol{S}\boldsymbol{\varepsilon}$.

Condition on $\boldsymbol{S} = \boldsymbol{s}$. The blocks $(\boldsymbol{s}\boldsymbol{X}, \boldsymbol{s}\boldsymbol{\varepsilon})$ and $(\boldsymbol{Q}\boldsymbol{X}, \boldsymbol{Q}\boldsymbol{\varepsilon})$ are jointly Gaussian. Their cross-covariances vanish: for feature columns j, ℓ,

$$\operatorname{Cov}((\boldsymbol{s}\boldsymbol{X})_{aj}, (\boldsymbol{Q}\boldsymbol{X})_{i\ell} \mid \boldsymbol{S} = \boldsymbol{s}) = \boldsymbol{\Sigma}_{j\ell}(\boldsymbol{s}\boldsymbol{Q})_{ai} = 0,$$

and the error-block cross-covariance is $\sigma^2 \boldsymbol{s}\boldsymbol{Q} = \boldsymbol{0}$; design–error cross-covariances vanish by independence. Thus the two blocks are conditionally independent given $\boldsymbol{S}$. In particular, after conditioning on the larger field $\mathcal{F}_{\boldsymbol{S}}$, the complementary block retains its Gaussian law conditional on $\boldsymbol{S}$. This law is allowed to depend on $\boldsymbol{S}$ through $\boldsymbol{Q}$.

Set $v = \boldsymbol{\beta}^* - \widetilde{\boldsymbol{\beta}}$. Since v is $\mathcal{F}_{\boldsymbol{S}}$-measurable and the summary is finite dimensional, the Doob–Dynkin lemma represents v as a measurable function of $(\boldsymbol{S}, \boldsymbol{S}\boldsymbol{X}, \boldsymbol{S}\boldsymbol{\varepsilon})$ almost surely. It is therefore fixed under the conditional law. The projected residual is

$$\boldsymbol{Q}\widetilde{\boldsymbol{u}} = \boldsymbol{Q}\boldsymbol{X}v + \boldsymbol{Q}\boldsymbol{\varepsilon}.$$

Given $\mathcal{F}_{\boldsymbol{S}}$, this vector is centered Gaussian, with covariance

$$(v^\top \boldsymbol{\Sigma} v)\boldsymbol{Q}\boldsymbol{Q}^\top + \sigma^2 \boldsymbol{Q}\boldsymbol{Q}^\top = R(\widetilde{\boldsymbol{\beta}})\boldsymbol{Q}.$$

The calculation includes $R(\widetilde{\boldsymbol{\beta}}) = 0$, when the conditional law is the point mass at zero. Both $\boldsymbol{Q}$ and $R(\widetilde{\boldsymbol{\beta}})$ are $\mathcal{F}_{\boldsymbol{S}}$-measurable, so the following expression defines a measurable conditional kernel. For every bounded Borel function φ, the regular conditional law satisfies

$$\mathbb{E}[\varphi(\boldsymbol{Q}\widetilde{\boldsymbol{u}}) \mid \mathcal{F}_{\boldsymbol{S}}] = \int_{\mathbb{R}^n} \varphi\left(\sqrt{R(\widetilde{\boldsymbol{\beta}})}\boldsymbol{Q}z\right) \mathrm{d}\mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_n)(z) \quad \text{almost surely.}$$

Thus, the Gaussian assertion in (3) is an identity of measurable conditional probability kernels.

For each fixed conditional value of $\boldsymbol{Q}$, the spectral theorem gives an orthonormal basis in which $\boldsymbol{Q}$ is diagonal with k ones and m zeros. The calculation depends only on the rank of $\boldsymbol{Q}$, so it does not require a measurable choice of basis. On the $\mathcal{F}_{\boldsymbol{S}}$-measurable event $\{R(\widetilde{\boldsymbol{\beta}}) > 0\}$, the squared norm of the displayed Gaussian vector divided by $R(\widetilde{\boldsymbol{\beta}})$ has the conditional χ^2_k law; on its complement, the criterion is zero. Equivalently, the conditional law of $\widehat{R}_c$ is that of $(R(\widetilde{\boldsymbol{\beta}})/k)Z$ for $Z \sim \chi^2_k$. Its conditional mean, second moment, and variance are respectively $R(\widetilde{\boldsymbol{\beta}})$, $(1 + 2/k)R(\widetilde{\boldsymbol{\beta}})^2$, and $2R(\widetilde{\boldsymbol{\beta}})^2/k$. These are finite for each fixed fitted summary. If $\mathbb{E}R(\widetilde{\boldsymbol{\beta}})^2 < \infty$, the same second-moment identity gives $\widehat{R}_c \in L^2$, so these conditional moments coincide with ordinary conditional expectation and variance. Finally, the continuous χ^2_k distribution has positive $\alpha/2$ and $1 - \alpha/2$ quantiles. Inverting the pivot on $\{R(\widetilde{\boldsymbol{\beta}}) > 0\}$ gives (4). □

D. Residual correction and sketch comparisons

This appendix proves Theorems 4.1–4.4 in main-text order, using the moment and trace calculations from Appendix B.

D.1. Proof of Theorem 4.1

Use the trace-normalized representative and coordinates in Sections B.1 and B.2.

D.1.1. Unobservable-estimator part of Theorem 4.1

Proof. By (37),

$$\Xi_1 = \boldsymbol{\varepsilon}^\top \boldsymbol{v} + \boldsymbol{\varepsilon}^\top \boldsymbol{\ell},$$
$$\Xi_2 = n\|\boldsymbol{q}\|^2 - \|\boldsymbol{v} + \boldsymbol{\ell}\|^2.$$

Since $\boldsymbol{u} + t_{\boldsymbol{\Sigma}}\boldsymbol{r} = \boldsymbol{\varepsilon} - \boldsymbol{v}$, direct expansion yields the exact Stein risk identity

$$\begin{aligned} &n\|\boldsymbol{q}\|^2 + \|\boldsymbol{\varepsilon}\|^2 - \|\boldsymbol{u} + t_{\boldsymbol{\Sigma}}\boldsymbol{r}\|^2 \\ &\quad = \Xi_2 + 2\Xi_1 + 2\boldsymbol{v}^\top\boldsymbol{\ell} + \|\boldsymbol{\ell}\|^2 - 2\boldsymbol{\varepsilon}^\top\boldsymbol{\ell}. \end{aligned} \tag{64}$$

The first two terms on the right have expected absolute value $O(\sqrt{n})$ by (38). Cauchy–Schwarz, (39), and $\mathbb{E}\|\boldsymbol{\varepsilon}\|^2 = n\sigma^2$ show that $\mathbb{E}|\boldsymbol{v}^\top\boldsymbol{\ell}| + \mathbb{E}|\boldsymbol{\varepsilon}^\top\boldsymbol{\ell}| = O(1)$ and $\mathbb{E}\|\boldsymbol{\ell}\|^2 = O(n^{-1})$. Finally, the case $k = 1$ of (15) gives $\mathbb{E}|\|\boldsymbol{\varepsilon}\|^2/n - \sigma^2| = O(n^{-1/2})$. Divide (64) by n, use $R = \sigma^2 + \|\boldsymbol{q}\|^2$, and take expectations to obtain the $\hat{R}_u$ bound in (5). □

D.1.2. Observable-estimator part of Theorem 4.1

Proof. The difference-of-squares identity gives

$$|\hat{R}_u - \hat{R}_o| \le \left| t_{\boldsymbol{\Sigma}} - \frac{d}{b} \right| B_{\boldsymbol{A}} \le \frac{B_{\boldsymbol{A}}\mathcal{T}_{\boldsymbol{A}}}{n}.$$

Equation (63) shows that its expectation is $O(n^{-1/2})$. Together with the bound for $\hat{R}_u$ from Section D.1.1, this proves the claim for $\hat{R}_o$. □

By Markov's inequality, a common constant $C < \infty$ can be chosen such that, for all sufficiently large n and every $\epsilon > 0$,

$$\mathbb{P}\left(|\widehat{R} - R| > \epsilon\right) \le \min\left\{1, \frac{C}{\epsilon\sqrt{n}}\right\}, \qquad \widehat{R} \in \{\hat{R}_u, \hat{R}_o\}.$$

D.2. Proof of Theorem 4.2

We retain the natural scalings of Section 4: $\boldsymbol{S}\boldsymbol{S}^\top = c_n\boldsymbol{I}_m$ for scaled orthogonal sketches and $S_{ki} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1/m)$ for dense Gaussian sketches. The calculations below also give a finite-sample expected risk and the Gaussian trace limits.

D.2.1. Scaled orthogonal case of Theorem 4.2

Proof. Condition on the projector $\boldsymbol{P}_{\boldsymbol{S}}$ and choose an orthogonal matrix that sends $\boldsymbol{P}_{\boldsymbol{S}}$ to $\mathbf{diag}(\boldsymbol{I}_m, \boldsymbol{0})$. Joint Gaussian rotational invariance reduces the problem to ordinary least squares based on the first m rows. Write the rotated design and noise as

$$\boldsymbol{Z} = \begin{pmatrix} \boldsymbol{Z}_1 \\ \boldsymbol{Z}_0 \end{pmatrix}, \qquad \boldsymbol{\varepsilon} = \begin{pmatrix} \boldsymbol{\varepsilon}_1 \\ \boldsymbol{\varepsilon}_0 \end{pmatrix},$$

where $\boldsymbol{Z}_1 \in \mathbb{R}^{m\times d}$. In this proof, define $\boldsymbol{u}_\perp \in \mathbb{R}^{n-m}$ as the coordinates of the complementary residual $\boldsymbol{Q}_{\boldsymbol{S}}\boldsymbol{u}$ in this orthonormal basis, so $\|\boldsymbol{u}_\perp\| = \|\boldsymbol{Q}_{\boldsymbol{S}}\boldsymbol{u}\|$. Put

$$\boldsymbol{q} = (\boldsymbol{Z}_1^\top\boldsymbol{Z}_1)^{-1}\boldsymbol{Z}_1^\top\boldsymbol{\varepsilon}_1, \qquad \boldsymbol{u}_1 = \boldsymbol{\varepsilon}_1 - \boldsymbol{Z}_1\boldsymbol{q}, \qquad \boldsymbol{u}_\perp := \boldsymbol{\varepsilon}_0 - \boldsymbol{Z}_0\boldsymbol{q}.$$

Then $R = \sigma^2 + \|\boldsymbol{q}\|^2$. Since $\boldsymbol{A} = c_n \boldsymbol{P}_{\boldsymbol{S}}$, we have $c_n t_{\boldsymbol{\Sigma}} = \mathrm{Tr}((\boldsymbol{Z}_1^\top \boldsymbol{Z}_1)^{-1})$, and hence

$$\hat{R}_u = \frac{(1 + c_n t_{\boldsymbol{\Sigma}})^2 \|\boldsymbol{u}_1\|^2 + \|\boldsymbol{u}_\perp\|^2}{n},$$
$$\hat{R}_o = \frac{\{m/(m-d)\}^2 \|\boldsymbol{u}_1\|^2 + \|\boldsymbol{u}_\perp\|^2}{n}.$$

Let $\boldsymbol{W}_1 = \boldsymbol{Z}_1^\top \boldsymbol{Z}_1$. Conditional on $\boldsymbol{Z}_1$, $\boldsymbol{q} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{W}_1^{-1})$. If $m > d+1$, the Wishart identity $\mathbb{E}(\boldsymbol{W}_1^{-1}) = \boldsymbol{I}_d/(m-d-1)$ therefore gives

$$\mathbb{E}R = \sigma^2 + \mathbb{E}\|\boldsymbol{q}\|^2 = \sigma^2 \left(1 + \frac{d}{m-d-1}\right). \tag{65}$$

Now assume additionally the proportional conditions in the theorem. The conditional Gaussian law gives $\boldsymbol{q}/\sigma \overset{\mathrm{d}}{=} \boldsymbol{W}_1^{-1/2} \boldsymbol{g}$, where $\boldsymbol{g} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_d)$ is independent of $\boldsymbol{W}_1$. The squared radius $U = \|\boldsymbol{g}\|^2 \sim \chi^2_d$ is independent of its direction and $\boldsymbol{W}_1$. By orthogonal invariance, the inverse quadratic form in this direction has the law of $(\boldsymbol{W}_1^{-1})_{11}$. The reciprocal is a Schur complement: the squared norm of the first Gaussian design column after projection off the remaining $d-1$ columns. It therefore has law χ^2_{m-d+1}. Writing this reciprocal as V, we have $U \perp V$ and

$$R \overset{\mathrm{d}}{=} \sigma^2 \left(1 + \frac{U}{V}\right), \qquad U \sim \chi^2_d, \quad V \sim \chi^2_{m-d+1}, \quad U \perp V.$$

Its inverse chi-square moments yield

$$\mathrm{Var}(R) = \frac{2\sigma^4 d(m-1)}{(m-d-1)^2(m-d-3)} = O(n^{-1}).$$

Each diagonal entry of $\boldsymbol{W}_1^{-1}$ has the law of the reciprocal of χ^2_{m-d+1}. Applying Cauchy–Schwarz to their centered sum gives

$$\mathbb{E}[c_n t_{\boldsymbol{\Sigma}}] = \frac{d}{m-d-1}, \qquad \mathrm{Var}(c_n t_{\boldsymbol{\Sigma}}) \le \frac{2d^2}{(m-d-1)^2(m-d-3)} = O(n^{-1}).$$

Moreover, $\boldsymbol{u}_1$ is the projection of $\boldsymbol{\varepsilon}_1$ onto the $(m-d)$-dimensional orthogonal complement of $\mathrm{col}(\boldsymbol{Z}_1)$. Thus, conditionally on $\boldsymbol{Z}_1$, $\|\boldsymbol{u}_1\|^2/\sigma^2 \sim \chi^2_{m-d}$. The pair $(\boldsymbol{Z}_0, \boldsymbol{\varepsilon}_0)$ is independent of $\boldsymbol{q}$, so, conditionally on $\boldsymbol{q}$, the coordinates of $\boldsymbol{u}_\perp = \boldsymbol{\varepsilon}_0 - \boldsymbol{Z}_0 \boldsymbol{q}$ are independent $\mathcal{N}(0, \sigma^2 + \|\boldsymbol{q}\|^2)$ variables. Hence $\|\boldsymbol{u}_\perp\|^2/(\sigma^2 + \|\boldsymbol{q}\|^2) \sim \chi^2_{n-m}$ conditionally on $\boldsymbol{q}$; when $m = n$, this is the degenerate zero-dimensional term. These identities and Chebyshev's inequality yield

$$c_n t_{\boldsymbol{\Sigma}} = \frac{d}{m-d} + O_{\mathbb{P}}(n^{-1/2}),$$
$$\|\boldsymbol{q}\|^2 = \frac{\sigma^2 d}{m-d} + O_{\mathbb{P}}(n^{-1/2}),$$
$$\|\boldsymbol{u}_1\|^2 = \sigma^2(m-d) + O_{\mathbb{P}}(\sqrt{n}),$$
$$\frac{\|\boldsymbol{u}_\perp\|^2}{n} = \frac{n-m}{n}(\sigma^2 + \|\boldsymbol{q}\|^2) + O_{\mathbb{P}}(n^{-1/2}).$$

These relations prove the scaled orthogonal risk limits in (7). For the GCV gap, note that

$$\frac{\|\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}\|^2}{n} \xrightarrow{\mathrm{p}} \sigma^2 \left\{(\rho - \gamma) + (1-\rho)\frac{\rho}{\rho - \gamma}\right\}.$$

Division by $(1-\gamma)^2$ and subtraction of $\mathcal{R}_{\mathrm{SO}}$ yield Δ_{SO} in (7). □

D.2.2. Dense Gaussian case of Theorem 4.2

Proof. Whiten the features and write the thin polar decomposition $\boldsymbol{Z} = \boldsymbol{U}\boldsymbol{R}$, where $\boldsymbol{R} = (\boldsymbol{Z}^\top \boldsymbol{Z})^{1/2}$. Choose a measurable orthogonal completion $[\boldsymbol{U}, \boldsymbol{V}]$ and define

$$\sqrt{m}\, \boldsymbol{S}[\boldsymbol{U}, \boldsymbol{V}] = [\boldsymbol{G}, \boldsymbol{H}], \qquad \boldsymbol{K} = (\boldsymbol{G}^\top \boldsymbol{G})^{-1} \boldsymbol{G}^\top \boldsymbol{H}, \qquad \boldsymbol{P}_{\boldsymbol{G}} := \boldsymbol{G}(\boldsymbol{G}^\top \boldsymbol{G})^{-1} \boldsymbol{G}^\top,$$

where the symbols $\boldsymbol{G}$ and $\boldsymbol{H}$ are local to this proof. Conditional on $\boldsymbol{Z}$, right-orthogonal invariance of the independent Gaussian sketch makes $[\boldsymbol{G}, \boldsymbol{H}]$ a standard Gaussian matrix whose law does not depend on $\boldsymbol{Z}$. Thus $\boldsymbol{G}$ and $\boldsymbol{H}$ are independent of one another and of $\boldsymbol{Z}$. Write $\boldsymbol{a} = \boldsymbol{U}^\top \boldsymbol{\varepsilon}$ and $\boldsymbol{c} = \boldsymbol{V}^\top \boldsymbol{\varepsilon}$. Conditional on $\boldsymbol{Z}$, the independently generated Gaussian sketch and noise make $\boldsymbol{G}, \boldsymbol{H}, \boldsymbol{a}, \boldsymbol{c}$ mutually independent, with $\boldsymbol{a}$ and $\boldsymbol{c}$ centered Gaussian vectors of covariance $\sigma^2 \boldsymbol{I}$ in their respective dimensions. Their joint conditional law does not depend on $\boldsymbol{Z}$, so these blocks are also jointly independent of $\boldsymbol{Z}$. Since $\boldsymbol{\varepsilon} = \boldsymbol{U}\boldsymbol{a} + \boldsymbol{V}\boldsymbol{c}$, in the $[\boldsymbol{U}, \boldsymbol{V}]$ coordinates,

$$\begin{aligned} \boldsymbol{u} &\equiv (-\boldsymbol{K}\boldsymbol{c}, \boldsymbol{c}), & \boldsymbol{A}\boldsymbol{u} &\equiv (\boldsymbol{0}, \boldsymbol{W}_\perp \boldsymbol{c}), \\ \boldsymbol{W}_\perp &:= \frac{1}{m} \boldsymbol{H}^\top (\boldsymbol{I}_m - \boldsymbol{P}_{\boldsymbol{G}}) \boldsymbol{H}, & b_{\boldsymbol{A}} &= \operatorname{Tr}(\boldsymbol{W}_\perp). \end{aligned} \tag{66}$$

To see the first identity, the sketched normal equations give $\boldsymbol{R}\boldsymbol{q} = \boldsymbol{a} + \boldsymbol{K}\boldsymbol{c}$; subtracting $\boldsymbol{Z}\boldsymbol{q}$ from $\boldsymbol{\varepsilon}$ leaves $(-\boldsymbol{K}\boldsymbol{c}, \boldsymbol{c})$ in these coordinates. The definition of $\boldsymbol{K}$ then gives $\boldsymbol{G}^\top(\boldsymbol{H} - \boldsymbol{G}\boldsymbol{K}) = 0$, which removes the upper coordinate of $\boldsymbol{A}\boldsymbol{u}$ and yields the second identity. Conditional on $\boldsymbol{G}$, choose an orthonormal basis for the range of $\boldsymbol{I}_m - \boldsymbol{P}_{\boldsymbol{G}}$. Multiplying $\boldsymbol{H}$ by that basis gives an $(m-d) \times (n-d)$ matrix $\boldsymbol{Y}$ of independent standard Gaussian entries, independent of $\boldsymbol{G}$, and $m\boldsymbol{W}_\perp = \boldsymbol{Y}^\top \boldsymbol{Y}$ in distribution. Consequently,

$$m b_{\boldsymbol{A}} \sim \chi^2_{(m-d)(n-d)}, \qquad \frac{b_{\boldsymbol{A}}}{n} \xrightarrow{\mathrm{p}} \frac{(\rho-\gamma)(1-\gamma)}{\rho}. \tag{67}$$

The second statement follows from the mean and variance of a chi-square variable. In particular, $d/b_{\boldsymbol{A}} \xrightarrow{\mathrm{p}} \tau_{\mathrm{G}}(\gamma, \rho)$, where

$$\tau_{\mathrm{G}}(\gamma, \rho) := \frac{\rho\gamma}{(\rho-\gamma)(1-\gamma)}.$$

Scalar distributional representations make the remaining residual limits explicit. On an auxiliary probability space, take independent variables

$$\begin{aligned} &T \sim \chi^2_{n-d}, && J \sim \chi^2_d, && V \sim \chi^2_{m-d+1}, \\ &U \sim \chi^2_{m-d}, && B \sim \chi^2_{n-d-1}. \end{aligned}$$

For all sufficiently large n, the following marginal distributional identities hold:

$$\begin{aligned} \|\boldsymbol{K}\boldsymbol{c}\|^2 &\overset{\mathrm{d}}{=} \sigma^2 \frac{TJ}{V}, & \|\boldsymbol{u}\|^2 &\overset{\mathrm{d}}{=} \sigma^2 T \left(1 + \frac{J}{V}\right), \\ \boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{u} &\overset{\mathrm{d}}{=} \frac{\sigma^2}{m} TU, & \|\boldsymbol{A}\boldsymbol{u}\|^2 &\overset{\mathrm{d}}{=} \frac{\sigma^2}{m^2} TU(U+B). \end{aligned}$$

No joint equality between the four displayed statistics is needed. For the first two identities, condition on $\boldsymbol{c}$. When $\boldsymbol{c} \neq \boldsymbol{0}$, choose an orthogonal $\boldsymbol{O} \in \mathbb{R}^{(n-d)\times(n-d)}$ with $\boldsymbol{O}^\top \boldsymbol{c} = \|\boldsymbol{c}\| \boldsymbol{e}_1$. Since $\boldsymbol{H}\boldsymbol{c} = \|\boldsymbol{c}\| \boldsymbol{H}\boldsymbol{O}\boldsymbol{e}_1$ and $\boldsymbol{H}\boldsymbol{O}$ has the same conditional Gaussian law as $\boldsymbol{H}$, $\boldsymbol{K}\boldsymbol{c}/\|\boldsymbol{c}\|$ has the Gaussian OLS coefficient law $\boldsymbol{g}/\sqrt{V}$, where $\boldsymbol{g} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_d)$ is independent of V and $\boldsymbol{c}$. The zero-vector case gives the same identities by continuity. Also, $\|\boldsymbol{c}\|^2 \overset{\mathrm{d}}{=} \sigma^2 T$. For the last two identities, use $m\boldsymbol{W}_\perp = \boldsymbol{Y}^\top \boldsymbol{Y}$ and rotate $\boldsymbol{c}$ to the first coordinate. The first column of $\boldsymbol{Y}$ has squared norm U. Conditional on that column, its inner products with the other columns are independent centered normal variables of variance U. Thus $\|\boldsymbol{Y}^\top \boldsymbol{Y}\boldsymbol{e}_1\|^2$ has law $U^2 + UB$, with B independent of U.

A chi-square variable with k_n degrees of freedom, divided by n, converges in probability to $\lim k_n/n$, since its variance is $2k_n/n^2$. Applying this fact to the preceding scalar laws gives

$$\frac{\|\boldsymbol{K}\boldsymbol{c}\|^2}{n} \xrightarrow{\mathrm{p}} \sigma^2 \frac{\gamma(1-\gamma)}{\rho-\gamma}, \qquad \frac{\|\boldsymbol{a} + \boldsymbol{K}\boldsymbol{c}\|^2}{n} \xrightarrow{\mathrm{p}} \sigma^2 \left\{ \gamma + \frac{\gamma(1-\gamma)}{\rho-\gamma} \right\}. \tag{68}$$

For the second limit, $\|\boldsymbol{a}\|^2/n \xrightarrow{\mathrm{p}} \sigma^2\gamma$ and the cross term vanishes in probability: independence and the inverse chi-square mean give

$$\mathbb{E}[(\boldsymbol{a}^\top \boldsymbol{K}\boldsymbol{c})^2] = \sigma^4 \frac{d(n-d)}{m-d-1} = O(n).$$

The three residual limits are

$$\begin{aligned}\frac{\|\boldsymbol{u}\|^2}{n} &\xrightarrow{\mathrm{p}} \sigma^2\frac{\rho(1-\gamma)}{\rho-\gamma},\\ \frac{\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{u}}{n} &\xrightarrow{\mathrm{p}} \sigma^2\frac{(\rho-\gamma)(1-\gamma)}{\rho},\\ \frac{\|\boldsymbol{A}\boldsymbol{u}\|^2}{n} &\xrightarrow{\mathrm{p}} \sigma^2\frac{(\rho-\gamma)(1-\gamma)(\rho+1-2\gamma)}{\rho^2}.\end{aligned} \tag{69}$$

These arguments include $\rho = 1$.

To control the population trace coefficient, put $\boldsymbol{W}_Z = \boldsymbol{Z}^\top\boldsymbol{Z}$ and $\boldsymbol{W}_G = \boldsymbol{G}^\top\boldsymbol{G}$. Polar coordinates give

$$t_{\boldsymbol{\Sigma}} = m\,\mathrm{Tr}(\boldsymbol{W}_G^{-1}\boldsymbol{W}_Z^{-1}), \qquad \mathbb{E}t_{\boldsymbol{\Sigma}} = \frac{md}{(m-d-1)(n-d-1)}.$$

Here the two Gram matrices are independent. A useful elementary bound, for any deterministic positive semidefinite $\boldsymbol{D}$, is

$$\mathbb{E}\left[\left\{\mathrm{Tr}(\boldsymbol{D}\boldsymbol{W}_G^{-1}) - \frac{\mathrm{Tr}(\boldsymbol{D})}{m-d-1}\right\}^2\right] \le \frac{2\{\mathrm{Tr}(\boldsymbol{D})\}^2}{(m-d-1)^2(m-d-3)}.$$

Indeed, diagonalize $\boldsymbol{D}$. Each unit-direction quadratic form in $\boldsymbol{W}_G^{-1}$ has law $1/\chi^2_{m-d+1}$, by the Schur complement and Gaussian orthogonal invariance. Cauchy–Schwarz bounds every covariance by the common variance $2/\{(m-d-1)^2(m-d-3)\}$, which proves the bound. Applying the same argument to $\boldsymbol{W}_Z$ gives

$$\mathbb{E}\{\mathrm{Tr}(\boldsymbol{W}_Z^{-1})\}^2 \le \frac{d^2}{(n-d-1)(n-d-3)}.$$

Conditional on $\boldsymbol{W}_Z$, split the centered trace coefficient into its fluctuation about $m\,\mathrm{Tr}(\boldsymbol{W}_Z^{-1})/(m-d-1)$ and the fluctuation of that conditional mean. The inequality $(x+y)^2 \le 2x^2+2y^2$ then yields

$$\begin{aligned}\mathbb{E}(t_{\boldsymbol{\Sigma}} - \mathbb{E}t_{\boldsymbol{\Sigma}})^2 \le{}& \frac{4m^2d^2}{(m-d-1)^2(m-d-3)(n-d-1)(n-d-3)}\\ &+ \frac{4m^2d^2}{(m-d-1)^2(n-d-1)^2(n-d-3)} = O(n^{-1}).\end{aligned}$$

Thus, together with (67),

$$\begin{aligned}t_{\boldsymbol{\Sigma}},\ \frac{d}{b_{\boldsymbol{A}}} &\xrightarrow{\mathrm{p}} \tau_{\mathrm{G}}(\gamma,\rho),\\ t_{\boldsymbol{\Sigma}} - \frac{d}{b_{\boldsymbol{A}}} &\xrightarrow{\mathrm{p}} 0.\end{aligned} \tag{70}$$

Finally, the whitened coefficient error is $\boldsymbol{q} = \boldsymbol{R}^{-1}(\boldsymbol{a}+\boldsymbol{K}\boldsymbol{c})$. The vector $\boldsymbol{a}+\boldsymbol{K}\boldsymbol{c}$ is independent of $\boldsymbol{Z}$. The same inverse-Gram directional law therefore gives

$$\|\boldsymbol{q}\|^2 \overset{\mathrm{d}}{=} \frac{\|\boldsymbol{a}+\boldsymbol{K}\boldsymbol{c}\|^2}{Q}, \qquad Q \sim \chi^2_{n-d+1},$$

where Q is independent of the vector on an auxiliary probability space. Since $Q/n \xrightarrow{\mathrm{p}} 1-\gamma > 0$, (68) gives

$$\|\boldsymbol{\Sigma}^{1/2}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*)\|^2 \xrightarrow{\mathrm{p}} \sigma^2\left\{\frac{\gamma}{1-\gamma} + \frac{\gamma}{\rho-\gamma}\right\}.$$

Adding σ^2 proves the risk limit.

Expanding either corrected risk estimate as

$$\frac{1}{n}\|\boldsymbol{u}+c\boldsymbol{A}\boldsymbol{u}\|^2=\frac{\|\boldsymbol{u}\|^2}{n}+2c\frac{\boldsymbol{u}^\top\boldsymbol{A}\boldsymbol{u}}{n}+c^2\frac{\|\boldsymbol{A}\boldsymbol{u}\|^2}{n}$$

and taking $c=t_{\boldsymbol{\Sigma}}$ or $c=d/b_{\boldsymbol{A}}$ proves the Gaussian risk limits in (7). For the GCV gap, combine

$$\mathrm{Tr}(\boldsymbol{X}\boldsymbol{M}^{-1}\boldsymbol{X}^\top\boldsymbol{A})=d$$

with (69) to obtain

$$\hat{R}_{\mathrm{GCV}}\xrightarrow{\mathrm{p}}\frac{\sigma^2\rho}{(\rho-\gamma)(1-\gamma)}.$$

Subtracting $\mathcal{R}_{\mathrm{G}}$ gives Δ_{G} in (7). □

D.3. Proof of Theorem 4.3

Proof. Condition on $\boldsymbol{S}$ and write $\boldsymbol{S}=\sqrt{c_n}\boldsymbol{F}$, with $\boldsymbol{F}\boldsymbol{F}^\top=\boldsymbol{I}_m$. As in Section D.2.1, whiten the feature coordinates and set

$$\begin{aligned}\boldsymbol{Z}_1&=\boldsymbol{F}\boldsymbol{X}\boldsymbol{\Sigma}^{-1/2}, & \boldsymbol{\varepsilon}_1&=\boldsymbol{F}\boldsymbol{\varepsilon},\\ \boldsymbol{q}&=\boldsymbol{\Sigma}^{1/2}(\hat{\boldsymbol{\beta}}_{\boldsymbol{S}}-\boldsymbol{\beta}^*)=(\boldsymbol{Z}_1^\top\boldsymbol{Z}_1)^{-1}\boldsymbol{Z}_1^\top\boldsymbol{\varepsilon}_1, & \boldsymbol{u}_1&=\boldsymbol{F}\boldsymbol{u}=\boldsymbol{\varepsilon}_1-\boldsymbol{Z}_1\boldsymbol{q}.\end{aligned}$$

The entries of $\boldsymbol{Z}_1$ are independent standard Gaussian variables, and $\boldsymbol{\varepsilon}_1\sim\mathcal{N}(\boldsymbol{0},\sigma^2\boldsymbol{I}_m)$ independently of $\boldsymbol{Z}_1$. The risk and the GCV criterion based on the m row-whitened sketch coordinates are

$$R=\sigma^2+\|\boldsymbol{q}\|^2, \qquad \hat{R}_{\mathrm{GCV},\boldsymbol{F}}:=\frac{\|\boldsymbol{F}\boldsymbol{u}\|^2/m}{(1-d/m)^2}=\frac{m}{k^2}\|\boldsymbol{u}_1\|^2.$$

These expressions remove the dependence on $\boldsymbol{\Sigma}$ and $\boldsymbol{\beta}^*$. Since $\|\boldsymbol{P}_{\boldsymbol{S}}\boldsymbol{u}\|^2=\|\boldsymbol{F}\boldsymbol{u}\|^2$, (6) gives

$$\hat{R}_o=\frac{q}{n}\widehat{R}_c+\frac{m}{n}\hat{R}_{\mathrm{GCV},\boldsymbol{F}}. \tag{71}$$

Given $\boldsymbol{Z}_1$, the Gaussian projections defining $\boldsymbol{q}$ and $\boldsymbol{u}_1$ are independent, with $\|\boldsymbol{u}_1\|^2/\sigma^2\sim\chi^2_k$. Since this conditional chi-square law does not depend on $\boldsymbol{Z}_1$ or $\boldsymbol{S}$, the residual sum of squares is independent of $(\boldsymbol{Z}_1,\boldsymbol{q},\boldsymbol{S})$. In particular, $\hat{R}_{\mathrm{GCV},\boldsymbol{F}}$ is independent of R.

To obtain the moments of R, write $\boldsymbol{W}_1=\boldsymbol{Z}_1^\top\boldsymbol{Z}_1\sim\mathrm{Wishart}_d(m,\boldsymbol{I}_d)$. The conditional Gaussian law of $\boldsymbol{q}$ gives the representation $\boldsymbol{q}/\sigma\overset{\mathrm{d}}{=}\boldsymbol{W}_1^{-1/2}\boldsymbol{g}$, where $\boldsymbol{g}\sim\mathcal{N}(\boldsymbol{0},\boldsymbol{I}_d)$ is independent of $\boldsymbol{W}_1$. The squared radius $U=\|\boldsymbol{g}\|^2\sim\chi^2_d$ is independent of its direction. Orthogonal invariance of $\boldsymbol{W}_1$ implies that the quadratic form of $\boldsymbol{W}_1^{-1}$ in this direction has the law of $(\boldsymbol{W}_1^{-1})_{11}$. The Schur-complement identity writes its reciprocal as the squared norm of the first design column after projection onto the orthogonal complement of the remaining $d-1$ columns. This projection has rank $m-d+1=k+1$, so the reciprocal has law χ^2_{k+1}. It is independent of U. Consequently,

$$\frac{R}{\sigma^2}\overset{\mathrm{d}}{=}1+\frac{U}{V}, \qquad U\sim\chi^2_d,\quad V\sim\chi^2_{k+1},\quad U\perp V.$$

For $k>3$, the chi-square moments yield

$$\begin{aligned}\frac{\mathbb{E}R}{\sigma^2}&=\frac{m-1}{k-1}, & \frac{\mathrm{Var}(R)}{\sigma^4}&=\frac{2d(m-1)}{(k-1)^2(k-3)},\\ \frac{\mathbb{E}\hat{R}_{\mathrm{GCV},\boldsymbol{F}}}{\sigma^2}&=\frac{m}{k}, & \frac{\mathrm{Var}(\hat{R}_{\mathrm{GCV},\boldsymbol{F}})}{\sigma^4}&=\frac{2m^2}{k^3}.\end{aligned}$$

It follows that $\mathbb{E}R^2=\sigma^4 e_{m,d}$. Independence of $\hat{R}_{\mathrm{GCV},\boldsymbol{F}}$ and R, together with $\mathbb{E}[\hat{R}_{\mathrm{GCV},\boldsymbol{F}}-R]=-\sigma^2 d/\{k(k-1)\}$, gives

$$\mathbb{E}[(\hat{R}_{\mathrm{GCV},\boldsymbol{F}}-R)^2]=\sigma^4 a_{m,d}.$$

Thus the three terms defining $a_{m,d}$ are the variance of the GCV risk estimate, the variance of R, and their squared mean difference, each divided by σ^4.

By Theorem 3.1,

$$\mathbb{E}[\widehat{R}_c - R \mid \mathcal{F}_{\boldsymbol{S}}] = 0, \qquad \mathbb{E}[(\widehat{R}_c - R)^2 \mid \mathcal{F}_{\boldsymbol{S}}] = \frac{2R^2}{q}.$$

The preceding moment formulas show that both estimation errors are square integrable, so their product is integrable by Cauchy–Schwarz. Both R and $\hat{R}_{\mathrm{GCV},\boldsymbol{F}}$ are $\mathcal{F}_{\boldsymbol{S}}$-measurable, so

$$\mathbb{E}[(\widehat{R}_c - R)(\hat{R}_{\mathrm{GCV},\boldsymbol{F}} - R)] = 0.$$

Subtracting R from (71) now gives

$$\hat{R}_o - R = \frac{q}{n}(\widehat{R}_c - R) + \frac{m}{n}(\hat{R}_{\mathrm{GCV},\boldsymbol{F}} - R).$$

Squaring this identity and taking expectations proves (8). Dividing its two expressions gives the exact ratio

$$\frac{\mathrm{MSE}(\hat{R}_o)}{\mathrm{MSE}(\widehat{R}_c)} = \frac{q^2}{n^2} + \frac{m^2 q\, a_{m,d}}{2n^2 e_{m,d}},$$

which proves (9). □

D.4. Proof of Theorem 4.4

Proof. Substituting $d = n\gamma_n$ and $m = n\rho_n$ into the exact MSE formulas from Theorem 4.3, then expanding their ratio in n^{-1}, gives

$$\begin{aligned}\frac{\mathrm{MSE}(\hat{R}_o)}{\mathrm{MSE}(\widehat{R}_c)} &= (1-\rho_n)\left(1 + \frac{2\gamma_n\rho_n}{\rho_n - \gamma_n}\right) \\ &\quad - \frac{5\gamma_n^2(1-\rho_n)}{2n(\rho_n-\gamma_n)^2} + O(n^{-2}),\end{aligned} \tag{72}$$

uniformly when (γ_n, ρ_n) stays in a compact subset of $\{(g,r) : 0 < g < r < 1\}$. The limiting ratio minus one equals

$$\frac{\rho\{3\gamma - (1+2\gamma)\rho\}}{\rho - \gamma}.$$

Its sign proves the limiting threshold and the stated MSE ordering.

For the local finite-sample crossover, suppose γ_n is bounded away from zero and one. The points $(\gamma_n, \rho_*(\gamma_n))$ then stay in a compact subset of the same interior domain. The expansion (72) also holds after one derivative in ρ_n, since the exact ratio is rational and its denominators stay away from zero after normalization. At $\rho = \rho_*(\gamma)$, the derivative of the leading ratio is

$$-\frac{3(1+2\gamma)}{2(1-\gamma)},$$

which is uniformly bounded away from zero on such compact sets. Apply the implicit function theorem to the exact ratio minus one, with parameters (n^{-1}, γ_n), at each point $(0, \gamma, \rho_*(\gamma))$. Its nonzero derivative gives a smooth local solution of the MSE equality. Compactness gives a common neighborhood and a common upper bound on n^{-1}; the negative derivative gives uniqueness in that neighborhood. The coefficient of n^{-1} in (72), evaluated at the limiting crossover, is $-5(1+2\gamma)/\{8(1-\gamma)\}$. Taylor expansion of the equality about $\rho_*(\gamma_n)$ gives a displacement of order n^{-1}; its coefficient is the negative ratio of the n^{-1} coefficient to the derivative, namely $-5/12$. The local solution has bounded second derivatives on a smaller compact neighborhood, so Taylor's theorem gives a uniform second-order remainder and proves (10). □